Confrontation Costs in Negotiations: Bargaining Under the Veil of a Screen *

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Abstract

Haggling frequently involves confrontation over conflicting objectives. In a series of experiments, I reveal that more than 40% of subjects forgo profitable in-person negotiations to avoid the unpleasant emotions arising from this confrontation. My findings demonstrate that digital bargaining significantly alleviates these emotions, halving negotiation avoidance and thus facilitating agreements that would otherwise be hindered. This challenges the conventional wisdom that face-to-face bargaining improves welfare. Moreover, I find that women, who face elevated confrontation costs, can especially benefit from bargaining electronically. A survey I run on a representative sample of the US population confirms the experimental conclusions and finds widespread use of digital negotiations.

Keywords: bargaining, conflict aversion, social pressure, image concerns, gender.

JEL codes: C78, C91, D91, J16

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1. Introduction

During the past decade, electronic negotiations have experienced a tremendous growth. By 2021, consumer-to-consumer (c2c) platforms such as Craigslist and Facebook Marketplace counted over a billion monthly active users in more than 70 countries. These platforms, where users bargain through an electronic chat, are commonly used to trade furniture, tools and vehicles, and in the US, they are already the most frequently used device for house rentals (Boeing, 2020; Besbris, Schachter and Kuk, 2021). Similarly, in the job market, it is not uncommon to start a negotiation by e-mail or LinkedIn, and people frequently use WhatsApp or Instagram to bargain over the price of services with contractors or other professionals.

According to the existing evidence, this technological change appears detrimental to welfare. In a meta-analysis of the effect of communication on cooperation, Balliet (2010) finds that face-to-face communication is more effective than written communication at fostering cooperation. Consistently, roughly 90% of Harvard Business Review subscribers report that face-to-face meetings are the most effective way to seal a deal (HBR Analytic Services, 2009). Since agreements generate gains from trade, this greater ease of reaching agreements in person has always been interpreted as evidence that face-to-face negotiations are welfare-improving (see Boothby, Cooney and Schweitzer 2023 for a review). However, what if some people find haggling in person too emotionally costly and thus avoid potentially profitable interactions?

Psychology studies have found that bargaining triggers feelings of anxiety and apprehension (Babcock et al., 2013; Bowles, Babcock and Lai, 2007; Brooks and Schweitzer, 2011). Yet, the question of how such feelings shape individuals' willingness to bargain remains unanswered. Here, I conduct three experiments and a survey to examine how the emotions produced by the conflict inherent in bargaining, which I denote *confrontation costs*, influence the decision to negotiate. By doing so, this paper is the first to study the effect of emotions on the decision to enter negotiations (see Lelieveld and van Dijk 2022 for a review). Moreover, I study how confrontation costs vary across communication channels and by gender, and how this variation affects negotiation avoidance.

In recent years, with the rapid development of online markets, an incipient literature studying online bargaining has emerged (see Backus, Blake and Tadelis 2022). However, previous papers exploring the differences between digital and in-person negotiations have always neglected both the decision to enter negotiations and confrontation costs (see reviews in Friedman and Belkin 2013 and Geiger 2020). Accounting for these new elements, I re-evaluate whether digital negotiations actually harm welfare. This paper is the first to go beyond the known effect of online marketplaces reducing search costs to explore how they can affect bargaining costs—and through them—negotiation engagement and transaction rates. Moreover, using a gender-balanced design, I explore the role that confrontation costs play in the gender gap in bargaining propensity—which explains part of the gender wage gap (e.g., Biasi and Sarsons, 2022)—and whether electronic communication reduces the gender difference in the willingness to bargain.

Experiment 1 begins with a simple bargaining environment. In the baseline treatment, participants must decide whether to engage in a negotiation to split a given amount of money, or whether to opt out. Subjects are randomly assigned to a face-to-face or electronic session, which varies the communication channel. To simplify the analysis, only one member of each bargaining pair (the decider) is allowed to

choose whether or not the negotiation takes place. If he chooses not to bargain, both members of the pair wait alone in front of their computers for three minutes and directly receive their outside options. If instead the decider chooses to negotiate, the pair will have three minutes to reach an agreement in a free-form negotiation. If no agreement is reached, both participants receive their disagreement payoffs, which coincide with their outside options. The negotiation pie is larger than the sum of the disagreement payoffs, so there are gains from trade, and all the information is public and symmetric. Thus, if bargaining is costless, opting in must be weakly dominant even for infinitely risk-averse deciders—as their outside option, which coincides with the disagreement payoffs, is secured.

I find a positive opt-out rate of roughly 40% face to face and 20% electronically, indicating that bargaining entails non-monetary costs. Moreover, the higher opt-out rate in person offers an important insight. In line with previous studies, when I analyze the content of negotiations, I find that participants are more likely to reach an agreement in person. However, once their greater reluctance to engage in bargaining is taken into account, the overall agreement rate does not differ significantly across communication conditions—and its point estimate is higher electronically. This means that, in my experiment, face-to-face negotiations do not enable better capture of gains from trade—and thus, do not lead to higher monetary welfare—challenging the conventional wisdom. Indeed, the greater proportion of subjects who choose to negotiate in the digital condition reveals that the share of deciders expecting positive utility from bargaining is actually higher electronically than in person.

Building on the observation that the proportion of subjects expecting positive utility from the negotiation is greater electronically, a natural question arises: why? Is it due to costs, or is it due to expected profits? Note that if deciders believe they are more persuasive electronically—and therefore expect to secure better deals—then they would engage in bargaining more often electronically than in person even if costs are the same. To distinguish between these two motives, I use a between-subjects design to introduce a second treatment that makes the terms of agreement a choice variable for the decider—keeping every other aspect identical to the baseline treatment. In this new treatment, deciders who choose to bargain engage in a negotiation in which they can impose how to split the pie at any point. Notably, I find that the opt-out rate is still significantly higher in person (32%) than electronically (17%). This reveals that even in a situation where deciders control their profits and do not *need* to exert any effort to persuade their counterparts, the confrontation component of the negotiation still makes them more reluctant to bargain face to face.

Once the potential of digital negotiations to lower confrontation costs and encourage negotiations is established, I examine how deciders respond when bargaining is mandatory. I infer the behavior of negotiation-avoiders by comparing the average outcomes in the baseline treatment against a treatment that forces participants to negotiate. I find that, on average, deciders in the forced treatment modulate their aggressiveness. They take the initiative less often than in the baseline and demand a smaller share of the pie, especially in person. Assuming that subjects who opt in under the baseline treatment would bargain similarly in the forced treatment, this suggests that subjects who would prefer to opt out adopt a softer haggling strategy—probably as a second-best alternative to mitigate confrontation costs. Crucially, this results in many deciders earning less than their disagreement payoffs in the face-to-face condition of the forced treatment, suggesting that the high agreement rate that previous studies observe in person could stem from participants' inability to opt out: Negotiation-avoiders,

when forced to bargain, are essentially caught in a *bargaining trap* (Schweighofer-Kodritsch, 2022). Unless we endogenize selection into bargaining, we cannot take the deal rate as a sufficient statistic for total welfare.

To benchmark the effect of image concerns on the opt-out choice, a dictator game with a discrete and costly exit option is conducted (see Dana, Cain and Dawes 2006; Lazear, Malmendier and Weber 2012). In this treatment, deciders can avoid a distributive situation analogous to the one posed in negotiations, but without any interaction (only a photo disclosure). The opt-out rate I find is 16%, indicating that image concerns are relevant but do not suffice to explain participants' reluctance to bargain, particularly in person. Additionally, to test whether the stronger aversion to face-to-face negotiations extends to any form of in-person interaction, I conduct a conversation treatment in which participants decide whether to meet and converse for a randomly allocated payment. I find an opt-out rate of 5% in person and 4% electronically, ruling out this hypothesis.

At the end of Experiment 1, all participants complete the TKI® (Kilmann and Thomas, 1977) and fill out a debriefing survey. The TKI® is the leading psychological test for assessing how individuals handle conflict. I find a positive correlation between a high conflict avoidance score and opting out of negotiations. This finding is further supported by the debriefing survey, in which most deciders report that the negative emotions associated with confronting their counterpart were their primary reason for opting out.[1]

To assess the robustness of the conclusions, in Experiment 2, I use a design similar to Experiment 1, with a few important modifications. This experiment follows a within-subject design in which deciders choose whether to opt in to or out of a face-to-face negotiation, and whether to opt in to or out of an electronic negotiation. Afterwards, the computer rolls a die: if it rolls a 1, their choice to opt in to or out of the face-to-face negotiation is implemented; if it rolls a 2, their choice to opt in to or out of the electronic negotiation is implemented; if it rolls a 3, they are forced to bargain face-to-face, and if it rolls a 4, they are forced to bargain electronically. Right after deciders make their choices and before they bargain, I adapt the chatbot developed in Chopra and Haaland (2023) to interview participants about the motivations behind those choices. Then, they observe the die roll and I elicit their beliefs about negotiation outcomes. Negotiations take place for those who bargain, and the experiment concludes with a second interview gathering additional information about participants' experiences as well as a debriefing survey similar to the one in Experiment 1. All other important features of Experiment 2 remain unchanged from Experiment 1.

The within-subject design allows me to classify participants into four types: always bargainers (opt in both electronically and in person, 49%), old-school bargainers (opt in face to face but opt out electronically, 5%), e-bargainers (opt in electronically but opt out in person, 29%), and never bargainers (opt out in both conditions, 17%). The interview responses confirm that confrontation costs are the main self-reported motive behind negotiation avoidance: around 70% of e-bargainers and never bargainers mention them to explain their choices. Likewise, deciders systematically associate favorable emotions with the communication channels in which they choose to bargain, and unfavorable emotions with

[1] Consistently, in the forced treatment, the intensity of the negative emotions reported by deciders is significantly greater in person than electronically—a pattern that does not extend to the conversation treatment.

those they prefer to avoid. The analysis of the beliefs and negotiation outcomes of those deciders who bargain unwillingly confirms the conclusions drawn from the forced treatment in Experiment 1. Negotiation avoiders trade off monetary payoffs against confrontation costs if they are forced to bargain: they prefer to lose money relative to their disagreement payoffs rather than confront and disagree.

To assess the external validity of the results, I run the baseline and asymmetric treatments of Experiment 1 in Madrid, reaching the same conclusions. Moreover, I ran Experiment 3 in Buenos Aires, which allows me to quantify the size of the costs and extend the results to a context with asymmetric information but more symmetric disagreement payoffs. In Experiment 3, participants are randomly assigned the role of buyers or sellers, and buyers can use their endowment to purchase a mug from the sellers in a five-minute negotiation. If no agreement is reached, buyers take their endowment home. I implement a Becker-DeGroot-Marschak procedure to elicit the minimum payment buyers would accept to avoid the negotiation, allowing me to use the difference between the buyer's endowment and the elicited payment to estimate (a lower bound for) the cost distribution. Participants are randomly assigned to either a face-to-face or electronic session. The results in Experiment 3 strongly reject the hypothesis that the cost distribution is the same across communication conditions. I find that while the median subject incurs no cost in electronic negotiations, in person he is willing to forgo nearly the equivalent of an hour's work at the minimum wage to avoid bargaining. Moreover, I find a positive association between the avoidance score on the TKI® and the elicited cost of bargaining, confirming that the conclusions of Experiments 1 and 2 are not specific to a particular scenario, but are consistent across different settings.

Finally, I further investigate the generalizability of the findings outside the lab and measure the prevalence of digital negotiations conducting a survey on negotiation habits in a representative sample of the US population. The results confirm that bargaining produces negative feelings in more than 80% of respondents. Moreover, they show that these feelings are more intense in person for nearly all of them (93%). Consequently, roughly half of the sample report they would bargain electronically for a desired item priced above their budget constraint, while only 20% would attempt to do so in person. When subjects describe their actual usage of c2c platforms, I find that over 80% use them, and 70% of the users report having engaged in digital bargaining. This indicates the widespread prevalence of electronic negotiations in the US, confirming that the present paper is delving into a major—yet understudied—phenomenon.

The analysis of the results by gender—both in the experiments and in the survey—also shows that women face higher confrontation costs, which makes them more reluctant to bargain than men. Moreover, although I find that both men and women are more reluctant to bargain in person, electronic communication proves effective at mitigating the gender gap in bargaining propensity, even if it does not eliminate it completely.

Related Literature—Experimental economics has long studied bargaining as a problem of surplus division, examining how focal points, fairness concerns or bargaining protocols shape negotiations (see Roth 1995 for a classic review). A central lesson from this literature is that even in simple environments with complete contracts, complete information, and clear gains from trade, agreements may fail to emerge, despite the prediction of canonical models (Nash, 1950; Kalai and Smorodinsky, 1975; Rubinstein, 1982). We know that such breakdowns could be linked to conflicting views about what

constitutes an acceptable agreement (e.g., Babcock and Loewenstein 1997). Nevertheless, the economics literature has paid very little attention to the emotional aspect of bargaining in general (Lelieveld and van Dijk, 2022), and to the negative feelings associated with conflict and disagreement in particular. This paper introduces the concept of confrontation costs and studies their effect on the decision to engage in negotiations and on negotiation outcomes.

A suitable anthropological background for this new concept of confrontation costs can be found in the Politeness Theory developed by Brown and Levinson (1987). These authors define "positive face" as *the need for one's wishes and desires to be recognized and appreciated*, and "negative face" as *an individual's need for freedom of action and choice*. Brown and Levinson propose that people suffer both from losing and from threatening others' face, and thus they try to avoid such emotionally costly situations. This desire to save face has been studied by economists in the context of charitable giving both under the denomination of "social pressure" (DellaVigna, List and Malmendier, 2012) and "avoiding the ask" (Andreoni, Rao and Trachtman, 2017). These papers show that individuals are willing to exert effort to avoid interactions in which they may be asked for a donation, thereby escaping a face-threatening situation. In this paper, I document that a similar phenomenon affects bargaining, extending the relevance of face-saving strategies beyond the scope of altruistic behavior, to contexts in which subjects avoid mutually beneficial interactions to avoid confrontation.[2]

Prior to my study, in the psychology literature, Miles (2010) has theoretically discussed the role that preserving face may play in the decision to bargain. Likewise, empirical studies have shown that negotiators can feel offended when they receive an aggressive offer (Schweinsberg et al., 2012; Lee et al., 2018) and that they suffer when their counterpart disapproves of their behavior (Chen and Kamei, 2018). Here, I bridge the gap, showing that these negative emotions that arise from confronting the other party deter individuals from engaging in negotiations.[3] Moreover, I find that this phenomenon is sensitive to the communication channel. Crucially, this introduces a new perspective to the study of digital negotiations, illustrating that the efficiency gains of bargaining in person (Greiner, Caravella and Roth, 2014; Backstrom et al., 2023) may vanish once we account for the bargaining decision and for confrontation costs.

Although the opt-out choice has been seldom considered in the bargaining literature, a notable exception is gender studies. Papers such as Babcock et al. (2003, 2013) and Bowles, Babcock and Lai (2007) have found that women are more reluctant to bargain than men—a tendency that contributes to the gender wage gap (e.g. Card, Cardoso and Kline, 2016; Biasi and Sarsons, 2022; Dreber, Heikensten and Säve-Söderbergh, 2022). This has sparked a debate on whether women should be encouraged to lean in (see Exley, Niederle and Vesterlund 2020 and Ashraf et al. 2020, for two different perspectives on the matter). My paper is the first to study how the communication channel affects the gender gap in bargaining propensity and how it affects negotiation outcomes.[4] Moreover, by examining the roots of

[2]More broadly, this also connects to the literature on information avoidance (see Golman, Hagmann and Loewenstein, 2017), which shows people forgo useful information when it can trigger negative emotions (e.g., avoid visiting the doctor).

[3]Using a different approach, recent studies have estimated the magnitude of bargaining initiation costs from transaction data on home appliances, mortgages, and cellphones, applying structural models (see Jindal and Newberry, 2018; Allen, Clark and Houde, 2019; Zhang, Manchanda and Chu, 2021, respectively).

[4]In a recent working paper, Greenberg and Petrie (2024) also compare gender differences in various communication conditions, but they do not endogenize the entry decision; they force participants to bargain.

women's reluctance to bargain, it informs interventions aimed at reducing the gender bargaining gap in the direction proposed by Recalde and Vesterlund (2023): "fixing the institutions" rather than "fixing women." According to my results, if we seek to encourage women to bargain, we should promote less confrontational and/or digital environments.

The following section presents the baseline treatment of Experiment 1. Next, Section 3 introduces the four additional treatments that dig deeper into the mechanisms, Section 4 explores the robustness of the results, Section 5 studies the results by gender and Section 6 explores the external validity of the findings. Section 7 concludes.

2. Experiment 1 - The Baseline

2.1. Design of the Baseline Treatment

In this treatment, participants are randomly assigned to an electronic or a face-to-face session. Within the session, they are randomly assigned the role X (decider) or Y (counterpart). They play a game with complete information. The decider must choose whether to bargain with his counterpart over how to split a pie P or whether to avoid the negotiation. If the decider chooses to bargain with the counterpart, they will have three minutes to negotiate freely. If they make a deal, each will receive the amount they arranged; if not, they will receive their disagreement payoffs (denoted by x for the decider, and y for the counterpart). If the decider chooses to avoid the negotiation, the decider and the counterpart will wait for three minutes in front of their computers and directly get their outside options, which coincide with their disagreement payoffs (x and y).[5]

Each participant takes part in four rounds. In each round, the decider is paired with a different anonymous counterpart and has to decide whether or not to opt in to the negotiation. As shown in Table 1, the amount of money to split and the disagreement payoffs differ across rounds, but the sum of the disagreement payoffs is always smaller than the amount they receive to split ($x + y < P$). This means that there are always gains to be made from trade.

TABLE 1

ROUND PAYOFFS

	Money to Split (P)	Decider Outside Option (x)	Counterpart Outside Option (y)	Gains from Trade (P-x-y)
Round A	1,800	1,400	200	200
Round B	1,800	1,200	200	400
Round C	1,800	1,200	400	200
Round D	1,500	1,200	100	200

Note: In Experiment 1, the decider chooses whether or not to bargain in four rounds (making the decision for the four rounds at the same time). In each round, the amount of money to split and/or the outside options vary. The first column reports the money to split in each round (P), the second column reports the decider's outside option (x), and the third column reports the counterpart's outside option (y). In all rounds the outside options coincide with the disagreement payoffs.

The timing of the experiment is as follows. First, instructions are read aloud (see a sample of instructions in Online Appendix F). Then, participants answer some control questions (see Online

[5]See Online Appendix Figure A.1 for a diagram which explains graphically the functioning of the experiment.

Appendix G). Once any doubts have been resolved and all participants have answered the control questions correctly, participants learn their role, which is maintained throughout the experiment. The four rounds are presented to deciders, who choose whether or not to bargain in each round. They make their decision for the four rounds simultaneously, before any negotiation takes place (see the decision screen in Online Appendix Figure A.2). Once all deciders have made their choices, round order is randomly determined and the first round starts. Those deciders that have chosen to bargain negotiate for three minutes with their counterparts. The rest wait for three minutes in silence in front of their computers. This process is repeated in subsequent rounds until the four rounds are over. Finally, all participants respond to a debriefing survey and to the TKI® psychological test, sign their receipt, and leave the lab.

A key aspect of the design is that for subjects with a genuine interest in others' wellbeing—what Cappelen et al. (2017) call intrinsic fairness concerns—opting in and splitting the money evenly dominates.[6] Likewise, in the absence of non-monetary bargaining costs, opting in is also the dominant strategy for self-regarding subjects. However, this might not be the case for self-regarding deciders who suffer from conflict. We know that the uneven outside options that the decider and his counterpart face generate multiple focal points where participants can coordinate. Crucially, when there are competing focal points, self-serving beliefs make conflicts and disagreements likely (Babcock and Loewenstein, 1997; Hennig-Schmidt et al., 2018; Karagözoğlu and Kocher, 2019; Ingersoll and Roomets, 2020). Thus, in this experiment, self-regarding deciders who suffer confrontation costs must choose between pushing for the only focal point that makes them better off (dividing the bargaining surplus in half), or avoiding conflict.[7] This enables me to identify the existence of non-monetary costs from the entry decision. Moreover, restricting the entry choice to deciders, who suffer the strongest tension between making profits and avoiding conflict, allows me to study this decision while avoiding the need to elicit beliefs about selection into negotiations.[8]

Electronic condition

Participants are randomly assigned to an electronic or face-to-face session. In the electronic sessions, negotiations are conducted in a digital chat on the lab computers (see a photo of the lab in Online Appendix Figure A.3). Negotiations are completely unstructured: any participant can make, withdraw

[6]The same is true for subjects who are not intrinsically fair-minded, but suffer guilt or shame if they do not do the right thing (Charness and Dufwenberg, 2006; Dillenberger and Sadowski, 2012).

[7]The work of Binmore, Shaked and Sutton (1989) identifies two distinct focal points in these type of negotiations: split-the-difference and deal-me-out. In the former, subjects split gains from trade equally, while in the latter, gains from trade go fully to the unfavored party. Moreover, the extensive literature on other-regarding behavior shows that dividing the pie in half (equal-split) represents a third possible focal point (see Cooper and Kagel 2016 for a survey). Thus, deciders only make money relative to their outside option in one of the focal points identified in the literature (see Cox et al. 2026 for a recent theory of moral reference points). This does not hold true for the counterpart, who can make money relative to their outside option in any of the focal points (i.e., he can avoid conflict and still make more than in his outside option, which would make his entry choice somewhat trivial).

[8]The fact that only one party can choose to start the negotiation resembles the situation of a buyer purchasing in a market with multiple sellers, who list their products and wait for potential customers. In a market with multiple sellers, the buyer observes the listing price of the first seller he encounters. Then, he can choose to open a negotiation, offering to share the search costs he could save if they reach a deal, or he can continue searching. Thus, in a physical bazaar, the buyer would decide between bargaining for a particular product or exploring other stands. In a c2c digital marketplace, the buyer would choose between bargaining for any of the currently listed items or waiting a few days until another seller lists a similar product at a lower price.

or accept offers at any point during the three minutes, and the content of the conversation is absolutely free. For a negotiation to be closed with an agreement, one of the parties has to accept the other's offer within the three minutes. If the parties reach an agreement before the three minutes are up, the chat remains open until the time is over; if the three minutes go by without an agreement, the negotiation ends and participants earn their disagreement payoffs.[9] To ensure that differences in behavior across communication conditions could not be attributed to identity revelation, the negotiation screen includes a photo of the participant's bargaining partner (see the bargaining screen in Online Appendix Figure A.4).

Face-to-face condition

In the face-to-face sessions, participants that choose to bargain are led from the lab to the bargaining room next to the lab, where they are seated in pairs (see a photo of the bargaining room in Online Appendix Figure A.5). The functioning of the negotiation is completely analogous to the electronic condition: those pairs who reach an agreement within the three minutes wait at the bargaining table until the time is over—and can continue the conversation or remain silent, while if the three minutes go by without an agreement, participants earn their disagreement payoffs. Like in the electronic condition, all negotiations are recorded.

2.2. Experimental Procedure

Between November 2021 and March 2022, 712 participants were recruited using ORSEE (Greiner, 2015). The experiment was conducted in the mobile lab of Universidad Torcuato Di Tella (Buenos Aires) using z-Tree (Fischbacher, 2007). 180 subjects were randomly assigned to the baseline treatment and each subject was randomly assigned to a face-to-face or electronic session. The average age of participants was 22.7 years, they had different backgrounds (business, medicine, no college...) and roughly 60% were female. Deciders' gender was balanced across communication conditions. The length of the experiment was roughly 90 minutes from the start until everyone had signed their receipt and left the lab, and average earnings per participant were 1,231 AR$, including a 400 AR$ show-up fee. To put these figures in context, at the time of the experiment, the average official exchange rate was 99.5 AR$ per US$ and the monthly Netflix subscription fee was 279 AR$ plus taxes.

2.3. Results of the Baseline Treatment

I define the avoidance rate as the share of rounds in which the decider chooses not to bargain. Table 2 presents the mean avoidance rate, the probability of agreement conditional on bargaining, and the average total payoff per pair. The results show a positive avoidance rate both electronically and face to face, indicating that bargaining entails non-monetary costs. Moreover, a comparison of the communication conditions reveals that participants are significantly more reluctant to bargain when the negotiation is in person (avoidance rate is 36.9% face to face vs 19.6% electronically; p=0.010).

I find that the probability of sealing the deal *conditional on bargaining* is significantly higher in person than electronically (95.2% vs 85.2%; p=0.029). Previous studies that neglected the opt-out decisions

[9]For a discussion on the advantages of unstructured bargaining protocols, see Camerer, Nave and Smith (2019) and Karagözoğlu and Kocher (2019).

TABLE 2

BASELINE TREATMENT

	Electronic	Face to Face	Difference
Avoidance (%)	19.6	36.9	-17.4**
	(3.9)	(5.4)	(6.6)
Agreement (%)	85.2	95.2	-10.0**
	(4.0)	(1.9)	(4.5)
Total Payoff ($)	6608.7	6518.2	90.5
	(43.8)	(49.9)	(66.4)
Num. Deciders	46	44	

Note: This table reports the average avoidance rate, the average rate of agreement for those negotiations that occur, the average total monetary payoff per pair expressed in Argentinian pesos, and the number of deciders in the baseline treatment for each communication condition. It also reports the difference between the two communication conditions. * p<0.1, ** p<0.05, *** p<0.01 correspond to the p-values of two-sided t-tests.

have interpreted this greater ease of reaching agreements in person as evidence of the superiority of in-person negotiations (see for example Valley, Moag and Bazerman, 1998; Valley et al., 2002). However, the greater reluctance to bargain that I also identify in face-to-face negotiations poses the question of which of the two effects prevails. Are in-person negotiations still superior once we account for the bargaining decision?

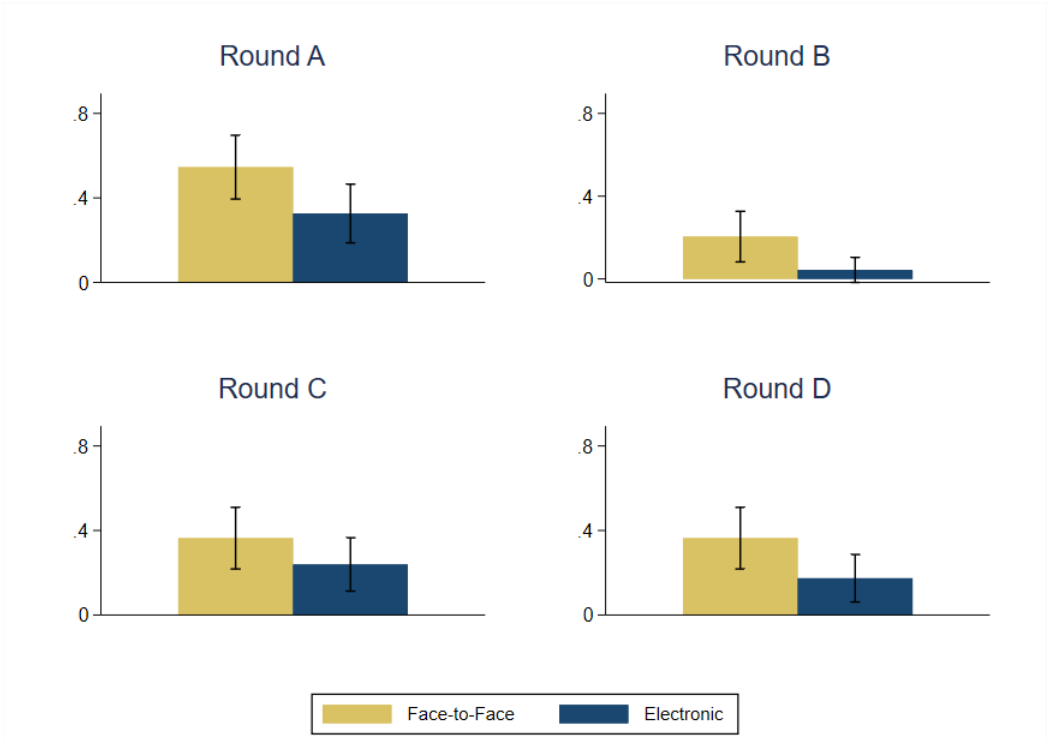
The comparison of the average total payoff per pair shows that in my experiment, the possibility of opting out of negotiations offsets the positive impact that bargaining in person has on agreements. When two participants make a deal, their combined payoff equals the negotiation pie (P), which is always larger than the sum of their outside options (which coincide with their disagreement payoffs, x and y). For each decider, I compute the sum of the payoffs earned by him and his counterpart across the four rounds. The average of this measure is 6,518 AR$ face to face, compared to 6,609 AR$ electronically (p=0.176). This indicates that in this experiment, an avoidance rate in person that almost doubles that of electronic negotiations weighs more than the ease of reaching agreements conditional on bargaining.[10]

Additional findings

Figure 1 shows the probability of avoidance by round and communication channel. Visual inspection suggests that participants are particularly reluctant to bargain in round A, where the distance between focal points is the greatest, and thus the likelihood for conflict is the highest (Babcock and Loewenstein, 1997). Meanwhile, participants are more keen to bargain in round B, in which the distance between focal points is lower and gains from trade are the highest. This suggests that when participants decide whether to opt in or out, they trade off the potential monetary gains that they can obtain from the negotiation with the expected non-monetary costs. Figure 2 shows the earnings distribution for those deciders who opted in and reached an agreement by round and communication

[10]One should note that if all deciders choose to bargain and all negotiations end with an agreement, the average total payoff per dyad would equal 6,900 AR$. This number comes from adding together the money to split (P) in the four rounds: 1,800 AR$, 1,800 AR$, 1,800 AR$ and 1,500 AR$. This figure does not correspond to the actual earnings of participants, who are paid for one round chosen at random plus a 400 AR$ show-up fee.

FIGURE 1
AVOIDANCE RATE BY ROUND



Note: This graph shows deciders' avoidance rate and their 95% confidence intervals in the baseline treatment, both in the face-to-face and electronic conditions for rounds A, B, C and D. A t-test comparison between the face-to-face and electronic condition yields a p-value of 0.036 in round A, 0.022 in round B, 0.2 in round C and 0.044 in round D.

condition. Most agreements fall in the surplus region, while some also fall to the left of the decider's disagreement payoff, with a mass point at the *equal-split*. This is consistent with previous literature. Regarding the differences by communication channel, the point estimate for decider's average earnings is always higher in person, but this difference is only significant in round A.

2.4. Discussion

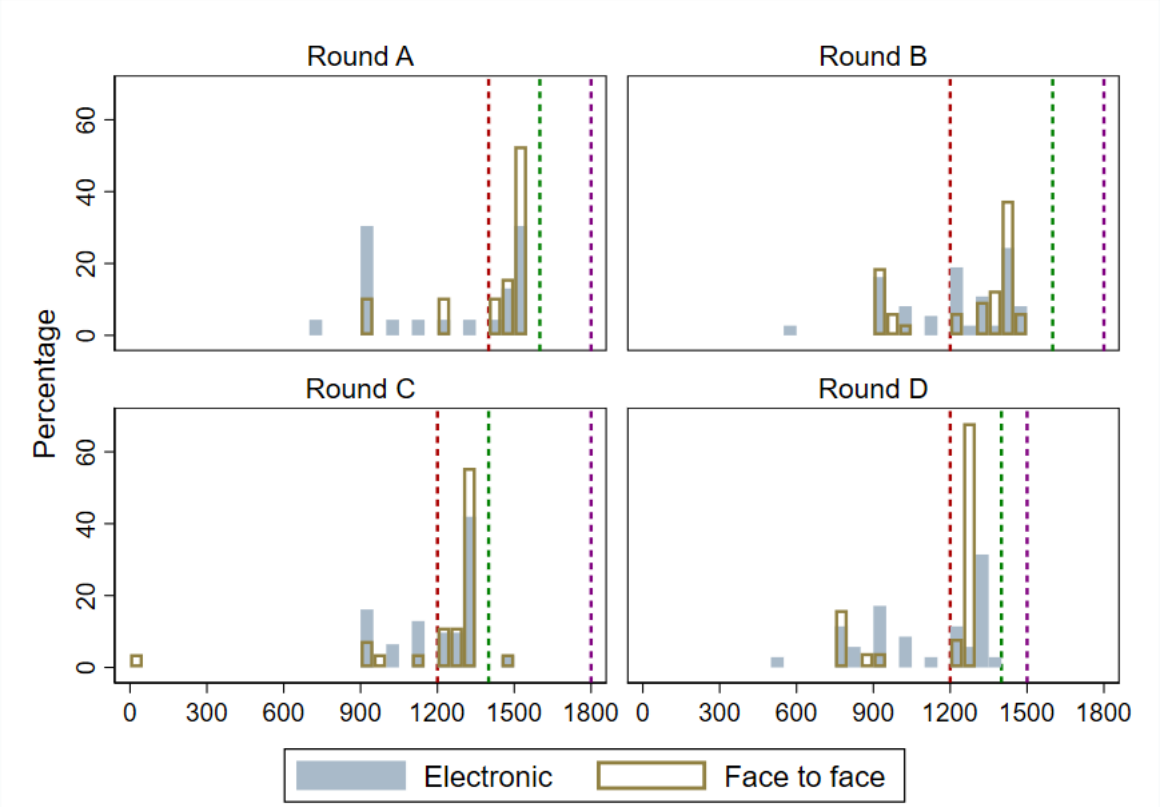
Taken together, the results of the baseline treatment confirm that there are multiple focal points in the negotiation, which creates room for conflict. They also show that some participants incur non-monetary costs associated with bargaining, making them particularly reluctant to bargain in person. Hence, in contrast with the general presumption, face-to-face negotiations are not welfare-improving in the context of my experiment. Far from it, as participants should only join the negotiation if they expect to gain positive utility, the greater entry rate in the digital condition implies that a higher proportion of deciders expect utility gains from bargaining electronically.

3. Experiment 1 - Mechanisms

The baseline treatment is designed to (1) assess the role of confrontation costs in bargaining, (2) examine whether electronic communication fosters negotiations, and (3) explore how this affects deal closure. To achieve this, it employs a synchronous, one-shot negotiation with public and symmetric

FIGURE 2

DECIDERS' EARNINGS AFTER AN AGREEMENT



Note: Distribution of deciders' earnings after an agreement in the baseline treatment. The red line denotes the decider's outside option (x), the purple line denotes the money to split (P) and the green line denotes the money to split minus the outside option of the counterpart ($P - y$). The outside options always coincide with the disagreement payoffs.

information, complete contracts, and a fixed duration. This setup excludes some competing factors, beyond confrontation costs, that might make in-person or digital negotiations preferable.[11] To further understand the role of confrontation costs, some additional treatments are conducted.

3.1. Analytical Framework

I hypothesize that bargaining aggressively raises potential earnings but also increases conflict. This makes some deciders feel uneasy, especially when they haggle face to face. Therefore, some of them prefer to secure a lower bound for profits—their outside option—and avoid confrontation by opting out of the negotiation, particularly in person.

The results in the baseline treatment are compatible with this explanation. However, they could also have other causes. Given the existence of non-monetary costs, the higher willingness to opt in electronically may be explained by deciders' beliefs. If deciders believe that they are more persuasive

[11]Note that in repeated interaction settings, negotiators may favor in-person encounters to build stronger relationships, and under informational asymmetries, face-to-face bargaining may help them form more accurate beliefs. Conversely, some asynchronicity in digital negotiations might allow negotiators to reflect and refine their strategies, and the paper trail created by written negotiations could be valuable when contracts are incomplete. The design of Experiment 1 closes these channels, using a one-shot, symmetric information negotiation, with fixed duration and complete contracts.

electronically and that this would yield them higher profits, they would opt in more frequently even if costs are the same across communication channels. Moreover, the essence of the costs may lie in the effort required to persuade the other party, rather than in the conflict inherent in haggling. By changing two features of the baseline design, I pinpoint the role that costs play in explaining differences by communication channel and further explore the nature of these costs.

If deciders had the final word in the negotiation—so the negotiation outcome became a choice variable for them—the need to persuade the counterpart would be removed, but the confrontational aspect of negotiations would be maintained. Under these circumstances, as I formalize in the model in Online Appendix C, any difference in the opt-out rate across communication channels must be explained by confrontation costs. I use this approach in the asymmetric treatment, which maintains the same features of the baseline, but gives the decider the right to impose the negotiation outcome.[12] Relatedly, if confrontation costs are relevant in explaining the reluctance to bargain, we should expect that if we force participants to bargain, some will trade off monetary payoffs and confrontation costs to maximize utility. This will result in negotiation avoiders adopting an accommodating behavior—as the model in Online Appendix C shows. The forced treatment, which removes the opt-out choice, tests this trade-off hypothesis.

Finally, I introduce two more treatments that test two additional explanations—which may operate alongside confrontation costs, or in isolation. In the dictator treatment, I benchmark the effect of image concerns, which may explain part of the reluctance to engage in bargaining.[13] In the conversation treatment, I explore the possibility that the resistance to bargaining face to face extends to other in-person interactions unrelated to bargaining. Table 3 summarizes all the treatments and their characteristics.

3.2. Asymmetric Treatment

In the asymmetric treatment everything works as in the baseline, but the decider can impose the terms of the agreement at any point during the negotiation.[14] Intuitively, those deciders who opt out in the asymmetric treatment do not want—or feel unable—to confront their counterpart to impose a deal that makes them better off. Perhaps even fearing they might succumb to their counterpart's demands to avoid conflict in a hypothetical negotiation.

[12] Given the existence of non-monetary bargaining costs, in the baseline treatment a higher propensity to bargain electronically may reflect both higher expected profits and lower costs. However, it could also reflect (i) lower variance in expected profits combined with bargaining costs and risk aversion; and/or (ii) strategic uncertainty aversion (Greiner, 2023). By making the terms of agreement a choice variable for the decider, the asymmetric treatment addresses these alternative channels when comparing communication modes. In this treatment, differences between in-person and digital opt-out rates should be attributed to costs. The uncertainty about the counterpart's hostility—and about one's own reaction to it—could still play a role in this treatment, but I interpret this as part of confrontation costs.

[13] I use the term "image concerns" to refer to a desire to be perceived as fair by others. This connects with the concept of social image (as opposed to self-image) in Bénabou and Tirole (2006) and with the concept of extrinsic fairness concerns (as opposed to intrinsic fairness concerns) in Cappelen et al. (2017). It can operate in distributive situations in the absence of any kind of verbal interaction or any form of non-verbal communication.

[14] In practical terms, in the face-to-face condition, it is not necessary for both parties to sign the answer sheet during the three minutes; only the decider's signature is required. Similarly, in the electronic condition, the counterpart does not have an *accept* button, which is replaced by a *confirm* button on the decider's screen. As in the baseline treatment, regardless of whether the deal is set before the three minutes are up, the chat remains open in the digital condition, and participants stay at the bargaining table in the face-to-face condition until time expires. An example of such an asymmetric negotiation could be a PhD student bargaining for funds with the PhD program director.

TABLE 3

SUMMARY OF TREATMENTS

	Entry Decision	Type of Interaction	Who Decides Final Allocation	Conditions	Participants
Baseline	Yes	Negotiation	Consensus	Electronic & F2F	180
Asymmetric	Yes	Negotiation	The Decider	Electronic & F2F	196
Forced	No	Negotiation	Consensus	Electronic & F2F	186
Dictator	Yes	No Interaction	The Decider	No Interaction	82
Conversation	Yes	Conversation	Random Device	Electronic & F2F	68

Note: This table summarizes the key features of each treatment. The first column records whether the decider can choose if he wants to enter the second phase of the experiment. The second column details the nature of this second phase and specifically the type of interaction between the decider and his counterpart in the second phase of the experiment. The third column specifies who chooses the final allocation of the pie (P). The fourth column states whether the conditions are electronic and/or face to face (F2F). The last column reports the number of participants. In a nutshell, in the asymmetric treatment, the decider can unilaterally impose the final allocation, while in the forced treatment, the decider is unable to opt out. Everything else remains the same as in the baseline. The dictator treatment mirrors the asymmetric treatment—the decider has the final word, but there is no interaction between the decider and his counterpart. In the conversation treatment, participants decide whether to converse for extra money.

Figure 3 illustrates the avoidance rate by communication channel in the baseline and asymmetric treatments. Although the point estimates for the avoidance rate in the asymmetric treatment are slightly smaller than in the baseline, the difference between communication conditions remains significant (32.2% face to face vs 17.4% electronically; p=0.010). This reveals that costs—not expected profits—can explain why participants are more reluctant to bargain in person. Moreover, it reveals that these costs are not related to the effort that persuasion takes: When the *need* to persuade is removed, but the confrontational aspect remains, deciders are still reluctant to bargain, especially in person.

Despite deciders having the final say in the negotiation, there are some interactions that end up with no agreement (agreement rates: 92.5% face to face, 83.9% electronically, p=0.027, see Online Appendix Table B.1). I find that the greater ease of reaching an agreement in person is again counterbalanced by a higher avoidance rate, resulting in similar monetary welfare across communication conditions (6,550 AR$ face to face vs 6,600 AR$ electronically, p=0.366). Furthermore, the better predisposition to bargain in the electronic condition reflects that a greater proportion of deciders expect to get positive utility from negotiations when they are digital, just as in the baseline treatment. When I look at the results of the negotiation, I find that the distribution of deciders' earnings after an agreement in the asymmetric treatment (Online Appendix Figure A.6) resembles that of the baseline.[15]

Altogether, the results from the asymmetric treatment reveal that confrontation costs explain participants' reluctance to bargain face-to-face, and confirm that in-person negotiations are not welfare-improving in my experiment.

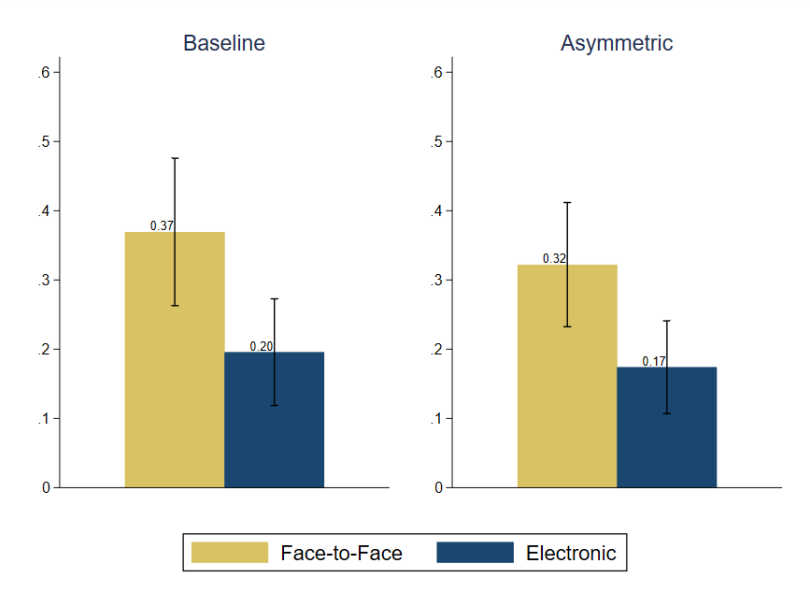
3.3. Forced treatment

The forced treatment removes the possibility of opting out, mimicking previous studies. However, given that the rest of the features of the design remain as in the baseline treatment, I maintain the labels "decider" and "counterpart" to name the participants with the highest and lowest disagreement

[15]This supports the notion that the interactions in the asymmetric treatment can be cataloged as negotiations, despite deciders having the final say.

FIGURE 3

AVOIDANCE RATE – BASELINE AND ASYMMETRIC TREATMENTS



Note: This figure shows the average avoidance rates by treatment and communication condition, along with their 95% confidence intervals. The t-test comparing the avoidance rate in the face to face and electronic condition yields a p=0.010 both in the baseline and asymmetric treatments.

payoff.

Table 4 shows the average sum of payoffs earned by the decider and his counterpart across the four rounds. These total payoffs are significantly higher face-to-face (6,815 face-to-face vs 6,622 electronic, p<0.001). This result stems directly from the higher agreement rate observed in person and aligns with previous studies, which have interpreted this as a sign of higher welfare. In this subsection, I explore whether this finding instead hinges on the trade-off hypothesis: if deciders are forced to bargain and some experience confrontation costs, they may adopt a soft haggling strategy as their second-best alternative to opting out.

TABLE 4

FORCED TREATMENT

	Electronic	Face to face	Difference
Agreement (%)	72.8	91.5	-18.7***
	(3.5)	(2.2)	(4.2)
Total payoff ($)	6621.7	6814.9	-193.2***
	(37.0)	(23.4)	(43.8)
Num. Deciders	46	47	

Note: This table reports the average rate of agreement for those negotiations that occur, the average total monetary payoff per pair expressed in Argentinian pesos, and the number of deciders in the forced treatment for each communication condition. It also reports the difference between the two communication conditions. * p<0.1, ** p<0.05, *** p<0.01 correspond to the p-values of two-sided t-tests.

A first approximation to the differences between the forced and baseline treatments shows that the agreement rate in the electronic condition is substantially lower in the forced treatment (72.8% in

the forced vs. 85.2% in the baseline, p=0.024). This is consistent with some deciders choosing not to bargain in the digital condition of the baseline treatment due to a perceived low probability of reaching a satisfactory deal. Consistently, when forced to bargain, they do not reach an agreement. Interestingly, when comparing the agreement rates across treatments in the face-to-face condition, I find that they are statistically equivalent (91.5% in the forced, 95.2% in the baseline, p=0.210). This is intriguing, as it suggests that some deciders who would not bargain in the baseline treatment are willing to reach an agreement when forced to negotiate. I digitize the content of the negotiations and explore how deciders adapt their bargaining strategies when they cannot opt out, to assess if reaching an (unsatisfactory) deal becomes their way out of conflict.

I focus on the first two decisions that deciders make in the negotiation: whether to make the opening offer and how much money to demand for themselves. I infer how those deciders who would rather opt out behave by comparing the average choices in the baseline and forced treatments. This approach rests on the premise that those deciders who would prefer to negotiate in the baseline treatment would not change their bargaining strategy in the forced treatment. Online Appendix Table B.2 shows an OLS regression of a dummy that takes value one if the decider opens the negotiation, on a dummy that takes value one in the forced treatment and value zero in the baseline (columns 1 and 2). It also shows a regression of the claim that the decider makes for himself if he opens the negotiation on that same dummy (columns 3 and 4). The results suggest that deciders are more reluctant to make the opening offer in the forced treatment. More importantly, if they are in person, when they make opening offers they are significantly less aggressive.[16] These results are consistent with the trade-off hypothesis: Deciders suffering confrontation costs choose a soft bargaining strategy as a second-best to opting out. This pattern emerges especially in face-to-face negotiations, when confrontation costs are higher.

To explore the impact that this has on negotiation outcomes, in columns 1 and 2 of Table 5, I show a regression of deciders' earnings on a dummy that takes a value of one in the forced treatment and zero in the baseline. The results in column 1 indicate that in person, deciders' average profits are significantly smaller in the forced treatment than in the baseline. In contrast, average profits across the baseline and forced treatments in the electronic condition are the same. Columns 3 and 4 replicate this same regression changing the dependent variable to a dummy that takes a value of one if the decider reaches an agreement in which he earns *less* than his disagreement payoff. Results reveal that, in the face-to-face condition, deciders in the forced treatment are almost 30 percentage points more likely to incur a loss relative to their disagreement payoffs. In the electronic condition, there is no difference in the proportion of deciders earning less than their disagreement payoffs. This also aligns with the trade-off hypothesis: due to confrontation costs, disagreement in person is particularly costly for some deciders, and as a result, when forced to bargain, they prefer to lose money and mitigate confrontation costs rather than dissent and earn their disagreement payoff.

Finally, a comparison of deciders' average earnings in the forced treatment across communication conditions shows that the point estimate is always higher electronically. This is the opposite of what we

[16] In columns 3 and 4, I focus on those deciders who open the negotiation, as their first demand is not affected by any prior offer made by their counterpart. I consider these observations to be more representative of deciders' bargaining strategy. If I take all deciders' first demands, the results remain qualitatively the same and statistically significant.

TABLE 5

DECIDER'S EARNINGS AFTER AN AGREEMENT

	(1) Decider's Profits	(2) Decider's Profits	(3) $\mathbb{1}$(Decider's Profits< x)	(4) $\mathbb{1}$(Decider's Profits< x)
Forced	-115.01**	12.93	0.29***	0.01
	(44.46)	(43.32)	(0.09)	(0.10)
Observations	276	260	276	260
Num. Clusters	85	88	85	88
Dep. Var. Mean	1165.18	1172.06	0.42	0.42
Sample	Face-to-Face	Electronic	Face-to-Face	Electronic

Note: Columns 1 and 2 show an OLS regression of deciders' earnings after an agreement on a dummy that takes a value of 1 in the forced treatment, and zero otherwise, controlling for round fixed effects. Column 1 only includes data from the face-to-face sessions, while column 2 only includes data from the electronic sessions. Columns 3 and 4 repeat the same OLS with a different dependent variable: a dummy which takes value one if deciders' earnings after an agreement are smaller than their disagreement payoffs. Again, column 3 only includes data from the face-to-face sessions, while column 4 only includes data from the electronic sessions. Errors clustered at the individual level. * p<0.1, ** p<0.05, *** p<0.01.

observe in the baseline treatment. Nevertheless, as in the baseline treatment, some of these differences are non-significant (see Online Appendix Figure A.7).

3.4. Complementary Mechanisms

The results in the baseline, asymmetric, and forced treatments are all consistent with the predictions listed in the analytical framework. They show that some agents incur confrontation costs, and that this is more common in person. In this section, I explore whether image concerns or a general aversion to in-person interactions are also playing a role in deciders' choices.

Dictator treatment

In the dictator treatment, deciders who choose to opt in see a screen with a picture of their counterpart, where they decide how to split the pie P. This decision is shared with their counterpart along with their photo. If instead the decider chooses to opt out, there is no second stage, both members of the pair receive their outside options (x and y), and no further information is revealed. Thus, in this treatment, there is no communication whatsoever—verbal or non verbal—no matter whether the decider opts in or out.[17]

The opt-out rate in the dictator treatment reflects the proportion of participants who do not want to enter a distributive situation—like the one posed in the negotiation treatments—to preserve a positive social image.[18] I find an average avoidance rate of 16% in the dictator treatment, which falls well below

[17] I do not run a dictator treatment where deciders make their decisions in front of their counterparts, as this would introduce non-verbal communication. Such communication would prevent the isolation of image concerns from confrontation costs, since a scornful glance may be as emotionally painful as harsh words. Still, Lazear, Malmendier and Weber (2012) find that playing the dictator game in person does not make any difference in the opt out rate, which may suggest the opposite.

[18] In the classical sense of Dana, Cain and Dawes (2006), the dictator treatment, as it is designed in this experiment, measures image concerns only partially. In their original experiment, participants pay a price to prevent the receiver from ever learning that a dictator game was played. This never happens in my experiment, where counterparts are always aware of the experimental details. However, we know that revelation of identity may affect image concerns (Charness and Gneezy, 2008). Hence, as negotiations are a distributive situation in which identity is revealed, I want to control for this baseline cost that could be present. This is precisely what the dictator treatment does. For surveys of the literature of image concerns, see Bursztyn and Jensen (2017) and Grossman and Van der Weele (2017).

the 36.9% opt-out rate in the face-to-face condition of the baseline treatment (p=0.002). Conversely, when I compare it with the avoidance rate in the electronic condition, I find a modest difference (19.6% baseline electronic vs 16.5% dictator; p=0.546). This suggests that digital interactions significantly mitigate the effect of confrontation costs, whose impact on top of image concerns would be modest for digital negotiations, but not for in-person interactions.[19]

Conversation treatment

Finally, to measure a possible general aversion to any in-person interaction, I introduce the conversation treatment. There, instead of bargaining, participants choose whether or not to join a conversation. If they opt out, the pair does not meet, they wait for three minutes in front of their computers and they earn their outside options (x and y). Conversely, if they opt in, they have three minutes to talk about any topic and some extra money is split between them via a lottery system that adds to their outside options. The amount of money split by the lottery coincides with the difference between the size of the pie and the sum of the outside options ($P - x - y$).[20] If the reluctance to engage in face-to-face negotiations extends to any in-person interaction, we should observe a different opt-out rate in the face-to-face and electronic conditions of the conversation treatment.

I find that a large majority of participants chose to meet, with no significant difference in the probability of skipping the conversation across communication channels (5.4% face-to-face vs. 4.2% electronically, p=0.669). Although these rates are not directly comparable to those in the negotiation treatments, they are low and provide no evidence of participants being generally reluctant to engage in face-to-face interactions.

3.5. Direct Evidence of the Mechanism

In this subsection, I discuss the results of the TKI® psychological test and the debriefing survey.

TKI® Psychological Test

The TKI® is the leading conflict style inventory in psychology. Developed by Kilmann and Thomas (1977), it has been commercialized by the Myers-Briggs Company for firms worldwide. It comprises 30 questions which give participants a score assessing how frequently they use the following conflict management strategies: accommodating, competing, collaborating, compromising and avoiding. The test evaluates whether the person filling out the questionnaire, when faced with conflict, has a tendency to concede to others' demands, compete to impose his view, find a collaborative solution, find

[19] Interestingly, although the 16% opt-out rate in the dictator treatment is also similar to the 17.4% opt-out rate in the electronic condition of the *asymmetric* treatment—and so are the average profits for the decider—in the dictator treatment it is more likely to leave the counterpart with less than the outside option (y). This occurs 12.6% of the time in the dictator treatment, while approximately 5.6% of the time in the electronic condition of the asymmetric treatment. This suggests that confronting electronically is still somewhat costly, especially if one transgresses certain norms. Maybe these boundaries could explain why over 15% of deciders in the electronic condition of the *asymmetric* treatment prefer not to impose the terms of the deal, leaving the negotiation with no agreement (see Online Appendix Table B.1).

[20] Unlike in other treatments, in the conversation treatment participants in both roles (deciders and counterparts) choose whether or not to enter into the conversation. If either party chooses not to enter into the conversation, both get their outside option and wait for three minutes. If both participants opt in to the conversation, a lottery determines how ($P - x - y$) is split between them and they meet for three minutes. As both deciders and counterparts choose whether to opt out, the sample size is roughly half that of the baseline treatment. This makes the conversation treatment not directly comparable with the baseline, but it enables me to explore whether people are averse to interpersonal interactions comparing the face-to-face and electronic conditions of the conversation treatment, while optimizing research resources.

a compromise or just avoid the conflict. It is administered online, and subjects get a score in each category. The individual score is also shown in relation to the historical record of all individuals who have ever taken the test, which provides me with the percentile of the participants' score relative to a large population.

If confrontation costs are relevant to explain why participants avoid negotiations, we should expect that those who opt out in the experiment have a high conflict *avoiding* score on the TKI®.[21] To test this hypothesis, I run the following linear probability model:

$$OptOut_{ir} = \beta_1 TKI_i + \theta Controls_{ir} + \epsilon_{ir} \quad (1)$$

The sample includes observations in the baseline and asymmetric treatments—the two negotiation treatments where participants can opt out. Errors are clustered at the decider level. The dependent variable $OptOut_{ir}$ denotes whether decider i opts out in round r, TKI_i denotes the percentile of the decider's avoidance score on the TKI® test, and $Controls_{ir}$ includes alternative measures of decider i's prosociality (including whether he chose an equal-split in the corresponding round, or whether he chose to be eligible for non-paid experiments when he signed up in ORSEE).[22]

Online Appendix Table B.3 shows that moving from the bottom to the top of the TKI® avoidance score distribution is associated with an increase of 14.2 percentage points in the probability of opting out in my experiment (see column 1; average avoidance rate 26.9%). This result is even more significant when I control for the communication channel (column 2) and holds when I use an alternative prosociality proxy (column 3). In columns 4 and 5, I split the sample between those in the face-to-face condition and those in the electronic condition. Notably, I find a stronger effect face to face, in which moving from the bottom to the top of the distribution is associated with an increase in the opt-out probability of roughly 65% of its unconditional mean. This supports the interpretation that the desire to avoid confrontation deters people from bargaining, particularly in person.[23]

Debriefing survey

At the end of the experiment, I ran a debriefing survey to dig deeper into the attitudes of participants toward negotiations and their motivations in the lab. In this survey, 77% of deciders report feeling anxious, nervous or awkward when they have to bargain. Among them, an overwhelming majority (93%) state these feelings are more intense in person. Regarding their reasons for opting out, 72%

[21] There is a previous study in psychology which has explored how the TKI® scores correlate with behavior *during* the negotiation (Amanatullah, Morris and Curhan, 2008). They examine how unmitigated communion—a psychological construct which captures an excessive concern for relationships—relates to accommodating behavior and economic outcomes in negotiations.

[22] I control for decider's prosociality because I assume that the desire to avoid conflict plays no role in the opt-out decision of intrinsically-fair-minded subjects, who join the negotiation to evenly split the pie (see Online Appendix C). I use two different measures to control for this. In columns 1, 2, 4 and 5 in Online Appendix Table B.3, I include a dummy that takes a value of 1 if the decider reached an equal-split agreement in the corresponding round, and 0 otherwise. In column 3, I use an alternative proxy. When subjects sign up in ORSEE, they choose whether to be eligible for unpaid experiments. Column 3 includes this *unpaid eligibility dummy* as a proxy to control for deciders' altruism.

[23] Controlling for gender slightly reduces the estimated coefficients on TKI avoidance. The estimates remain significant in columns 2 and 4, but not in 1 and 3. Moreover, in Online Appendix Table B.4, I show the correlation between the opt-out choice and the score in the other four conflict management strategies. I find a *negative* and significant correlation between avoiding the negotiation and scoring high on collaborating and compromising in the TKI®.

of deciders who took part in the baseline treatment indicated that arguing to secure more than their outside option and/or ending the negotiation with a disagreement would have made them feel uneasy.

In the forced treatment, participants also rated at the end of the experiment how nervous and how relaxed they felt *before* the negotiation, and how awkward and how comfortable they felt *during* the negotiation, using a 1 to 7 Likert scale. Online Appendix Table B.5 shows that deciders reported feeling more nervous, less comfortable and significantly more awkward in the face-to-face condition than in the electronic condition. This occurs despite the fact that we have seen in Section 3.3 that deciders moderate their level of aggressiveness when they are forced to bargain in person—probably seeking to mitigate confrontation costs. Notably, this result reverses in the conversation treatment, where feelings are, if anything, more negative electronically. This corroborates that it is haggling and discussing in person that arouses negative feelings, not just interacting.

3.6. Discussion

The results in Section 3 show that electronic communication fosters negotiations through a reduction in confrontation costs. Moreover, they show that the greater rate of agreement that previous studies have found in face-to-face negotiations could be the consequence of some subjects choosing a particularly accommodating strategy to avoid conflict. These findings confirm that electronic negotiations can be welfare improving, challenging the general wisdom.

4. Experiment 2 - Robustness

To analyze the robustness of the conclusions that I reach in the first experiment, I run a second experiment with a slightly different structure. I use a within-subject design, where deciders have to make a decision to opt in to or out of each type of negotiation, in person and digital. Moreover, with some probability, I force them to bargain irrespectively of their choices. This allows me to classify deciders into four behavioral classes: those who want to bargain irrespectively of the communication channel (always bargainers), those who only want to bargain in person (old-school bargainers), those who only want to bargain electronically (e-bargainers) and those who do not want to bargain in any case (never bargainers). Besides, although I maintain most of the features of Experiment 1, I make a few important changes: the difference between the decider's and the counterpart's disagreement payoffs (x and y) is reduced, I measure participants' beliefs before the negotiations, and I elicit deciders' explanations of their choices using qualitative interviews. This enables me to test the robustness of the main result and to explore the mechanisms with a different set of empirical strategies than in the first experiment.

4.1. Experimental Design

Participants in Experiment 2 are randomly assigned the role of deciders or counterparts. There is only one negotiation round lasting five minutes. Deciders choose whether they want to bargain or not in case the negotiation is in person, and whether they want to bargain or not in case the negotiation is electronic. Afterwards, the computer rolls a die: if it rolls a 1, their choice to opt in to or out of the face-to-face negotiation is implemented; if it rolls a 2, their choice to opt in to or out of the electronic negotiation is implemented; if it rolls a 3 they are forced to bargain face-to-face and if it rolls a 4 they

are forced to bargain electronically. The money to split (P) is 24,000 AR$ and the outside options, which again coincide with the disagreement payoffs, are 16,000 AR$ (x) for the decider and 5,000 AR$ (y) for the counterpart (see Online Appendix Sections H and I for an English version of the instructions and control questions).[24]

To gain a better understanding of deciders' choices, during the experiment I conduct two chatbot-led interviews, adapting the procedure developed by Chopra and Haaland (2023). The first interview is conducted after deciders choose whether to opt in to or out of each type of negotiation, and it asks participants about the motivation behind their choices.[25] The second interview is conducted after negotiations take place and asks participants how they felt after learning the outcome of the die roll and during the negotiation (the latter only for those who bargained; see Online Appendix Sections D.1 and D.4 for details on the configuration of the chatbots).

After all participants learn the result of the die—and in the case of counterparts, after they learn the choice made by the decider—I also gather participants' beliefs about their expected profits in case of agreement and about the probability of agreement itself.[26] Besides, after the second interview, I run a short debriefing close ended survey similar to the one in Experiment 1.

The experiment was run in oTree (Chen, Schonger and Wickens, 2016) in the mobile lab of Universidad Torcuato Di Tella in October 2025. Sessions lasted around 90 minutes, 294 subjects participated and they earned 16,490 AR$ on average, including a 5,000 AR$ show-up fee.

4.2. Interview Content Classification

To analyze the content of the interviews, I follow the best practices proposed in Haaland et al. (2025). First, I manually create a rationales codebook. This codebook encompasses all the motives given by deciders in the first interview to explain their choice. Then, using this rationales codebook, I assign to each interview the corresponding codes (if any). I also identify the emotions that each subject associates with face-to-face negotiations (f), with digital negotiations (d), and with negotiations in general (g). Finally, I validate the coding of rationales and emotions with a trained research assistant–who was unaware of the experimental hypothesis–and with gpt-4o (see Bermejo et al., 2025, for evidence on the accuracy of LLM text coding). In Online Appendix Sections D.2 and D.3 I provide further details explaining the procedure to create the codebook, to code the data, and to validate the coding.

The rationales codebook includes 20 different non-mutually exclusive motives given by deciders to explain their choice to enter negotiations or not (see Online Appendix D.2). Two rationales map onto confrontation costs: "conflict avoidance" and "fear of conceding under pressure" (the latter reflecting the trade-off hypothesis, by which some participants prefer to lose money rather than argue or end the negotiation in disagreement). These rationales are distinguished from others such as "lack of persuasion ability", "avoid developing empathy" or "image concerns". I code rationales in the data based solely on the explanations provided by the interviewees. Explanations that are not consistent with the

[24]Between 2021 and 2025, Argentina suffered an exacerbated inflation process. Thus, payoffs were updated considering the price level of October 2025, when the experiment was run. In October 2025, when Experiment 2 was conducted, the exchange rate was around 1,450 AR$/USD, vs. the 100 AR$/USD of November 2021.

[25]It asked deciders about what they did and why, and it asked counterparts about what they would have done and why.

[26]To keep the experiment as simple as possible, this elicitation phase was not incentivized. Beliefs were elicited using a slider (see Online Appendix Figure A.8).

experimental rules are coded as "Poor understanding of the experimental instructions".[27]

The dictionary of emotions mentioned by participants includes 64 terms. These emotions are then classified as "favorable" or "unfavorable" depending on their impact on the negotiation (see Online Appendix D.3). This is more convenient for the analysis than the standard valence classification: a positive emotion (empathy) may be unfavorable for maximizing gains in a distributive negotiation, while a negative emotion (coldness), may be favorable. In the analysis, the emotions associated with in-person negotiations encompass both the emotions coded as general (g) and those specifically coded as face-to-face (f). Similarly, the emotions associated with electronic negotiations encompass both the emotions coded as general (g) and those specifically coded as digital (d).

In the second interview, I code participants' feelings after learning whether they were bargaining and how, and I code whether during the negotiation the feelings they experienced were better, worse or as expected (see Online Appendix D.5 for further details).

4.3. Results

In Experiment 2, 46% of deciders choose to avoid the negotiation if it is in person and 22% if it is electronic. These numbers are in line with Experiment 1. A paired t-test shows that the difference across communication channels is highly significant ($p<0.001$). Moreover, among willing bargainers (i.e. deciders who chose to opt-in in the communication channel where they actually bargained), the difference in the agreement rate by communication condition is small and not statistically significant: 80% electronically vs. 86% in person ($p=0.399$). This reinforces the notion that once we account for the decision to enter the negotiation, there are settings where face-to-face communication is not welfare improving: deciders try to bargain more often electronically, and conditional on trying, they are not significantly less likely to reach agreements.

Regarding the distribution of decider types, I find that 49% of deciders are always-bargainers, 5% are old-school bargainers, 29% are e-bargainers and 17% are never-bargainers. I analyze the five pieces of evidence I gather—the beliefs, the first interview, the negotiation outcomes, the second interview and the close ended survey—to explore the mechanism that explains deciders' choices.

First Interview

Participants take the first interview right after deciders make their choice to bargain or not in each communication channel, and before the negotiations take place. Table 6 provides the Top 3 rationales given by each type of decider to explain their choice, along with the relative frequencies. The two rationales associated with confrontation costs—"conflict avoidance" and "fear of conceding under pressure"—are both in the top 3 for e-bargainers and never bargainers. Notably, 74% of e-bargainers and 65% of never bargainers mention at least one of them to explain their decision (see the relative frequency of every rationale in Online Appendix Figure A.9). This confirms that, according to de-

[27] In the subsequent analysis of the interview content, I exclude 3 deciders who showed a poor understanding of the experimental instructions during the interview and 6 who reported in the interview a decision which did not match their actual choice. Notice that one participant may give more than one rationale to explain his decision. For example, a never bargainer may indicate: '*I hate conflict, so taking into account that I am already making good money, I think it is in my own interest to opt out. I am very risk averse.*'. According to this answer, three rationales would be coded: "self-interest", "risk aversion" and "conflict avoidance".

TABLE 6

TOP 3 RATIONALES BY DECIDER TYPE

Always Bargainer		Old-School Bargainer		E-Bargainer		Never Bargainer	
Rationale	Freq.	Rationale	Freq.	Rationale	Freq.	Rationale	Freq.
Don't Leave Money on the Table	0.44	Uncomfortable with Electronics	0.86	Conflict Avoidance	0.50	Self-Interest	0.57
Self-Interest	0.42	Minimize effort, Maximize agility	0.43	Fear of Conceding Under Pressure	0.38	Conflict Avoidance	0.48
Nothing to Lose	0.29	Experience Seeking	0.29	Stranger Aversion	0.36	Fear of Conceding Under Pressure	0.39

Note: This table shows the relative frequency of the 3 most common rationales for each decider type. See Online Appendix Figure A.9 for the complete list of rationales' frequencies for each type. See Online Appendix D.2 for the rationales' definitions.

ciders' explanations, confrontation costs are the main reason that explains negotiation avoidance in the experiment.

The analysis of the emotions that each behavioral type associates with negotiations confirms the strong relationship between emotions and choices. Table 7 shows the type of emotion that each behavioral type associates with in-person and with digital negotiations. Always bargainers are the ones who associate more favorable emotions with negotiations, old-school bargainers and e-bargainers associate favorable emotions with their most preferred communication channel and unfavorable emotions with their least preferred, and never bargainers associate unfavorable emotions with both types of negotiations (see Online Appendix Table B.6 for the detailed emotions).

TABLE 7

EMOTIONS BY DECIDER TYPE - FIRST INTERVIEW

	Always Barg.		Old-School Barg.		E-Bargainer		Never Barg.	
	Fav	Unfav	Fav	Unfav	Fav	Unfav	Fav	Unfav
Face-to-face	.63	.37	.67	.33	.09	.91	.08	.92
Electronic	.75	.25	.43	.57	.62	.38	.11	.89

Notes: This table shows the proportion of favorable/unfavorable emotions that each decider type associates with each type of negotiation in the first interview. See Online Appendix Table B.6 for the detailed emotions.

Beliefs

The theory described in the analytical background predicts that subjects with self-regarding preferences who suffer confrontation costs will trade off monetary payoffs and non-monetary costs if they cannot avoid bargaining. To explore this hypothesis, I exclude subjects with other-regarding preferences and compare the expected payoffs of always bargainers, e-bargainers and never bargainers by communication channel. I consider that a participant has other-regarding preferences if he enters a negotiation voluntarily and expects to divide the pie in halves.[28] Otherwise, I consider that he has self-regarding preferences.

[28] As the theory predicts, most subjects with other regarding preferences are always bargainers (79%). The rest are old-school bargainers and e-bargainers, who avoid in person/digital negotiations for different reasons.

Consistent with the theory, I find that on average, always-bargainers always expect to make more than their 16,000 AR$ disagreement payoff: 17,325 AR$ electronically and 16,913 AR$ face-to-face (no significant differences across channels, t-test p-value=0.342). E-bargainers expect to gain something with respect to their disagreement payoff if they bargain electronically, but they expect to lose money in person: 17,437 AR$ digitally and 14,444 AR$ face to face (t-test p-value=0.004). While never-bargainers never expect to do better than their disagreement payoff: they expect 14,625 AR$ electronically and 13,620 AR$ in person (t-test p-value=0.639). Consistently with the trade off hypothesis, this shows that on average, subjects who bargain against their will prefer to lose money rather than confront and disagree with their counterparts. The high correlation between deciders' expectations and actual payoffs (77%) suggests that this analysis of beliefs may actually be indicative of deciders' bargaining strategies.

It is important to note that in Experiments 1 and 2 deciders can always disagree and secure their disagreement payoffs. Thus, although persuasion ability may affect expected profits—that is, how much one expects to earn *above* the disagreement payoff—it should not explain expectations of earning *less* than the disagreement payoff. For willing negotiators, such expectations must be the product of intrinsic fairness concerns (Cappelen et al., 2017). For unwilling bargainers, it should imply that they anticipate accepting an avoidable loss to mitigate confrontation costs.[29]

Negotiation Strategies & Outcomes

The within subject design enables me to explore if the conclusions that I reach studying the bargaining strategies in the forced treatment of the first experiment hold in Experiment 2 under weaker assumptions. Namely, I can test if self-regarding deciders who prefer to avoid bargaining use a soft bargaining strategy as a second best. I find that in person, when always bargainers make the opening offer, they claim a significantly larger share for themselves than e-bargainers and never bargainers (see Online Appendix Table B.7). Likewise, electronically, when never bargainers make the opening offer they claim a significantly lower share than always bargainers and e-bargainers. This confirms that participants who bargain against their will take a particularly accommodating bargaining strategy.

Focusing on negotiation outcomes, I find that there is no significant difference in the average realized profits of always bargainers by communication channel (16,316 AR$ electronically vs. 16,065 AR$ in person, p-value=0.55). By contrast, there is a significant difference in the earnings of e-bargainers (16,600 AR$ electronically vs. 13,500 AR$ in person, p-value=0.0008). Never bargainers always make on average less than their disagreement payoffs: 15,750 AR$ electronically and 14,200 AR$ in person (t-test: p-value=0.241). This is consistent with the conclusions of the analysis of beliefs and reinforces the notion that negotiation avoiders face a trade off between making money and minimizing confrontation costs. They prefer to lose money rather than confront and get their disagreement payoffs.[30]

[29]The reasoning goes as follows: By revealed preferences, we know that unwilling negotiators expect negative utility from bargaining. If, in addition, they expect earnings below their disagreement payoffs when they are forced to bargain, then they must expect that the utility loss associated with conceding is smaller than the utility loss associated with confronting and ending the negotiation in disagreement.

[30]The within-subject design also allows me to explore whether the assumption required in the analysis of the forced treatment holds. Namely, that the differences I find between the baseline and forced treatments in Experiment 1 are not driven by willing bargainers changing their behavior. I find that willing bargainers do not change their bargaining strategy when the die is 1 or 2 vs. when it is 3 or 4: there is no significant difference either in expected earnings or in the first demands.

Second Interview

After the bargaining phase, participants take a second interview. Online Appendix Table B.9 shows the emotions that each behavioral type felt after learning whether they had to bargain in person, electronically, or not at all. Never bargainers are relieved when they learn they are not bargaining and suffer when they learn they are bargaining. E-bargainers and old-school bargainers show mixed feelings, consistent with the fact that they mostly bargain in their preferred communication channel, but not always. Always bargainers—who already knew they were bargaining for sure—experience the most favorable emotions after learning the way in which they are bargaining.[31]

Close-ended Survey

At the end of the experiment, I use a short close-ended survey resembling that of Experiment 1. In Online Appendix Table B.11, I regress how awkward/comfortable deciders felt during the negotiation on counterparts' expected profits. I find that in person, deciders feel significantly more awkward and less comfortable the higher their counterpart's expected profits are, especially when they bargain against their will. This is consistent with conflict being an increasing function of the distance between participants' claims. Interestingly, this relationship is not significant in electronic negotiations, which reinforces the notion that digital negotiations can successfully alleviate confrontation costs.

4.4. Discussion

Altogether, the results of Experiment 2 confirm the conclusions of Experiment 1. The negative emotions stemming from the conflict inherent in haggling deter some people from engaging in profitable negotiations. Moreover, bargaining electronically instead of in person can alleviate those negative feelings, encouraging some participants to bargain. This indicates that the general belief that negotiating in person improves welfare is not true in all instances. Notably, if negotiation avoiders are forced to bargain, they trade off confrontation costs against monetary profits to maximize utility.

5. Gender Analysis

This section explores whether there are any gender differences in bargaining propensity both in experiments 1 and 2, what explains these differences, and how the communication channel affects the opt-out choice by gender. Then, it briefly explores negotiation outcomes by gender and discusses the results, connecting them with previous literature.

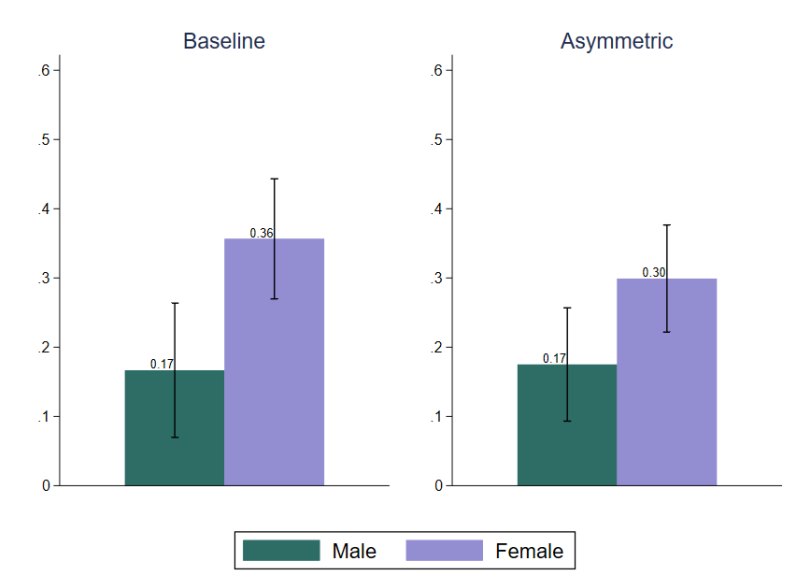
Turning to outcomes, I find that willing bargainers earn *more* when the die is 3 or 4 (see Online Appendix Table B.8). This means that in the forced treatment of Experiment 1, average deciders' earnings in person are significantly *lower* than in the baseline, *despite the fact* that those who bargain voluntarily may actually be doing *better*. This implies that in the forced treatment, those who bargain against their will are the ones making less than their disagreement payoffs, which reinforces the trade-off hypothesis.

[31] Notice that always bargainers and never bargainers may have preferences for one channel or another. This could explain that the former sometimes experience negative emotions (when they bargain in their least preferred channel), and the latter sometimes experience positive ones (when they do *not* bargain in their least preferred channel). On a different note, when I inspect the emotions felt by negotiators *during* the negotiation, I find that they tend to be slightly better than what they expected, but marginally (see Online Appendix Table B.10).

5.1. Experiment 1

In Experiment 1, deciders' gender was balanced across treatments and communication conditions. This enables me to explore the effect of gender on the decision to bargain independently from the effect of the communication channel. Figure 4 shows the avoidance rate in the baseline treatment by gender. The results show that the avoidance rate is significantly greater among female deciders (35.6% for females vs 16.7% for males; p=0.005). This confirms the notion that women are more reluctant to bargain than men. Yet, what role do confrontation costs play in this difference? Various literature surveys and meta-analyses have shown that women are more risk-averse (see Croson and Gneezy 2009 and Eckel and Grossman 2008) and more generous than men (Croson and Gneezy, 2009; Bilén, Dreber and Johannesson, 2021), while some studies have found that they are also less prone to overconfidence (Niederle and Vesterlund, 2007; Van Veldhuizen, 2022). These differences should be irrelevant for the opt-out choice in the absence of non-monetary costs—when opting in is always dominant. However, if some deciders incur costs, then differences in self-confidence, risk aversion or generosity could lead to differences in the opt-out rate. I use the asymmetric treatment, the debriefing survey, and the psychological test to explore the role that confrontation costs play in women's reluctance to negotiate.

FIGURE 4
AVOIDANCE RATE – GENDER



Note: This figure shows the average avoidance rates in the baseline and asymmetric treatments for both genders, along with their 95% confidence intervals. The t-test comparing the avoidance rate between men and women yields a p=0.005 in the baseline treatment and p=0.031 in the asymmetric treatment.

Figure 4 shows that while the avoidance rate for male deciders in the asymmetric treatment is 17.5%, it rises to 30% for female deciders (p=0.031). It is important to note that in this treatment, the decider has full control over how to split the pie. Thus, self-confidence in one's persuasion skills and risk aversion should not play any role in the opt-out choice. Likewise, as I show later in Section 5.3, in this treatment male deciders keep a greater or equal portion of the pie for themselves, which is not

consistent with them opting in more frequently due to generosity.[32] This supports the notion that it is differences in confrontation costs between men and women that explain women's greater reluctance to bargain.

This conclusion is supported by participants' responses about their negotiation habits in the debriefing survey. First, analyzing the full sample across all treatments, 81.2% of female deciders versus 70.9% of male deciders report experiencing negative emotions when they bargain (p=0.039). Second, the reported feelings at the end of the forced treatment show that female deciders felt significantly less relaxed before the negotiation (p=0.080) and less comfortable during the negotiation (p=0.004) than men—a pattern that does not hold in the conversation treatment (see Online Appendix Table B.12).[33]

Given that confrontation costs are an important driver of gender differences, and that electronic communication has proven effective in reducing those costs, I explore whether the communication channel affects the gender gap in bargaining propensity. Online Appendix Figure A.11 shows the average opt-out rate by treatment, gender and communication condition. In the baseline treatment, electronic communication significantly encourages women to bargain (p=0.005), while the effect is smaller and non-significant for men (p=0.402). Consequently, women are significantly more reluctant to bargain than men in person (p=0.011), whereas this difference becomes smaller and non-significant electronically (p=0.140). This indicates that electronic communication may attenuate the gender gap in negotiation avoidance, consistent with electronic negotiations reducing confrontation costs. I nonetheless take this result with caution, as the sample partitions mechanically reduce sample size and statistical power, and the results in the asymmetric treatment maintain the same directionality but not the pattern of statistical significance.

5.2. Experiment 2

In the second experiment, deciders choose what to do electronically and in person. Figure 5 shows the average opt-out rate by gender and communication channel. For women, electronic communication strongly augments bargaining propensity. In person, almost 60% of female deciders avoid bargaining, while electronically this number goes down to 26% (p<0.001). This effect is more modest—although still weakly significant—for men: 24% of male deciders avoid bargaining in person, while only 15% electronically (p=0.096). Consequently, women are significantly more reluctant to bargain than men in person (p<0.001), while the gap is reduced electronically (p=0.085). This reinforces the notion that electronic communication can alleviate gender differences in bargaining propensity, although it does not erase them completely.

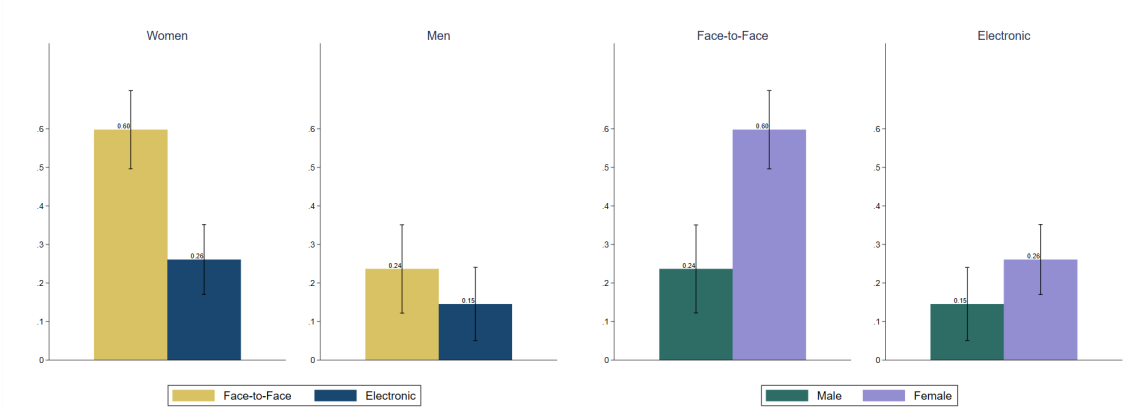
Regarding the interview content, the analysis of emotions in the first interview shows that female deciders associate more unfavorable emotions than male deciders with both in person and electronic negotiations (see Online Appendix Table B.15). This is consistent with women choosing to avoid negotiations more often than men. Regarding the motives given by those who prefer to avoid (some)

[32] It is worth recalling that deciders who aim for an equal split do not confront their counterparts—who are probably satisfied with such a deal. This means that if men were systematically more generous than women, this should lead them to opt in more frequently, irrespectively of confrontation costs. This is not consistent with the data (see Section 5.3 for a discussion of agreements by gender).

[33] I also analyse the opt out rate in the conversation and the dictator treatment by gender and find no significant differences, which reinforces the important role of confrontation costs explaining the gender entry gap vis-à-vis other motives.

FIGURE 5

EXPERIMENT 2 – GENDER AND COMMUNICATION CHANNEL INTERACTIONS



Note: Average opt-out rate by gender and communication condition in Experiment 2. Bars show the 95% confidence intervals.

negotiation(s), Online Appendix Figure A.13 shows the frequencies of each rationale for male and female deciders. Around 70% of male and female e-bargainers mention at least one of the motives associated with confrontation costs ("conflict avoidance" and "fear of conceding under pressure") to avoid in person negotiations. In the case of never bargainers, this number is similar for female deciders and lower for male deciders (60%), though the small size of the male subsample (n=5) cautions against making any strong interpretations of this difference. In sum, women avoid negotiations more often, and confrontation costs are the most frequent self-reported motive.

5.3. Negotiation Outcomes by Gender

Online Appendix Figure A.10 shows deciders' earnings after an agreement by round, treatment and gender in Experiment 1. To explore whether there are any statistically significant differences, I run an OLS regression, controlling for the communication channel and including round fixed effects, with standard errors clustered at the decider level. In the baseline and asymmetric treatments, where deciders can decide to opt in to or out of the negotiation, I find no significant differences in agreed earnings by deciders' gender (see columns 1 and 2 in Online Appendix Table B.13). Likewise, in the forced treatment, where bargaining is mandatory, I find no statistically significant difference either (see column 3). To further assess the robustness of these results, I repeat the same regression in Experiment 2, dividing the sample between those who bargain against their will, and those who bargain willingly (see Online Appendix Table B.14). Again, I find no significant differences in either case, although the coefficients for the involuntary sample are imprecisely estimated.

5.4. Discussion

The results in Experiments 1 and 2 confirm that women are more reluctant to bargain than men. Moreover, I find that this is true even in a setting where the disagreement payoffs equal the outside option, which makes bargaining the weakly dominant strategy (in monetary terms), irrespective of risk aversion. This is an important difference with previous studies (see for example Bowles, Babcock and Lai 2007 and Exley, Niederle and Vesterlund 2020). This feature of the design, along with the

asymmetric treatment, the debriefing survey, the psychological tests and the interviews, allows me to identify confrontation costs as an important driver of gender differences in bargaining propensity. Moreover, I find that electronic communication, which mitigates confrontation costs, can be effective in reducing the gender gap—even if it does not disappear completely.

Regarding negotiation outcomes, the analysis in sections 3.3 and 4.3 shows that forcing deciders into negotiations harms them. This is in line with the results in Exley, Niederle and Vesterlund (2020), which advocate against forcing women to lean in. However, in contrast with their findings, I do not find that forcing women into negotiations is particularly harmful relative to men. When I compare deciders' earnings by gender in voluntary and in involuntary negotiations, I do not find any significant gender differences (for a review of the differences in bargaining outcomes based on negotiators' gender, see Hernandez-Arenaz and Iriberri 2019 and Bowles, Thomason and Macias-Alonso 2022).

6. External validity

This section explores whether the conclusions that we have reached in the experiments generalize to other populations, whether they extend to a different type of negotiation and whether they align with the survey responses from a representative sample of the US.

6.1. The Subject Pool

Participants in Experiment 1 (and Experiment 2) belong to a specific subject pool: individuals living in Buenos Aires who have signed up through ORSEE. Still, it is worth noting how closely their results on the TKI® psychological test align with those of other populations. The avoidance scores at the 10th, 25th, 50th, 75th, and 90th percentiles in the Experiment 1 sample correspond to the scores of individuals at the 12th, 22nd, 49th, 78th, and 88th percentiles in the universe of subjects who have ever taken the TKI® on the Myers-Briggs platform—which includes thousands of individuals in multiple institutions. This suggests that they are neither more nor less inclined to avoid conflict than the broader population.

Reassuringly, the robustness of the effect in a different population is confirmed in a sample of 406 participants from Universidad Carlos III (Madrid, Spain), who took the baseline and asymmetric treatments of Experiment 1. Consistent with the results from Buenos Aires, the results in Madrid show that deciders are more reluctant to bargain in person than electronically, that female deciders are more reluctant to bargain than male deciders and that confrontation costs underlie these differences.[34]

6.2. Experiment 3 – A Different Type of Negotiation

Experiments 1 and 2 feature a relatively abstract distributive negotiation with complete information. In Experiment 3, 134 new participants are recruited in Buenos Aires with two goals: First, to explore whether results extend to a more realistic setting with imperfect information and more symmetric disagreement payoffs; second, to estimate the full cost distribution. In this experiment, subjects are randomly assigned to the role of buyer or seller and to a face-to-face or electronic session. Both buyers and sellers receive a 600 AR$ show-up fee. In addition, sellers receive a mug that they can sell. Buyers receive an additional 1,000 AR$ endowment, which they can use along with their show-up fee

[34] I discuss the details of the Madrid experiment and its outcomes further in Online Appendix E.

to buy the mug (see a picture of the mug in Online Appendix Figure A.12). I elicit the price that buyers are willing to pay to avoid a five-minute negotiation over the mug using a Becker-DeGroot-Marschak (BDM) procedure.[35]

The experiment was run in z-Tree (Fischbacher, 2007). The instructions were read aloud, including five examples to help subjects understand the BDM procedure (see the instructions in Online Appendix J). After reading the instructions and before starting the experiment, participants had to correctly answer some control questions to ensure that they properly understood the procedure (see Online Appendix K). To maximize observations, all participants decided as buyers before they learned their actual role. The mug that participants could trade was a commemorative mug of Argentina's national soccer team, and it could be found in some small brick-and-mortar shops as well as on MercadoLibre, a popular online marketplace. I chose this mug to make disagreement payoffs more symmetric than in Experiments 1 and 2, and to ensure that different participants would have different valuations of the item (see Online Appendix Figures A.14 and A.15). This heterogeneous subjective value—which is shared by many items that are traded outside the lab—introduces asymmetric information into the design, hampering the emergence of focal points. Along with a possible endowment effect, this allows for confrontation costs to kick in due to participants' self-serving biases (Babcock and Loewenstein, 1997).[36]

Figure 6 depicts the cumulative distribution function of the elicited bargaining costs in person and electronically. Visual inspection of the graph shows that the cost distribution in the face-to-face condition first-order stochastically dominates the distribution in the electronic condition. Almost 70% of the subjects are willing to pay something to avoid bargaining if the negotiation is face-to-face (vs. 40% electronically) and among those who are willing to pay, the modal price is 700 AR$ or more (vs. 200 AR$ in the electronic condition).[37] Moreover, while the median cost in the electronic condition is 0 AR$ and the average cost is 161 AR$, these figures rise to 300 AR$ (median) and 327 AR$ (average) in the face-to-face condition. A formal Mann-Whitney U test (Mann and Whitney, 1947) strongly rejects that both distributions are the same ($p<0.001$).[38]

To explore the relationship between bargaining costs and conflict avoidance in Experiment 3, I run an OLS regression of the price participants pay to avoid the negotiation on the TKI® avoidance

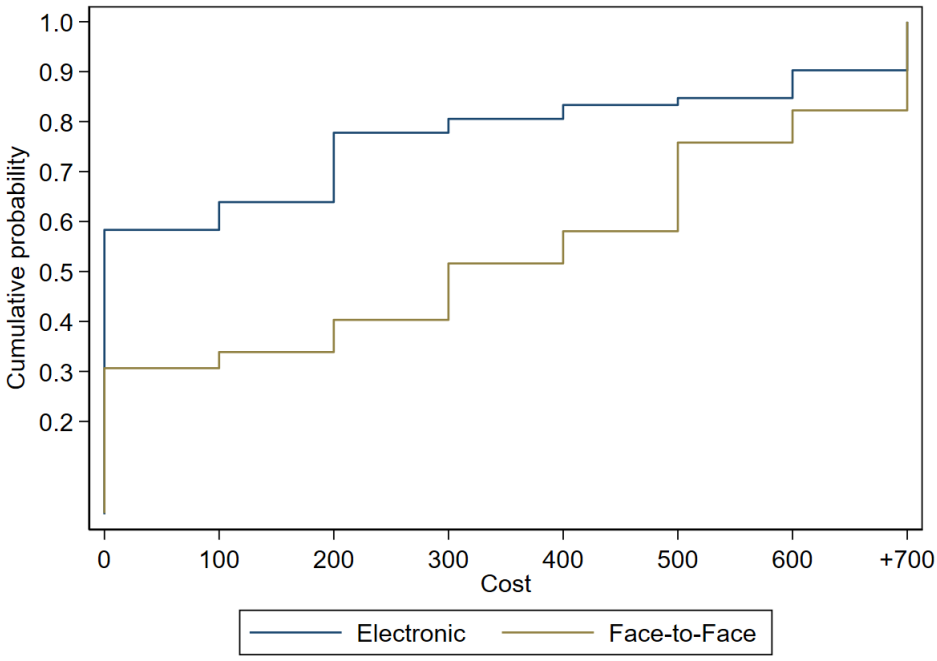
[35]The BDM works as follows. Buyers must choose whether to meet the seller and bargain for five minutes or whether to opt out. If they choose to bargain, they can make a deal, pay the price, and go home with the mug; or they can make no deal and leave with their full 1,600 AR$ endowment. If they opt out, they bring home an amount of money that varies depending on the outcome of a 10-sided die. Buyers must choose whether to opt in or out for each possible outcome of the die (see Online Appendix Table B.16). This allows me to use the difference between 1,600 AR$ and the minimum payment the buyer accepts for opting out as a lower bound for his bargaining costs (in the example of Online Appendix Table B.16, 300 AR$). The experimental protocol did not force participants to make consistent choices—they could opt out for a low value of the outside option and opt in for a higher value. Reassuringly, only 10% of participants made such inconsistent choices, suggesting that the instructions were well understood overall.

[36]The sessions were run between November 28 and December 2, 2022, right at the beginning of the Soccer World Cup in the Torcuato Di Tella University mobile lab. The average age of subjects was 29 years old, only 23.13% were students or former students of Di Tella and 53.4% of participants were women. Online Appendix Figure A.15 shows some actual listings of the commemorative mug I used.

[37]During Experiment 3, the average official exchange rate was 160 AR$/US$.

[38]Previous studies have found that some subjects may have difficulties understanding the BDM procedure. However, this is particularly troublesome when the error term is related to the treatment effect (see Plott and Zeiler, 2005; Cason and Plott, 2014; Bull et al., 2019). In this case, there are no obvious reasons to think that the mistakes participants could make in Experiment 3 should be systematically related to the communication channel.

FIGURE 6
NON-MONETARY COSTS OF BARGAINING



Note: This figure shows the cumulative distribution function for the elicited non-monetary costs of bargaining. The Mann-Whitney U test comparing both distributions yields a p=0.000.

score, controlling for participants' valuation of the mug (see Column 3 in Online Appendix Table B.17). Moving from the bottom to the top of the distribution on the TKI® avoidance score is associated with an increase of roughly 130 AR$ in the elicited avoiding-price (average price 238 AR$, s.d. 267 AR$). This correlation suggests that confrontation costs are behind participants' reluctance to bargain also in Experiment 3. Consistently, when I ask participants about their bargaining habits, almost 70% declare that they suffer confrontation costs and among them, the overwhelming majority (91.1%) declare those are higher in-person.[39]

6.3. Prolific Survey – A US Representative Sample

Finally, to explore the relevance of confrontation costs beyond the lab and measure the prevalence of haggling in digital marketplaces, I conducted a survey on negotiation attitudes using a sample of 400 subjects representative of the US census.[40] Table 8 confirms that confrontation costs extend to the general population: 84% of respondents reported experiencing unpleasant feelings when bargaining, a percentage that is significantly higher among women (76.8% of men vs. 90.3% of women, p<0.001).

[39] Among the 134 participants, only 60 got to bargain for the mug (34 electronically and 26 in person). Due to this limited sample size, the comparisons of bargaining outcomes lack statistical power. For completeness, I report the point estimate for the proportion of successful negotiations (69% face-to-face vs 53% electronic) and the agreed price (624 AR$ face-to-face vs 539 AR$ electronic). A tentative interpretation of these figures could suggest that subjects need to be more confident that their offer will be acceptable to engage in face-to-face negotiations, which is consistent with confrontation costs.

[40] The sample was taken from Prolific, and it is representative of the US census in terms of age, sex and ethnicity. A transcription of the survey can be found in Online Appendix L. The survey was conducted in October 2023.

Moreover, the vast majority of respondents (92.9%) stated that these feelings are more intense when bargaining in person. Consistently, barely 20% of respondents would bargain for a desired item priced above their budget constraint if they have to do it face-to-face, while almost 50% would do so electronically. This notion is confirmed in Online Appendix Table B.19, which shows the likelihood of respondents haggling over an item that is *slightly* beyond their budget constraint (average probability on a scale 1 to 5: 3.89 electronic vs 2.94 face to face, p<0.001). Notably, while women reported being more reluctant to bargain than men in person (3.26 men vs 2.63 women, p<0.001), they are equally likely to bargain electronically (3.91 men vs 3.87 women, p=0.737).

TABLE 8

Panel A: Bargaining Costs

	All	Men	Women	p-value
Emotional Cost	83.75	76.80	90.29	0.00
Cost Higher Face to Face	92.89	92.74	93.03	0.91

Panel B: Bargaining Choice

	Percentage
Never Try to Bargain	52.75
Always Try to Bargain	18.25
Bargain Only Electronically	27.00
Bargain Only Face to Face	2.00

Note: Panel A shows the percentage of Prolific respondents who report experiencing negative feelings (anxiety, awkwardness, nerves...) when they bargain, and the percentage who report that the intensity of these feelings is stronger face to face, along with the p-value of the t-test of the difference. Panel B shows the responses to a question that asks participants whether they would bargain for a good that they like, but that is well beyond their budget constraint.

As for the digital bargaining habits of the US population, the results of the survey show that more than 80% of the respondents have used c2c digital marketplaces where buyers can digitally haggle over listed items—including Craigslist, Facebook Marketplace and other similar platforms (see Online Appendix Table B.18).[41] Moreover, three quarters of those users report actively using the chat to try to obtain discounts from the seller. This indicates that the adoption of c2c platforms is currently widespread among the US population and that haggling in these platforms is rather common. Crucially, regarding sellers' attitudes, I find that the ample majority of respondents (97%) reported that they would not restrain potential buyers from contacting them electronically, even if they could (see Online Appendix Table B.20).

Taken together, the results of the Prolific survey suggest that the experimental findings describe the behavior of the average American citizen. Most people find it costly to confront others in negotiations, but they are much more inclined to do so if they can do it electronically. This is facilitated—among other devices—by c2c platforms, which are used widely in the US. The comparison of Panel B of Table 8 with the results of Experiment 2 shows that in the lab I am probably estimating a lower bound for negotiation avoidance: in the survey there are fewer always bargainers and more never bargainers. This is consistent with the fact that subjects in the lab are typically motivated to "live the experience".

[41] If we include eBay, where for some listings bargaining is possible, this number rises to almost 90% of the sample. For papers studying negotiation dynamics in eBay, see Backus et al. (2020) and Backus et al. (Forthcoming).

Indeed, it is consistent with the finding that more than 20% of deciders classified as always bargainers in Experiment 2 justify their choice with an "experience seeking behavior" (see Online Appendix Figure A.9), reinforcing the idea that I may be estimating a lower bound for the effect of confrontation costs in negotiation avoidance.

7. Conclusion

This paper explores the existence and nature of bargaining costs, as well as the impact of the communication channel on bargaining propensity. It shows that the negative emotions triggered by the confrontational aspect of haggling can explain why some people avoid negotiations, why women do so more often than men, and why bargaining propensity is higher in electronic settings. Crucially, it also shows that a greater reluctance to bargain in person, coupled with heightened confrontation costs, challenges the common belief that in-person negotiations are necessarily welfare-improving.

The conclusions of this paper highlight an overlooked consequence of the expansion of electronic bargaining, showing that it may promote negotiations and transactions that would otherwise be hindered, while reducing bargaining costs. Interestingly, this introduces a different perspective on the desirability of digital versus in-person interactions. Although many social researchers have argued that the possibility of hiding behind an electronic device generates toxicity and too much confrontation on platforms such as *X* or online newspapers (Baek, Wojcieszak and Delli Carpini, 2012; Coe, Kenski and Rains, 2014; Bor and Petersen, 2022), in negotiations, the story could be rather different. It is precisely the reduction in confrontation costs enabled by the digital environment that may help people engage in mutually beneficial encounters they would otherwise miss.[42]

Regarding women's particular reluctance to engage in conflict, my results are consistent with findings from psychologists in other domains (Brewer, Mitchell and Weber, 2002; Davis, Capobianco and Kraus, 2010; Thomas, Fann Thomas and Schaubhut, 2008), suggesting an explanation for the results in Small et al. (2007) and in Leibbrandt and List (2015). When workers are informed that the salary is negotiable—which reduces contextual ambiguity in Leibbrandt and List (2015)—or the interaction is framed as an opportunity to ask (in Small et al., 2007), opting in becomes less confrontational and thus more attractive to women. In addition, my paper shows that the possibility of bargaining electronically could also reduce the gender gap. These findings inform interventions oriented at "fixing the institutions" to close the gender gap in bargaining propensity (Recalde and Vesterlund, 2023; Kumar et al., 2026).[43]

Regarding external validity, the consistent results that I obtain in five different samples (three experiments in Buenos Aires involving 712, 294 and 134 participants, one in Madrid with 406 participants

[42] Consistent with this hypothesis, Shelegia and Sherman (2022) show that although 40% of retail stores in Vienna accept bargaining and are willing to offer discounts to those who haggle, only 6% of customers report doing so. Brañas-Garza et al. (2023) suggest that the usefulness of social interactions preceding negotiations might lie precisely in easing the negative feelings associated with bargaining in person. Zeng, Dasgupta and Weinberg (2014) and Huang (2020) argue that the development of platforms offering vehicles at fixed prices is a response to some customers' aversion to interacting with salespersons. Finally, Gneezy et al. (2017) show that electronic interactions enhance accuracy in the transmission of negative information.

[43] See Cullen, Pakzad-Hurson and Perez-Truglia (2025) for a new working paper testing in the field different interventions to increase bargaining propensity.

and a survey in the US with 400 respondents) suggest that the conclusions are probably valid for most Western populations. Arguably, this may not be the case for some Eastern countries, or for other cultures where the social norms guiding negotiations are different (Zhao, Chen and Li, 2020; Lange et al., 2023; Lu, 2023). This poses an interesting question for future research.

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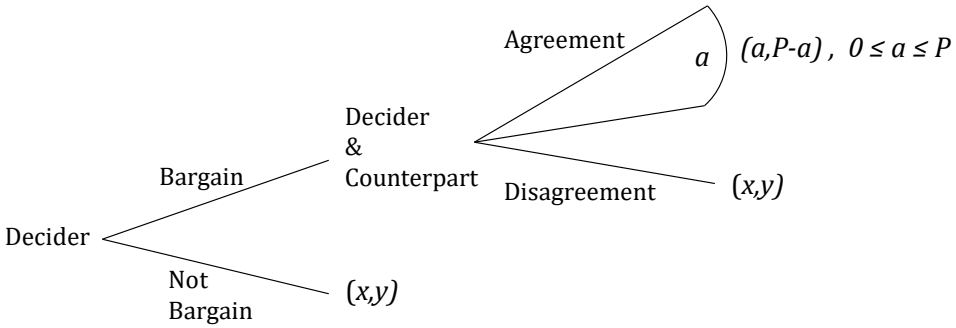
Online Appendix

Confrontation Costs in Negotiations: Bargaining Under the Veil of a Screen

Online Appendix A Additional figures

FIGURE A.1

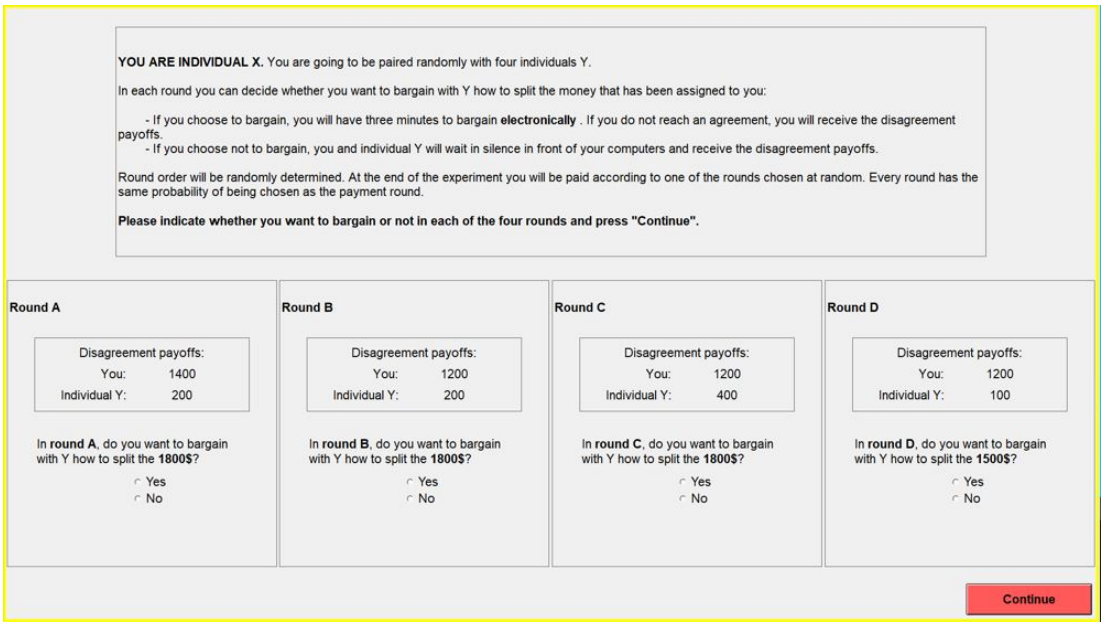
EXPERIMENT 1 – DESIGN



Note: This diagram summarizes the functioning of the experiment. In the first node, the decider chooses whether or not to bargain. If the decider chooses to bargain, the decider and his counterpart bargain for three minutes. If they do not reach an agreement, they get their disagreement payoffs. If the decider chooses not to bargain, both members of the pair wait for three minutes in front of their computers and directly get their outside options, which coincide with their disagreement payoffs.

FIGURE A.2

DECISION SCREEN

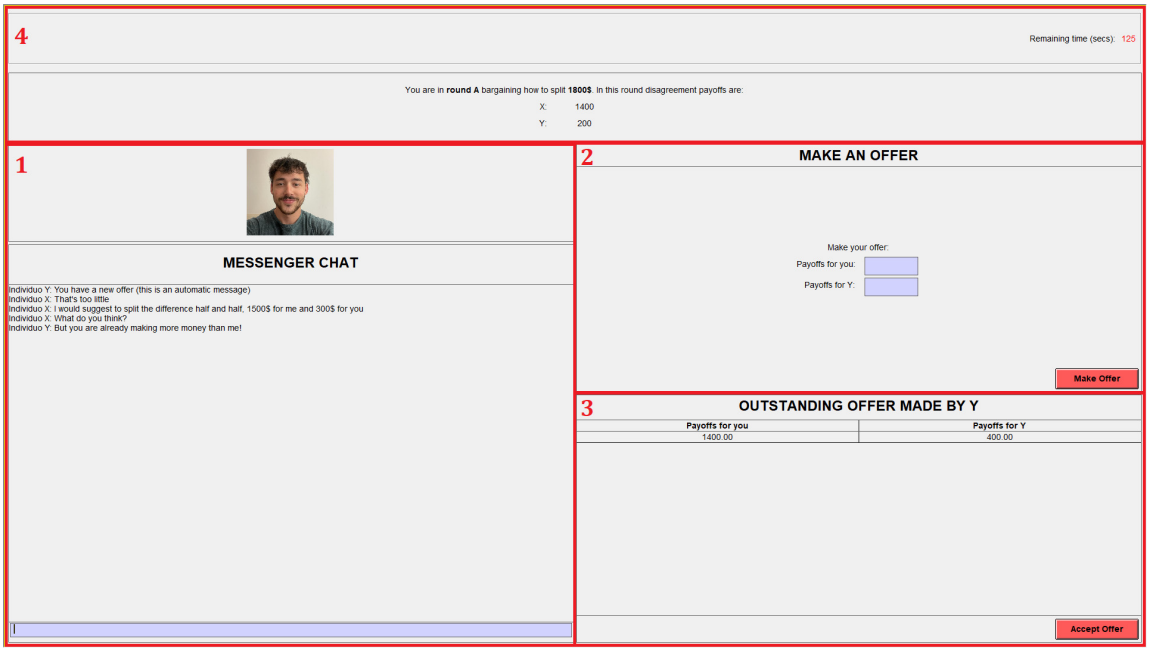


Note: Screenshot of an English translation of the decision screen in the baseline treatment. Decision screens in other treatments were adapted accordingly. Individual X is the decider and Individual Y is his counterpart.

Figure A.3
Di Tella Lab



Figure A.4
Electronic Chat



Note: Screenshot of an English translation of the negotiation screen in the baseline treatment. Individual X is the decider and Individual Y is his counterpart. In the bottom left of the screen, there is a chatbox (element 1) that includes a photo of the participant's bargaining partner. In the right part of the screen, there is a box in which participants can make offers (element 2), and right below, there is a box in which participants can see and eventually accept the offers made by the other party (element 3). An agreement is closed when one party accepts the offer of the other. In the upper part of the screen, there is information about the time remaining, the current round, the size of the pie, and the disagreement payoffs (element 4).

Figure A.5

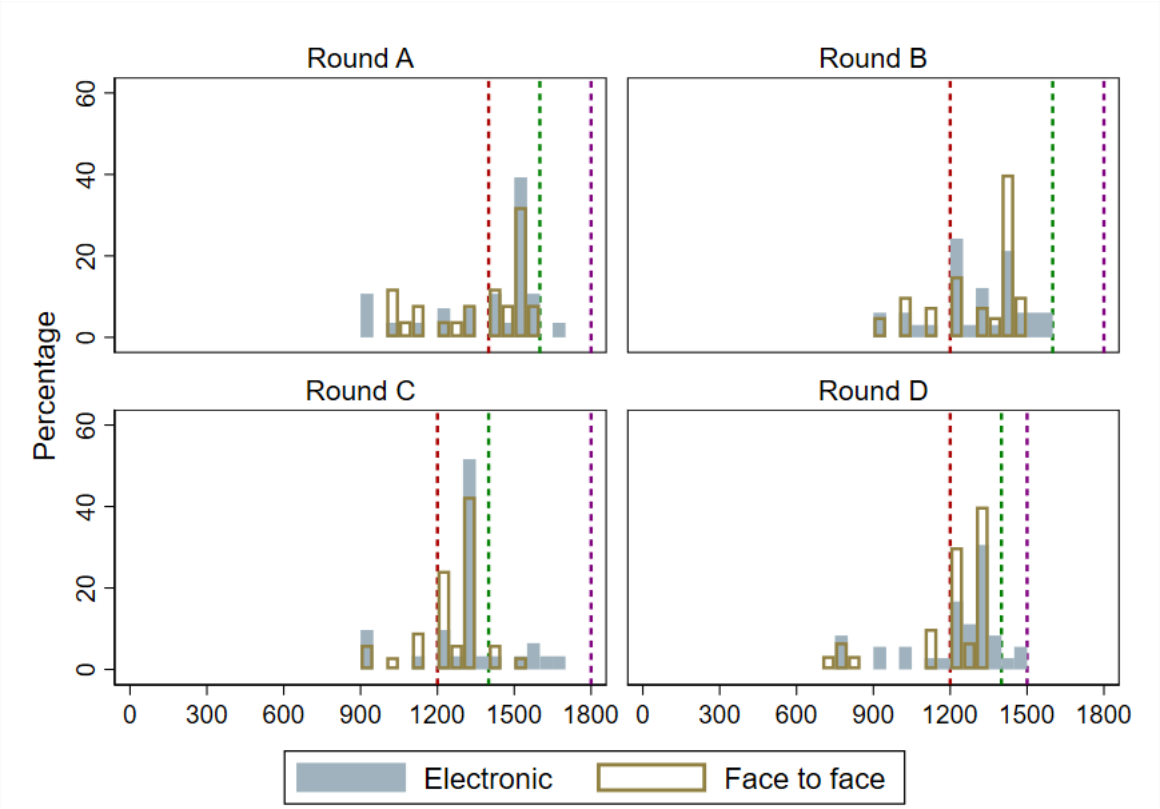
Negotiation Room



Note: Picture of a pair bargaining in the face-to-face condition. The elements on the negotiation table resemble those in the electronic chat. There is a chronometer, which indicates the time remaining, a recorder, which registers the negotiation, and an answer sheet. The answer sheet indicates the round, the size of the pie, and the disagreement payoffs. If participants make a deal, they must sign this answer sheet indicating the terms of the agreement.

FIGURE A.6

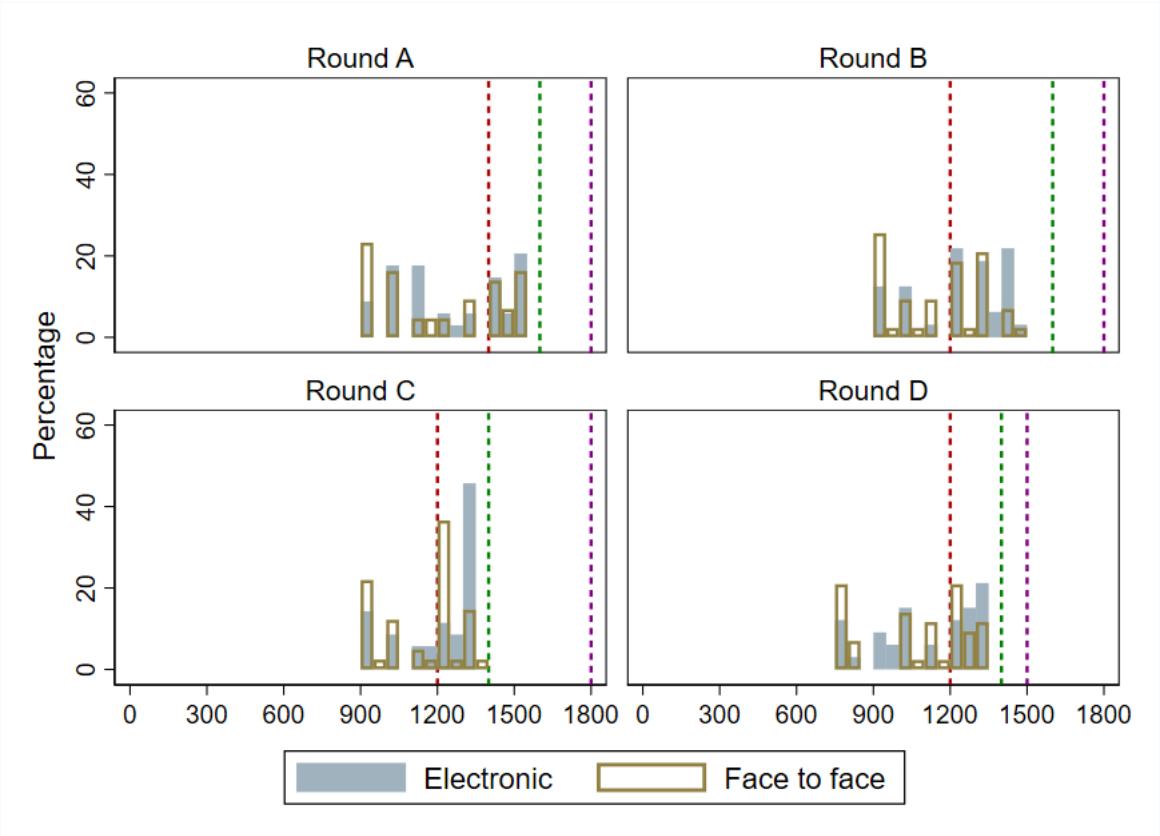
DECIDERS' EARNINGS AFTER AN AGREEMENT – ASYMMETRIC TREATMENT



Note: Distribution of deciders' earnings after an agreement in the asymmetric treatment. The red line denotes the decider's outside option (x), the purple line denotes the money to split (P) and the green line denotes the money to split minus the outside option of the counterpart ($P - y$). The outside options coincide with the disagreement payoffs.

FIGURE A.7

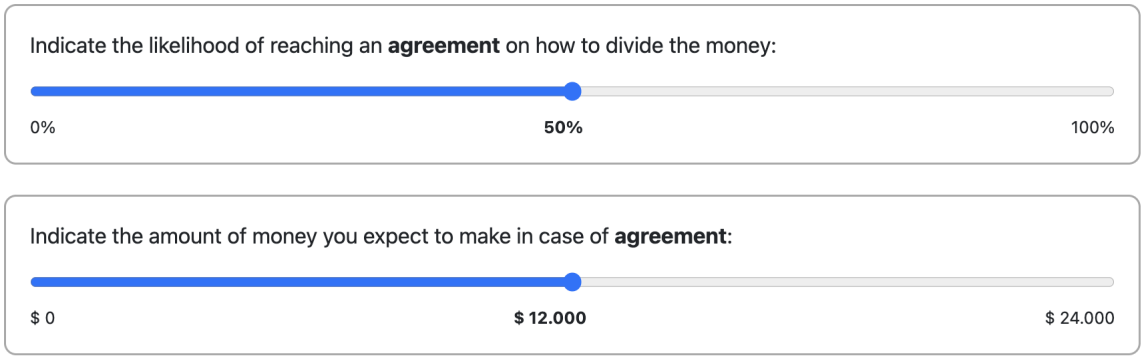
DECIDERS' EARNINGS AFTER AN AGREEMENT – FORCED TREATMENT



Note: Distribution of deciders' earnings after an agreement in the forced treatment. The red line denotes the decider's outside option (x), the purple line denotes the money to split (P) and the green line denotes the money to split minus the outside option of the counterpart ($P - y$). The outside options coincide with the disagreement payoffs.

FIGURE A.8

BELIEFS ELICITATION SCREEN - EXPERIMENT 2



Note: After learning the throw of the die, participants indicate their beliefs about the probability of reaching an agreement and the money they expect to make in case they reach an agreement.

Figure A.9

Deciders' Rationales



Note: This figure shows the percentage of always bargainers (N=66), old-school bargainers (N=7), e-bargainers (N=42) and never bargainers (N=23) that mention each rationale to explain their choice. I exclude 6 deciders who reported a decision which does not match their registered decision and 3 deciders who showed a poor understanding of the experimental instructions. The definition of each rationale can be found in Online Appendix D.2.

FIGURE A.10

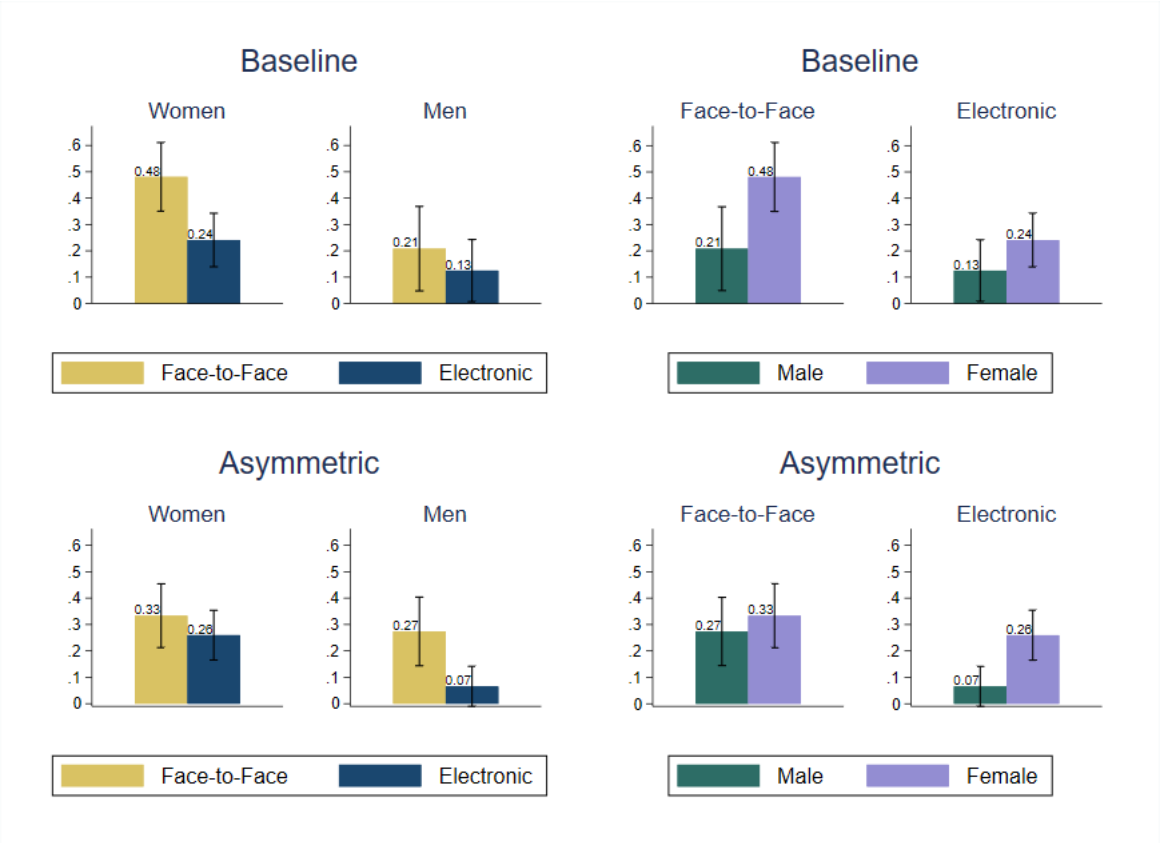
AVERAGE DECIDER EARNINGS AFTER AN AGREEMENT BY GENDER



Note: Average decider earnings after an agreement in the baseline, asymmetric and forced treatments reported by round and by gender. Bars show the 95% confidence intervals.

Figure A.11

Experiment 1 – Gender and Communication Channel Interactions



Note: Average opt out rate by treatment, gender and communication condition in Experiment 1. Bars show the 95% confidence intervals.

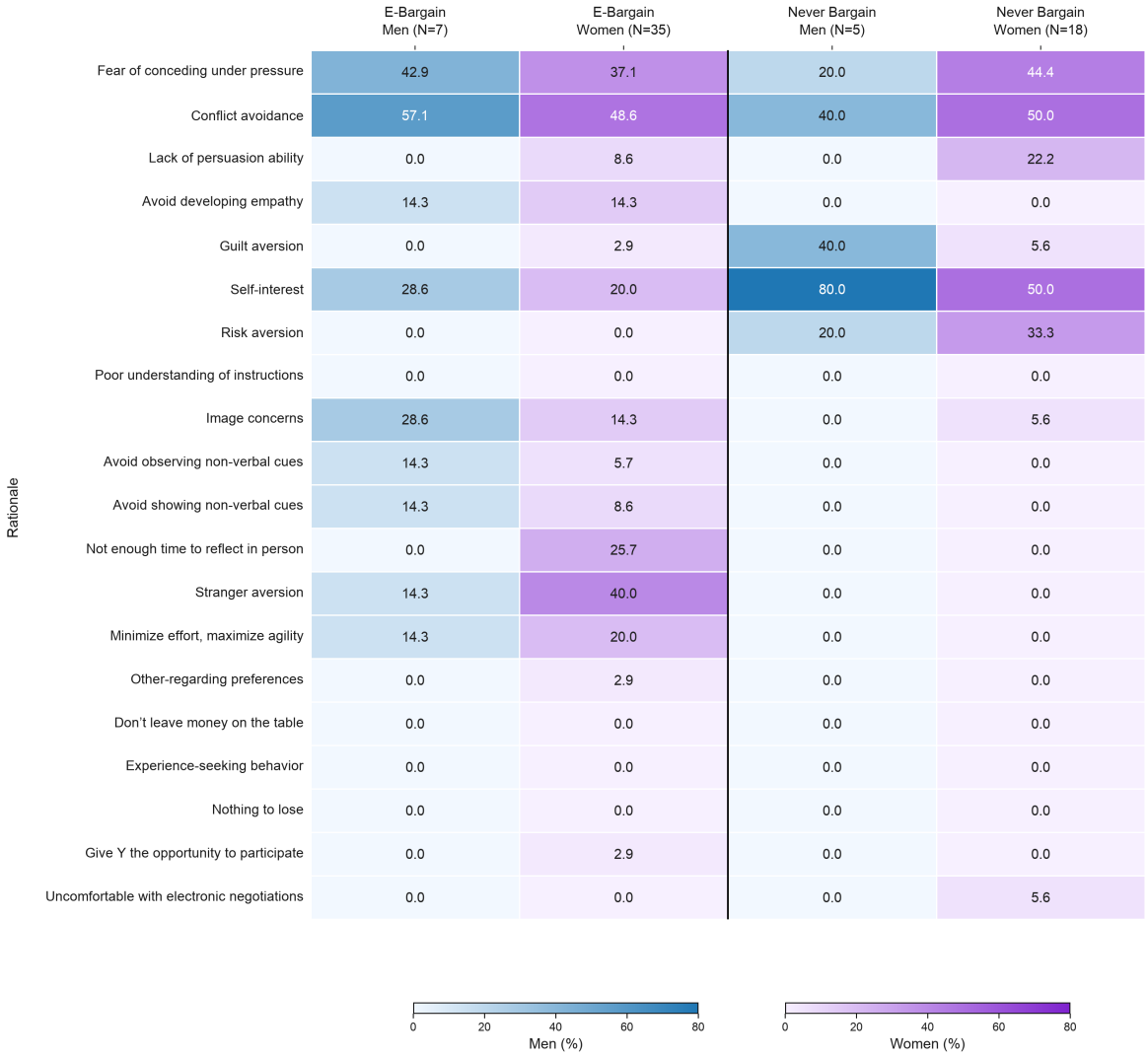
Figure A.12

"La Scaloneta" Mug



Figure A.13

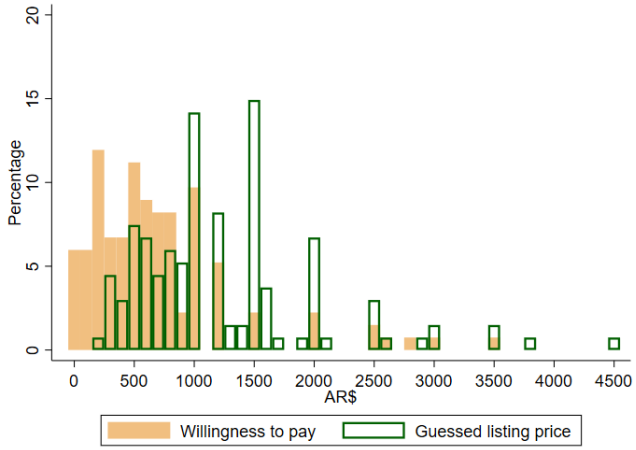
Deciders' Rationales by Gender



Note: This figure shows the percentage of male and female e-bargainers and never bargainers that mention each rationale to explain their choice. I exclude 6 deciders who reported a decision which does not match their registered decision and 3 deciders who showed a poor understanding of the experimental instructions. The definition of each rationale can be found in Online Appendix D.2.

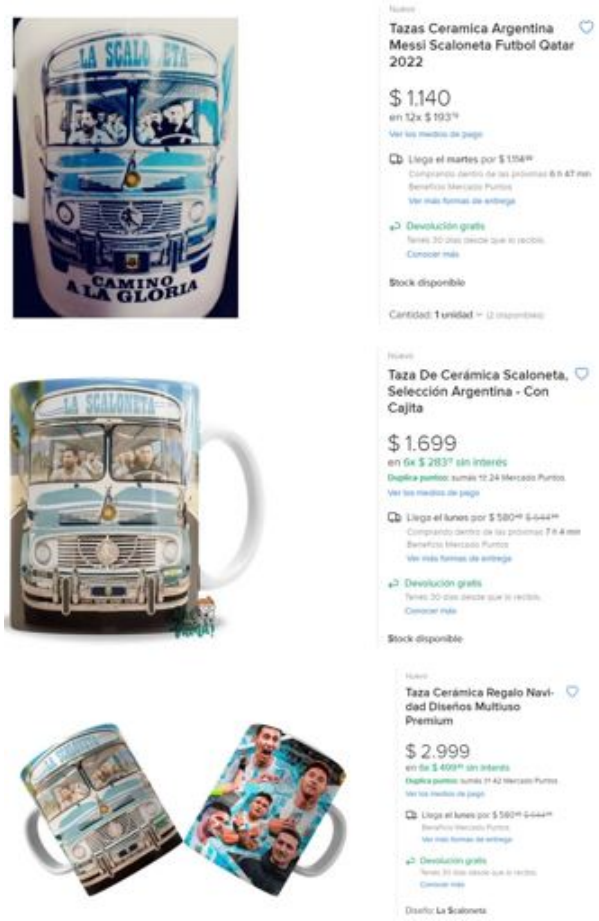
FIGURE A.14

WILLINGNESS TO PAY FOR THE MUG AND GUESSED LISTING PRICE



Note: This graph shows the distribution of participants' willingness to pay for the mug, along with their guess of the mug's listing price on MercadoLibre. Online Appendix Figure A.15 shows some actual listings in MercadoLibre.

Figure A.15
Mug Advertisements



Note: Advertisements of similar mugs on the website of MercadoLibre, Argentina's largest online marketplace. The mug in the upper advertisement costs 1,140 AR$ plus 1,114 AR$ delivery costs, the second has a price of 1,699 AR$ plus 580 AR$ delivery costs, and the mug in the third advertisement costs 2,999 AR$ plus 580 AR$ delivery costs. All mugs can also be picked up from brick-and-mortar shops.

Online Appendix B Additional tables

Table B.1

Asymmetric Treatment

	Electronic	Face to Face	Difference
Avoidance (%)	17.4	32.2	-14.8**
	(3.4)	(4.5)	(5.6)
Agreement (%)	83.9	92.5	-8.6**
	(3.3)	(2.0)	(3.8)
Total Payoff ($)	6,600.0	6,550.0	50.0
	(36.6)	(41.1)	(55.0)
Num. Deciders	46	52	

Note: This table reports the average avoidance rate, the average rate of agreement for those negotiations that occur, the average total monetary payoff per pair expressed in Argentinian pesos, and the number of deciders in the asymmetric treatment for each communication condition. It also reports the difference between the two communication conditions. * p<0.1, ** p<0.05, *** p<0.01 correspond to the p-values of two-sided t-tests.

Table B.2

Bargaining Strategy - Negotiation Openings

	(1) Decider Opens	(2) Decider Opens	(3) Decider's 1st Claim	(4) Decider's 1st Claim
Forced	-0.11	-0.17*	-148.58***	51.02
	(0.10)	(0.09)	(53.37)	(57.20)
Observations	295	331	136	161
Num. Clusters	138	158	67	79
Dep. Var. Mean	0.46	0.49	1204.26	1256.76
Sample	Face-to-Face	Electronic	Face-to-Face	Electronic

Note: In columns 1 and 2, I regress a dummy that takes a value of 1 if the decider opens the negotiation and zero if it is the counterpart who opens the negotiation, on a dummy that takes a value of 1 in the forced treatment and zero in the baseline. Column 1 uses the sample of face-to-face negotiations, and column 2 the sample of electronic negotiations. In columns 3 and 4 I change the dependent variable for the first demand of those deciders who open the negotiation. Errors are clustered at the participant level. * p<0.1, ** p<0.05, *** p<0.01.

Table B.3

TKI Avoidance Score

	(1) Opt-Out	(2) Opt-Out	(3) Opt-Out	(4) Opt-Out	(5) Opt-Out
TKI Avoiding	0.142*	0.153**	0.123*	0.223*	0.0739
	(0.0782)	(0.0753)	(0.0736)	(0.123)	(0.0821)
Face-to-Face		0.153***	0.158***		
		(0.0429)	(0.0422)		
Observations	744	744	744	380	364
Clusters	186	186	186	95	91
Sample	Full	Full	Full	Face	Elec.
Control	Equal split	Equal split	Eligibility	Equal split	Equal split

Note: OLS regression of a dummy that takes a value of 1 if an individual decides to avoid a negotiation on the individual's TKI avoidance score percentile, expressed on a scale from 0 to 1. Columns 1, 2, 4 and 5 control for altruism by including a dummy that takes a value of 1 if the decider implemented a 50/50 agreement; column 3 instead uses a dummy that takes a value of 1 if the decider indicated that he was eligible for non-remunerated experiments when he signed up in the Universidad Torcuato Di Tella recruiting platform for experimental subjects. Column 1 includes the full sample. Column 2 includes the full sample and a control that takes a value of 1 if the negotiation is face to face. Column 3 repeats the same specification as Column 2, but with the alternative altruism control. Column 4 includes only face-to-face negotiations. Column 5 includes only electronic negotiations. Standard errors are clustered at the individual level. * p<0.1, ** p<0.05, *** p<0.01.

Table B.4

TKI – Other Conflict Management Strategies

	(1) Opt-Out	(2) Opt-Out	(3) Opt-Out	(4) Opt-Out
TKI Score	0.00180	-0.15537**	-0.16410*	0.10330
	(0.077)	(0.077)	(0.087)	(0.079)
Face-to-Face	0.14906***	0.14121***	0.15600***	0.15393***
	(0.044)	(0.043)	(0.043)	(0.043)
Observations	744	744	744	744
Clusters	186	186	186	186
Measure	Compete	Compromise	Collaborate	Accommodate

Note: OLS regression of a dummy that takes a value of 1 if an individual decides to avoid a negotiation on a dummy that takes a value of 1 if the individual implements a 50/50 agreement, a dummy that takes value of 1 if the negotiation is face to face, and the TKI score percentile, expressed on a scale from 0 to 1, on: column 1 - compete, column 2 - compromise, column 3 - collaborate, column 4 - accommodate. Standard errors are clustered at the individual level. * p<0.1, ** p<0.05, *** p<0.01.

Table B.5

Debriefing Survey – Reported Feelings

Panel A: Forced Treatment			
	Electronic	Face-to-Face	Difference
Awkward	2.09	2.72	-0.64*
	(0.20)	(0.27)	(0.34)
Relaxed	4.43	4.49	-0.05
	(0.28)	(0.27)	(0.39)
Nervous	2.98	3.40	-0.43
	(0.25)	(0.27)	(0.37)
Comfortable	5.61	5.15	0.46
	(0.25)	(0.23)	(0.34)
Num. Deciders	46	47	
Panel B: Conversation Treatment			
	Electronic	Face-to-Face	Difference
Awkward	1.77	1.53	0.24
	(0.23)	(0.31)	(0.39)
Num. Participants	30	17	

Note: After the experiment, participants in the forced treatment rated on a 1 to 7 Likert scale how nervous and how relaxed they felt before the negotiation, how awkward and comfortable they felt during the negotiation, while participants in the conversation rated how awkward they felt during the conversation. Conversation ratings are oriented using the binary discomfort response: values above the midpoint are reversed for "No" and values below it for "Yes". Columns 1 and 2 report the average responses to the debriefing survey in the forced treatment (panel A) and in the conversation treatment (panel B), with standard errors in parentheses. Column 3 reports the difference between the average response in the electronic condition (col. 1) and the average response in the face-to-face condition (col. 2). * p<0.1, ** p<0.05, *** p<0.01 indicate the significance level of a t-test exploring whether this difference is different from zero. In this table, I excluded one session which used a 1 to 5 Likert scale, but if I include it, results remain qualitatively the same.

Table B.6
Top 5 Emotions (Total) by Bargainer Type

Always Bargainer

Face-to-face		Electronic	
Emotion	Share	Emotion	Share
Calm	0.18	Calm	0.21
Confidence	0.11	Confidence	0.15
Empathy	0.10	Coldness	0.10
Comfort	0.09	Comfort	0.09
Nervousness	0.09	Excitement	0.08
Num. Subjects	66	Num. Subjects	66
Num. Emotions	141	Num. Emotions	126

Old-School Bargainer

Face-to-face		Electronic	
Emotion	Share	Emotion	Share
Fun	0.17	Anxiety	0.14
Anxiety	0.11	Coldness	0.14
Confidence	0.11	Aggressiveness	0.07
Empathy	0.11	Ambition	0.07
Excitement	0.11	Awkwardness	0.07
Num. Subjects	7	Num. Subjects	7
Num. Emotions	18	Num. Emotions	14

E-Bargainer

Face-to-face		Electronic	
Emotion	Share	Emotion	Share
Awkwardness	0.23	Comfort	0.19
Nervousness	0.19	Nervousness	0.15
Anxiety	0.09	Calm	0.11
Pressure	0.09	Confidence	0.11
Embarrassment	0.07	Coldness	0.08
Num. Subjects	42	Num. Subjects	42
Num. Emotions	145	Num. Emotions	73

Never Bargainer

Face-to-face		Electronic	
Emotion	Share	Emotion	Share
Awkwardness	0.25	Awkwardness	0.27
Anxiety	0.15	Anxiety	0.16
Fear	0.15	Nervousness	0.08
Embarrassment	0.08	Embarrassment	0.05
Nervousness	0.08	Fear	0.05
Num. Subjects	23	Num. Subjects	23
Num. Emotions	40	Num. Emotions	37

Notes: Each panel reports the top 5 emotions by bargainer type using total emotion counts. Face-to-face combines general and face-to-face contexts; Electronic combines general and digital contexts. Num. Subjects denotes the number of deciders in each bargainer type after applying the sample restrictions. Of the 147 unique deciders in the original Excel file, 6 were excluded because their reported decision did not match their registered decision and 3 were excluded due to poor understanding of instructions, leaving 138 valid deciders. Num. Emotions denotes the total number of emotion mentions in each category in the long emotion dataset, which excludes 18 additional valid deciders with no recorded emotions.

Table B.7

Deciders' Opening Offers

	(1) Opening Offer		(2) Opening Offer
Always Barg.	1,040	Never Barg.	-2,051
	(743)		(1,886)
Face-to-Face	-3,826***	Electronic	2,020*
	(1,017)		(1,042)
Always Barg. × Face-to-Face	3,835**	Never Barg. × Electronic	-2,353
	(1,491)		(2,199)
Dep. Var. Mean	16,692	Dep. Var. Mean	16,692
p-value: $\beta_2 + \beta_3$	0.994	p-value: $\beta_2 + \beta_3$	0.864
Observations	39	Observations	39

Notes: OLS regression with robust standard errors. The dependent variable is the opening offer made by those always bargainers, e-bargainers and never bargainers who open the negotiation. The regressors are the decider type, the communication channel and an interaction between these two. Standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table B.8

Deciders' First Demands, Expected Earnings and Earnings in Case of Agreement

	(1) First Demands	(2) Expected Earnings	(3) Earnings
Die 3 or 4	-103.09	-53.68	1091.80**
	(658.80)	(554.31)	(523.33)
Observations	84	96	79
Dep. Var. Mean	16,119	15,618	14,951
Sample	Willing Bargainers	Willing Bargainers	Willing Bargainers

Note: This table reports the difference in expected earnings, first demands and effective earnings after an agreement, of willing bargainers, when the die is 3 or 4.

Table B.9

Emotions After Learning the Result of the Die

Participants Who Bargained

	Positive	Negative	Neutral	N
Always Bargainers	0.70	0.14	0.16	70
Old-School Bargainers	0.50	0.33	0.17	6
E-Bargainers	0.52	0.34	0.14	29
Never Bargainers	0.33	0.33	0.33	9

Participants Who Did Not Bargain

	Positive	Negative	Neutral	N
Old-School Bargainers	.	.	1.00	1
E-Bargainers	0.23	0.31	0.46	13
Never Bargainers	0.56	0.19	0.25	16

Note: In the second interview participants were asked how they felt after learning the result of the die. I classify their responses in three categories: positive, negative or neutral.

Table B.10

Emotions Before and During the Negotiation

Emotions Before Negotiation			
	Positive	Negative	Neutral
Decider	0.61	0.22	0.17
Counterpart	0.57	0.30	0.14
Emotions During Negotiation			
	Better	Worse	Same
Decider	0.34	0.16	0.50
Counterpart	0.43	0.30	0.27

Note: In the second interview participants were asked how they felt after learning the result of the die. I classify their responses in three categories: positive, negative or neutral. Those that bargained, are also asked whether their feelings in the negotiations were better, worse or as expected.

Table B.11

Deciders' Feelings and Counterpart's Expected Earnings

Panel A. Face-to-Face Negotiations

	Voluntary		Involuntary	
	Awkward	Comfortable	Awkward	Comfortable
	(1)	(2)	(3)	(4)
Counterpart's Beliefs	0.106**	-0.012	0.420**	-0.348***
	(0.041)	(0.052)	(0.149)	(0.103)

Panel B. Electronic Negotiations

	Voluntary		Involuntary	
	Awkward	Comfortable	Awkward	Comfortable
	(1)	(2)	(3)	(4)
Counterpart's Beliefs	0.033	-0.081	-0.145	0.025
	(0.058)	(0.053)	(0.074)	(0.076)

Note: OLS regression of the deciders' feelings during the negotiation on their counterparts' expected earnings. Deciders' feelings are reported at the end of the experiment in the debriefing survey, using a 1 to 5 scale. Counterparts' expected earnings are reported after learning the result of the die and the decider's choice for that result, and are expressed in thousands of AR$. A negotiation is involuntary if for that communication channel, the decider chose to opt out. A negotiation is voluntary if for that communication channel the decider chose to opt in. Errors are robust to heteroskedasticity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.12

Debriefing Survey – Gender

Panel A: Forced Treatment			
	Male	Female	Difference
Awkward	2.23	2.47	-0.25
	(0.27)	(0.22)	(0.35)
Relaxed	4.89	4.19	0.69*
	(0.30)	(0.25)	(0.39)
Nervous	2.80	3.39	-0.59
	(0.29)	(0.24)	(0.38)
Comfortable	5.97	5.04	0.94***
	(0.21)	(0.23)	(0.31)
Num. Deciders	35	57	
Panel B: Conversation Treatment			
	Male	Female	Difference
Awkward	1.79	1.63	0.16
	(0.33)	(0.23)	(0.40)
Num. Participants	19	27	

Note: After the experiment, participants in the forced treatment rated on a 1 to 7 Likert scale how nervous and how relaxed they felt before the negotiation, how awkward and comfortable they felt during the negotiation, while participants in the conversation rated how awkward they felt during the conversation. Conversation ratings are oriented using the binary discomfort response: values above the midpoint are reversed for "No" and values below it for "Yes". Columns 1 and 2 report the average responses to the debriefing survey in the forced treatment (panel A) and in the conversation treatment (panel B), with standard errors in parentheses. Column 3 reports the difference between the average response of male deciders (col. 1) and female deciders (col. 2). * p<0.1, ** p<0.05, *** p<0.01 indicate the significance level of a t-test exploring whether this difference is different from zero. In this table, I excluded one session which used a 1 to 5 Likert scale, but if I include it, results remain qualitatively the same.

Table B.13

Experiment 1 - Deciders' Earnings By Gender

	(1) Deciders' Earnings	(2) Deciders' Earnings	(3) Deciders' Earnings
Female	-33	-14	-20
	(52)	(35)	(36)
Num. Obs.	230	253	302
Num. Clust.	81	93	91
Dep. Var. Mean	1,195	1,268	1,152
Treatment	Baseline	Asymmetric	Forced

Note: OLS regression of the earnings of the deciders after an agreement on the deciders' gender. I control for the communication channel and include round fixed effects. Standard errors in parentheses, clustered at the decider level. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table B.14

Experiment 2 - Deciders' Earnings By Gender

	(1) Decider's Earnings	(2) Decider's Earnings
Women	126	833
	(549)	(937)
Num. Obs.	79	18
Dep. Var. Mean	14,951	13,861
Sample	Voluntary	Involuntary

Note: OLS regression of the earnings of the deciders after an agreement on the deciders' gender. I control for the communication channel. Robust standard errors in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table B.15

Emotions By Gender - Experiment 2 - First Interview

	Men		Women	
	Fav	Unfav	Fav	Unfav
Face-to-face	.52	.48	.23	.77
Electronic	.68	.32	.54	.46

Notes: This table reports the share of favorable and unfavorable emotion mentions by gender, pooling deciders across bargainer types. Face-to-face includes emotions associated with negotiations in general (g) and with face-to-face negotiations (f). Electronic includes emotions associated with negotiations in general (g) and with digital negotiations (d).

Table B.16

Experiment 3 – Buyer's Choice

Die	Choice Set	Bargain	Opt Out
10	Bargain, or opt out and get 1,800		X
9	Bargain, or opt out and get 1,700		X
8	Bargain, or opt out and get 1,600		X
7	Bargain, or opt out and get 1,500		X
6	Bargain, or opt out and get 1,400		X
5	Bargain, or opt out and get 1,300		X
4	Bargain, or opt out and get 1,200	X	
3	Bargain, or opt out and get 1,100	X	
2	Bargain, or opt out and get 1,000	X	
1	Bargain, or opt out and get 900	X	

Note: This table summarizes the choices that the buyer has to make in Experiment 3. The lower bound of the cost for a participant who makes such a choice would be 300 AR$.

Table B.17

Non-Monetary Cost of Bargaining

	(1) Cost	(2) Cost	(3) Cost
Face-to-Face	166.3***	148.6***	156.0***
	(44.48)	(47.06)	(46.51)
Female		35.21	28.27
		(46.44)	(46.02)
Will. to Pay		0.0442	0.0558
		(0.0499)	(0.0476)
Guessed Price		-0.0191	-0.0131
		(0.0370)	(0.0365)
TKI Avoiding			1.299*
			(0.750)
Observations	134	133	133

Note: OLS regression of the elicited cost of bargaining in Experiment 3 on the communication mode, controlling for the gender of the decider, his willingness to pay for the mug, the price he guesses the mug has in the market, and his avoidance score, expressed as the percentile of the participant's score in the historical distribution of TKI® scores. Heteroskedasticity-robust standard errors in parentheses. * p<0.1, ** p<0.05, *** p<0.01.

Table B.18

Panel A: c2c Usage

	Percentage
c2c User	80.75
c2c and/or eBay User	88.50

Panel B: Times Asked for a Discount

	Percentage of Users
Between 1 and 5	48.30
Between 6 and 10	9.29
More than 10	15.17
None	27.24

Note: Panel A shows the percentage of Prolific respondents that reported using c2c marketplaces and c2c marketplaces and/or eBay. Panel B shows how many times users of c2c platforms have used the chat to ask the seller for a discount in the last five years.

Table B.19

Panel A: Bargaining Probability

	Face to Face	Electronic	p-value
Bargaining Prob.	2.94	3.89	0.00

Panel B: Bargaining Probability by Gender

	Men	Women	p-value
Bargaining Prob. (Electronic)	3.91	3.87	0.74
Bargaining Prob. (Face to Face)	3.26	2.63	0.00

Note: Average self-reported bargaining probability for an item that is *slightly* beyond the budget constraint on a scale from 1 to 5, and p-value of the difference.

TABLE B.20

Panel A: c2c Usage as Sellers

	Percentage
Between 1 and 5	44.25
Between 6 and 10	7.25
More than 10	19.75
Never	28.75

Panel B: Sellers' Preferred Method of Contact

	Percentage
Only Phone Calls	3.25
Only Electronic	59.00
Both	37.75

Note: Panel A reports how many items respondents have listed on any c2c platform in the last five years. Panel B shows their preferred contact method in the event that they list something to sell.

Online Appendix C A simple model for the entry decision

The following reduced form model illustrates in a simple manner the decider's first stage. This model aims to capture the relevant trade-offs at entry and to derive some testable implications for the existence of confrontation costs. The model does not aim to cover the whole negotiation process (for an extension of the Rubinstein 1982 alternating offer bargaining model which includes haggling costs and yields similar predictions, please see Schweighofer-Kodritsch 2022).

C.1 Baseline treatment

In the baseline treatment, the payoffs for decider i if he reaches an agreement with his counterpart are represented by a reduced form function that encompasses the whole negotiation process: $g(d, \theta)$, where d represents how much the decider demands for himself, $\theta \in \{\theta_{f2f}, \theta_e\}$ denotes the communication channel, and there is a stochastic component. I assume that $g(d, \theta)$ is increasing in how much the decider demands in the negotiation, such that $\frac{\partial g(d,\theta)}{\partial d} > 0$, and that deciders never get more than what they demand ($g(d, \theta) \leq d$).

For some deciders, bargaining entails confrontation costs. Decider i's confrontation costs are non-decreasing in d and θ for all subjects, and they are increasing for some:

$$C_i(d, \theta) = c(d) \cdot \theta \cdot \epsilon_i \tag{2}$$

where $\epsilon_i \in [0, \infty)$ is larger than zero for those subjects who suffer costs. For them, I assume that all offers that demand more than *what is fair* are costly. The fair share for the decider is represented by k, which arguably should be somewhere between half of the pie and the decider's outside option, $k \in \left[\frac{P}{2}; x\right]$, so:

$$c(d) = 0 \text{ if } d \leq k,$$

$$c(d) > 0 \text{ and } c'(d) > 0 \text{ and } c''(d) \geq 0 \text{ if } d > k$$

Some deciders might derive some utility from the payoffs of their counterparts. Therefore, a general formulation for their utility function is:

$$u_i\left(v_x, v_y\right)$$

where the first argument denotes the decider's own payoffs, while the second denotes the payoffs of his counterpart. I assume that the derivative of the utility function is always positive with respect to its first argument, non-negative with respect to the second, and strictly greater for the first argument than for the second whenever $v_x < k$. These assumptions are standard and mean that deciders prefer getting more money to getting less, that they do not intrinsically enjoy their counterparts losing money, and that they may enjoy giving up some money in favor of their counterparts, but only to the point that they deem fair.

I let $P(\text{Agr} \mid d, \theta)$ be the ex-ante probability of agreement conditional on the decider's demand and the communication channel, and I assume that there exists a $d = x + \zeta$, such that $\lim_{\zeta \to 0} P(Agr|d, \theta) > 0$. Deciders know this probability, their epsilon and theta, and they have accurate beliefs about the stochastic component in $g(d, \theta)$.

With these ingredients, the participation constraint becomes as follows:

$$\max_d \; P(Agr|d, \theta) \cdot \{\mathbb{E}[u_i(g(d, \theta), P - g(d, \theta))|d, \theta] - u_i(x, y)\} - C_i(d, \theta) \geq 0 \tag{3}$$

where x and y denote the decider's and his counterpart's outside options, respectively.

Proposition 1

Suppose that, for every decider i and communication channel θ, there exists a demand $\widehat{d}$ such that

$$P(\text{Agr} \mid \widehat{d}, \theta) > 0 \quad \text{and} \quad \mathbb{E}[u_i(g(\widehat{d}, \theta), P - g(\widehat{d}, \theta)) \mid \widehat{d}, \theta] > u_i(x, y).$$

If decider i chooses not to bargain, then bargaining must be costly for him: $\epsilon_i > 0$.

Proof

If $\epsilon_i = 0$, then $C_i(d, \theta) = 0$ for all d. By assumption, there exists a demand $\widehat{d}$ for which the probability of agreement is strictly positive and the expected utility conditional on agreement exceeds $u_i(x, y)$. Therefore,

$$\max_d \{P(\text{Agr} \mid d, \theta) \left[\mathbb{E}[u_i(g(d, \theta), P - g(d, \theta)) \mid d, \theta] - u_i(x, y)\right]\}$$

$$\geq P(\text{Agr} \mid \widehat{d}, \theta) \left[\mathbb{E}[u_i(g(\widehat{d}, \theta), P - g(\widehat{d}, \theta)) \mid \widehat{d}, \theta] - u_i(x, y)\right] > 0$$

The entry condition is thus strictly satisfied, so decider i strictly prefers to bargain. Consequently, a decider who chooses not to bargain must have $\epsilon_i > 0$.

■

Under the assumption in Proposition 1, a positive opt-out rate in the baseline treatment indicates the existence of confrontation costs.

C.2 Asymmetric treatment

In the baseline treatment, if there were any differences in the probability of opting out by communication channel, they could be explained both by costs and by expected profits, the latter being a function of deciders' ability to persuade counterparts. To isolate the effect of costs, I examine behavior in the asymmetric treatment.

Remember that in the asymmetric treatment, the decider can choose how to split the pie. Therefore, both reaching an agreement and the earnings after the agreement become choice variables for him. This modifies his participation constraint as follows:

$$\max_{d} \; u_i(d, P-d) - C_i(d,\theta) \geq u_i(x,y) \tag{4}$$

Proposition 2

If a decider i is willing to bargain electronically but not in person, it must be because confrontation costs are larger face to face: $\theta_{f2f} > \theta_e$.

Proof

By definition, for decider i, argmax $u_i(d, P-d) \neq k$; otherwise he would always bargain. Following Proposition 1, we also know that $\epsilon_i \neq 0$, so we can also assume WLOG that $\epsilon_i = 1$. This makes costs $C_i(d,\theta) = c(d)\cdot\theta$.

Let me denote the optimal demands as d_{f2f} for the face-to-face condition and d_e for the electronic condition. This makes decider i's participation constraint:

$$\begin{gathered} u_i(d_e, P-d_e) - c(d_e)\cdot\theta_e \geq u_i(x,y) \\ u_i(d_{f2f}, P-d_{f2f}) - c(d_{f2f})\cdot\theta_{f2f} < u_i(x,y) \end{gathered}$$

Notice that d_e is an available choice in the face-to-face condition. Consequently, we know that:

$$\begin{gathered} u_i(d_e, P-d_e) - c(d_e)\cdot\theta_e \geq u_i(x,y) > u_i(d_{f2f}, P-d_{f2f}) - c(d_{f2f})\cdot\theta_{f2f} \geq \\ u_i(d_e, P-d_e) - c(d_e)\cdot\theta_{f2f} \end{gathered}$$

From this, it follows that $\theta_{f2f} > \theta_e$.

■

Considering that participants are randomly assigned to the electronic or face-to-face conditions, ϵ_i is iid across communication channels. Therefore, if the avoidance rate in the asymmetric treatment is higher in person, it must be because there are subjects who would bargain electronically, while they would avoid it in person. Following Proposition 2, this shows that confrontation costs must be higher face to face.

C.3 Forced treatment

Finally, while in the baseline and asymmetric treatments deciders must choose whether to engage in a negotiation, the forced treatment poses a different maximization problem. Instead of choosing whether to opt in, deciders must directly choose their first offer:

$$\max_{d} P(Agr|d,\theta)\cdot\mathbb{E}[u_i(g(d,\theta), P-g(d,\theta))|d,\theta] + (1-P(Agr|d,\theta))\cdot u_i(x,y) - C_i(d,\theta) \tag{5}$$

Proposition 3

Let d_j^* and d_{-j}^* denote the unique demands that solve the maximization problem in equation 5 for two deciders facing the same communication channel. Suppose that, when entry is voluntary, decider j strictly prefers not to bargain, whereas decider $-j$ weakly prefers to bargain. Assume that $c(d)$ is non-negative and non-decreasing, that $P(\text{Agr} \mid d,\theta)$ is non-increasing in d, and that

$$\frac{\partial\mathbb{E}[u_j(g(d,\theta), P-g(d,\theta)) \mid d,\theta]}{\partial d} \geq \frac{\partial\mathbb{E}[u_{-j}(g(d,\theta), P-g(d,\theta)) \mid d,\theta]}{\partial d}$$

for all d. If $d_j^* < d_{-j}^*$ and $P(\text{Agr} \mid d_{-j}^*,\theta) > 0$, then $\epsilon_j > \epsilon_{-j}$.

Proof

Since $d_j^* < d_{-j}^*$ and the probability of agreement is non-increasing in the demand,

$$P(\text{Agr} \mid d_j^*, \theta) \geq P(\text{Agr} \mid d_{-j}^*, \theta) > 0.$$

For decider j, the expected utility from bargaining, net of confrontation costs and relative to $u_j(x, y)$, is strictly negative at every feasible demand. Moreover, uniqueness of his optimum implies that he strictly prefers d_j^* to d_{-j}^*. Dividing these negative gains by their respective probabilities of agreement preserves this strict preference: the gain at d_{-j}^* is strictly lower and is divided by a weakly smaller positive probability. Subtracting the resulting expressions and cancelling $u_j(x, y)$ gives:

$$\begin{aligned} &\mathbb{E}[u_j(g(d_{-j}^*, \theta), P - g(d_{-j}^*, \theta)) \mid d_{-j}^*, \theta] \\ &\quad - \mathbb{E}[u_j(g(d_j^*, \theta), P - g(d_j^*, \theta)) \mid d_j^*, \theta] \\ &< \theta\epsilon_j \left[\frac{c(d_{-j}^*)}{P(\text{Agr} \mid d_{-j}^*, \theta)} - \frac{c(d_j^*)}{P(\text{Agr} \mid d_j^*, \theta)} \right]. \end{aligned}$$

For decider $-j$, the gain from bargaining relative to $u_{-j}(x, y)$ is non-negative at d_{-j}^*. Uniqueness of his optimum implies that this gain is strictly greater than the gain at d_j^*. Dividing each gain by its respective probability of agreement again preserves the strict preference for d_{-j}^*, because the gain there is non-negative and its probability of agreement is weakly smaller. Rearranging gives:

$$\begin{aligned} &\mathbb{E}[u_{-j}(g(d_{-j}^*, \theta), P - g(d_{-j}^*, \theta)) \mid d_{-j}^*, \theta] \\ &\quad - \mathbb{E}[u_{-j}(g(d_j^*, \theta), P - g(d_j^*, \theta)) \mid d_j^*, \theta] \\ &> \theta\epsilon_{-j} \left[\frac{c(d_{-j}^*)}{P(\text{Agr} \mid d_{-j}^*, \theta)} - \frac{c(d_j^*)}{P(\text{Agr} \mid d_j^*, \theta)} \right]. \end{aligned}$$

Integrating the assumed inequality between expected marginal utilities from d_j^* to d_{-j}^* shows that the expected utility increment on the left-hand side of the first inequality is weakly greater than the corresponding increment in the second. Combining the two inequalities therefore yields:

$$\begin{aligned} &\theta\epsilon_j \left[\frac{c(d_{-j}^*)}{P(\text{Agr} \mid d_{-j}^*, \theta)} - \frac{c(d_j^*)}{P(\text{Agr} \mid d_j^*, \theta)} \right] \\ &> \theta\epsilon_{-j} \left[\frac{c(d_{-j}^*)}{P(\text{Agr} \mid d_{-j}^*, \theta)} - \frac{c(d_j^*)}{P(\text{Agr} \mid d_j^*, \theta)} \right]. \end{aligned}$$

The common bracketed term is non-negative because $c(d)$ is non-negative and non-decreasing and the probability of agreement is non-increasing. The strict inequality rules out a zero value. Since $\theta > 0$, dividing by the common positive factor gives $\epsilon_j > \epsilon_{-j}$.

■

The inequality between the derivatives of expected utility conditional on agreement with respect to the demand, d, formalizes the assumption that decider j is not more generous than decider $-j$. Proposition 3 compares a decider who would strictly prefer to avoid bargaining (j) with one who would weakly prefer to bargain ($-j$). Under the remaining assumptions, if j demands less when forced to bargain, then he must have a higher confrontation-cost parameter ($\epsilon_j > \epsilon_{-j}$).

Let's analyze whether it is reasonable to assume that the average decider in the forced treatment is not more generous than the average decider who opts in in the baseline treatment. Consider that fully generous deciders, whose optimal demand in the absence of confrontation costs is $d = k$, incur no confrontation costs at that demand and thus opt in. There are no obvious reasons to expect that deciders who opt out in the baseline treatment (the *avoiders*) should be systematically more generous than those who opt in. If, despite this, we observe that when everyone is forced to bargain, deciders' demands are on average smaller, this pattern is consistent with avoiders making concessions to mitigate confrontation costs. Intuitively, those deciders who would abstain from bargaining to avoid confrontation may instead demand less money when forced to bargain. This would push the average demand down in the forced treatment relative to the baseline.

Online Appendix D Interviews of Experiment 2

This Online Appendix provides further details on the interviews of Experiment 2.

D.1 First Interview – Prompts and Technical Details

This interview uses the same parametrization and temperature as in Chopra and Haaland (2023), and maintains the instructions for the summary agent, the probing agent and the security agent largely unchanged. Here I provide the English translation of the instructions given to the topic agent. I focus on the interview structure for the deciders. The interview for the counterparts follows the same structure, with the particularity that counterparts' decisions are hypothetical (details for the counterparts can be sent upon request).

First Question

All the interviews with the deciders start with the same opening question:

```
In this interview, we would like to gather some additional information about
the decision you made on the previous screen.  || To that end, we first ask
you to tell us again what decision you made.  || Specifically, we want to know
whether you decided to negotiate or not to negotiate when the negotiation is
in person, and whether you decided to negotiate or not to negotiate when the
negotiation is electronic.  || Please be specific and tell us what decision
you made in each case.  || Thank you very much.
```

Topic 1

After the opening question, the chat asks deciders about the reasons behind their choice. It asks them three times to dig deep into the motivations, taking a completely agnostic standpoint. Here I reproduce the Topic with the instructions to the AI-interviewer:

```
The interview you are about to conduct is part of an academic research
project in behavioral economics.  In this project, we want to understand
why people decide to enter a negotiation, or avoid it, and how the communication
channel of the negotiation affects this decision.  To better understand
this question, we have designed the following experiment.  The participants
you will be interviewing have to choose whether they want to enter a five-minute
face-to-face negotiation, and whether they want to enter a five-minute
electronic negotiation.  Participants have to decide what they want to
do in each case—face-to-face and electronically, whether to enter the
negotiation or not.  In both cases, if they choose to negotiate, they
will have five minutes to decide how to split 24,000 Argentine pesos with
another person.  In the face-to-face negotiation they will have to discuss
it in person, and in the electronic negotiation they will have to discuss
it via chat.  If they do not reach an agreement within the five minutes,
the interviewees will receive 16,000 pesos, and the person they negotiated
with will receive 5,000 pesos (as you will notice, 16,000 pesos plus 5,000
pesos equals 21,000 pesos, an amount lower than the 24,000 pesos available
to split in the negotiation).  If, on the other hand, participants decide
not to negotiate, they will wait five minutes in silence in front of their
screen, and will receive 16,000 pesos themselves, while the person they
were paired with will receive 5,000 pesos.  Given that:  1.  the amount
they receive if they choose not to negotiate is the same as the amount
they receive if they fail to reach an agreement, and 2.  the amount to
be split (24,000 pesos) is greater than the sum of 16,000 pesos plus 5,000
pesos, economic theory predicts that all participants should attempt to
negotiate in both cases, face-to-face and electronically, since they would
not be risking anything.  Our goal is to understand why, in this context
where negotiating is weakly dominant, some participants prefer not to
negotiate.  To help you conduct the interview as effectively as possible,
I will now outline four important clarifications.  The first clarification
concerns the design of the experiment.  Participants have to decide what
to do if the negotiation is electronic, and what to do if the negotiation
is face-to-face.  They make this decision before knowing whether the negotiation
will be face-to-face or electronic.  That is, at this point in the interview,
participants only know what they themselves have chosen, but they do not
yet know what form the negotiation will take.  The second clarification
is that you should bear in mind that in this experiment, in terms of monetary
payoffs, the optimal strategy is always to negotiate (for the reasons
already explained).  Thus, any unwillingness to negotiate on the part
of participants must be explained by non-monetary reasons or by a misunderstanding
of the instructions.  Please keep this in mind throughout the interview:
in monetary terms, negotiating is the optimal choice.  So if a participant
argues that it is in their interest not to negotiate, ask them why, and
```

```
point out that they could earn more money by negotiating (always address
participants in plain terms:  do not use technical language like “optimal”
or refer to economic theory, bear in mind that you are not speaking with
an expert).  The third clarification is that non-monetary reasons—which
probably explain why some participants do not want to negotiate—are just
as valid as monetary reasons—which may explain why others choose to negotiate.
Do not judge participants’ responses.  Our goal is to understand their
motivations, whatever those may be, not to judge them.  Finally, the fourth
and last clarification is that in this experiment participants do not
have to choose between negotiating electronically or face-to-face.  It
is not one or the other.  Participants have to decide whether or not they
want to negotiate electronically, and whether or not they want to negotiate
face-to-face (i.e., they have to make two independent choices).  Therefore,
if a participant were to argue that they “opted out in person, because
they preferred to negotiate electronically instead of face-to-face,” you
should point out that they could have chosen to negotiate both ways, electronically
and face-to-face.  Since the two are not mutually exclusive (i.e., you
need to understand why they chose to negotiate in only one way rather
than both ways).  In summary:  I want participants to explain the reasons
behind their decision, whatever that decision might be.  With particular
emphasis on understanding the motivations of those who preferred not to
negotiate, but also understanding the motivations of those who chose to
always negotiate.  Bearing in mind that, in monetary terms, there is always
more money to be made by negotiating.  Furthermore, I want your interview
to remain focused on the interviewee at all times—I do not want you to
ask them what they think other people in their position would have done.
Focus on the interviewee; do not ask them about third parties.
```

Topic 2

In Topic 2, I introduce the concept of confrontation costs to the AI-interviewer, and request him to explore their role in a non-leading way. It asks the interviewee two times about this issue. Here I reproduce the Topic with the instructions to the AI-interviewer:

```
The authors of the research project for which you are conducting this interview
define “confrontation costs” as the negative emotions that arise from the conflict
inherent in any negotiation.  These emotions include anxiety, nervousness,
discomfort, apprehension, etc., and arise from the anticipation of potential
conflict (or from the conflict itself).  Next, you should explore the role
that “confrontation costs” play in the decision made by the participant.  I
want you to make a superb effort to do this in a subtle way, without leading
the interviewee in any direction.  In fact, you should never refer to the term
“confrontation costs” directly.  Remember that the interview must always remain
focused on the interviewee, not on what they think might happen to others in
their situation.
```

Closing Questions

Every interview finishes with the same two closing questions:

```
Is there anything else you would like to share?

Taking into account everything we have discussed, could you summarize, in one
sentence, the decision you made on the previous screen and why you made it?
```

D.2 First Interview – Rationales

This section defines each code, describes the creation of the rationales codebook, explains the labeling process, discusses the robustness of the labeled dataset and discusses the robustness of the analysis in the paper.

Rationales Codebook Description

These are the 21 codes that emerged from reading the interviews to explain participants’ choices:

1. Fear of conceding under pressure: The interviewee fears conceding to the counterpart’s demands; reports difficulty saying “no”; expresses concern about behaving in an accommodating way; and/or states that disagreements cause him/her discomfort. The explanation focuses on the discomfort—or the difficulty—that he/she experiences when being assertive and/or on a desire to avoid confronting the counterpart’s persuasion strategies.

2. Lack of persuasion ability: The interviewee explicitly states that he/she would not know how to persuade the other party and/or refers to his/her poor persuasion skills. The explanation is centered on a (lack of) ability.

3. Avoid developing empathy: The interviewee explicitly states that he/she wanted to avoid developing empathy with his/her counterpart. The explanation centers on his/her intrinsic empathic feelings, which he/she fears may be aroused during an encounter. The focus of the argument is internal: once he/she puts a face to someone, that person's wellbeing enters his/her utility function with a higher weight.

4. Guilt aversion: The interviewee mentions a concern about feeling guilty for not sharing the money evenly in the negotiation and/or about making the counterpart feel bad.

5. Conflict avoidance: The interviewee explicitly mentions a desire to avoid conflicts, feeling vulnerable, and/or feeling uneasy about the reactions the counterpart might have.

6. Self-interest: The interviewee explicitly states that he/she made the decision based on self-interest.

7. Risk aversion: The interviewee explicitly refers to a desire to avoid risks and/or to play it safe to justify his/her decision.

8. Poor understanding of instructions: The interviewee's explanation is clearly incongruent with the rules of the experiment.

9. Image concerns: The interviewee expresses a concern about his/her self- or social-image.

10. Avoid observing non-verbal cues: The interviewee refers to observing the counterpart's body language as a negative feature that deters him/her from bargaining in person. The way the interviewee phrases the argument clearly indicates that the problem is observing non-verbal cues.

11. Avoid showing non-verbal cues: The interviewee refers to the fact that the counterpart can observe one's body language as a negative feature that deters him/her from bargaining in person. The way the interviewee phrases the argument clearly indicates that the problem is showing non-verbal cues.

12. Not enough time to reflect in person: The interviewee explains his/her decision by noting that in-person negotiations do not provide enough time to reflect—or arguing that electronic negotiations give him/her more time to reflect.

13. Stranger aversion: The interviewee rationalizes his/her decision based on the discomfort he/she feels when interacting with a stranger.

14. Minimize effort, maximize agility: The interviewee emphasizes that he/she wants the negotiation to be fast and/or quick and/or agile and/or to minimize effort.

15. Other-regarding preferences: The interviewee decides to join the negotiation because he/she wants to divide the pie in halves. It is a fairness argument.

16. Don't leave money on the table: The interviewee explains his/her decision by noting that there are gains to be made from trade. It is an efficiency argument.

17. Experience-seeking behavior: The interviewee highlights a desire to live the full experience and/or a desire to have fun and/or curiosity about the negotiation to rationalize his/her decision.

18. Nothing to lose: The interviewee highlights that he/she has his/her outside option guaranteed.

19. Give Y the opportunity to participate: The interviewee highlights that he/she wants to give Y the chance to bargain and/or that he/she considers giving Y this opportunity the fair thing to do.

20. Uncomfortable with electronic negotiations: The interviewee distrusts and/or dislikes electronic interactions and/or argues that electronic negotiations do not give him/her the tools needed to conduct a successful negotiation.

21. Poor understanding of the interview (only for counterparts): The interviewee's explanation does not answer the question he/she is asked. For example, he/she answers what he/she would have done had he/she been Individual X.

Rationales Codebook Creation

I followed the best practices proposed in Haaland et al. (2025) to create the codebook of rationales. First, a trained research assistant and I read all deciders' interviews. This includes 147 interviews and 2,371 questions and responses. I defined *rationale* as "how the participant explains his/her decision to bargain or not to bargain in the experiment." I asked the RA to ignore the instances in which the interviewee talks about contexts unrelated to the experiment, or about the potential behavior of participants other than himself. We separately read the interviews, code each response with a code of our own and give a definition for the code. Then, we discussed together each response until we unified codes and definitions. Finally, we kept the 20 most representative codes and added one additional code, which may only apply to counterparts' interviews.

Rationales Data Coding

After creating the codebook, a second RA went through all the interviews. He was given the codebook, the definition of what a rationale is, along with the following instructions:

- The following codes identify the reasons given by interviewees to rationalize their choices in the decision screen. Each interviewee's answer may include one or more codes. Please note that, in some cases, the difference between two codes may be subtle. Select two or more codes only if they all clearly apply to the interviewee's answer. Otherwise, think carefully about which code(s) properly describe the interviewee's argument and restrict your answer to those.
- Sometimes, the interviewee's answers are vague, very convoluted, or unclear. Please leave these observations blank; do not try to guess.
- If you think that new rationales are needed to encompass some of the explanations given by the deciders, or that any definition of the existing rationales should be adjusted, please note down your proposal.

Then, the three of us met, reviewed again the 2,371 questions and responses one by one to reach a final consensus on the codes that correspond to each answer–if any, and refined the codebook definitions. No new codes were added in this process. The two RAs repeated the process for the 147 counterparts' interviews, which include 2,077 questions and responses (note that in the analysis that can be found in the paper, I am only exploiting the deciders' answers). I revised a subset of these answers. Then, we collapse all the information at the interview level.

Robustness of the Rationales Coding

To explore the robustness of the labeled data, an independent RA, who was unaware of the research hypothesis, and gpt-4o, labeled it again. I use the F1-score to assess the robustness of the coding. The F1-score is the harmonic mean of precision and recall. Precision measures how many of the rationales identified by the coder (gpt-4o or independent RA) are correct, while recall measures how many of the correct rationales (from the labelled dataset) the coder successfully identifies. All scores range from 0 to 1, with higher values indicating better performance. The average F1-score of the independent RA was 0.87 and that of gpt-4o was 0.75. These numbers are considered very satisfactory (Bermejo et al., 2025).

Robustness of the Rationales Analysis

I repeat the analysis in the paper using only the responses given to Topic 1, before I introduce the concept of confrontation costs to the AI-interviewer. All the conclusions remain qualitatively unchanged.

D.3 First Interview - Emotions

Besides coding the rationales, I also coded the emotions that participants associate with negotiations in the first interview. This section provides the full list of emotions, describes the coding process, discusses the robustness of the coding and discusses the robustness of the analysis in the paper.

List of Emotions

Table D.21 lists all the emotions identified in the first interview. It also indicates whether they are classified as favorable or unfavorable for bargaining.

TABLE D.21

LIST OF EMOTIONS

Emotion	Favorable	Emotion	Favorable	Emotion	Favorable
Aggressiveness	1	Security	1	Consciousness	1
Ambition	1	Selfishness	1	Contempt	0
Anger	0	Serenity	1	Curiosity	1
Annoyance	0	Shame	0	Boredom	0
Anxiety	0	Shyness	0	Defeatism	0
Apathy	0	Stress	0	Determination	1
Awkwardness	0	Suspicion	0	Discomfort	0
Calm	1	Tension	0	Effusiveness	1
Challenge	0	Tranquility	1	Embarrassment	0
Coldness	1	Trust	1	Empathy	0
Comfort	1	Uncertainty	0	Excitement	1
Compassion	0	Upset	0	Expectation	1
Competitiveness	1	Vulnerability	0	Fatigue	0
Concentration	1	Worry	0	Fear	0
Confidence	1	Optimism	1	Freedom	1
Hope	1	Pity	0	Frustration	0
Impatience	0	Positivity	1	Fun	1
Indifference	0	Pressure	0	Greed	1
Insecurity	0	Protection	1	Guilt	0
Intimidation	0	Restlessness	0	Laziness	0
Irritation	0	Judgement	0	Nervousness	0
Happiness	1				

Coding of Emotions

To code the emotions, two research assistants and I independently read the 147 deciders' interviews, which include 2,371 questions and responses. Moreover, two research assistants independently read the 147 counterparts' interviews, which include 2,077 questions and responses (note that in the analysis that can be found in the paper, I am only exploiting the deciders' answers). Codes were decided after two rounds of debate about each individual answer.[44]

These were the instructions to code emotions: List the emotions that the participant associates with negotiations (not with avoiding negotiations). Take into account the following considerations:

- Sometimes, the chatbot asks the participant about how others might feel in negotiations. Restrict your attention to the emotions that the participant associates with negotiations when he/she is talking about himself/herself.
- Sometimes, the chatbot asks participants how they feel/have felt in situations that are unrelated to the experiment. Do not pay attention to these emotions, focus on the answers that are linked to the experiment.
- Sometimes, the chatbot asks participants how they would feel in a hypothetical situation in the experiment: e.g., how they would feel if they bargained with someone of the opposite sex or how they would feel if their counterpart got angry. Do not pay attention to these emotions, only to the emotions that apply to the experimental negotiation in general.
- Sometimes, the interviewee expresses his/her emotion "by absence": e.g., I was not nervous. As a coder, identify the emotion that best describes what the interviewee is expressing: e.g., calm.
- Sometimes, when asked about his/her emotions towards bargaining, the interviewee answers in

[44] In the case of deciders, one RA and I first discussed the answers; we then discussed them again with a second RA. In the case of counterparts, the two RAs had a round of debate, then they had a second round of debate with me for a subset of the answers.

an indirect way: e.g., I feel I can perform very well in a negotiation. As a coder, identify the emotion that best describes what the interviewee is expressing: e.g., confidence.

- Usually, the interviewee refers to emotions in general. Please note these emotions by indicating (g) in parentheses. When the interviewee is only talking about emotions associated with face-to-face or with digital negotiations, please note (f) or (d) respectively. Remember that we only care about the answers that apply to the interviewee's feelings towards the negotiation in this experiment.
- Sometimes, the chatbot asks about differences in how the interviewee feels electronically and face-to-face. Please note an emotion only when the interviewee clearly indicates experiencing this emotion in an absolute sense. For example, one may feel more uncomfortable bargaining electronically than in person while still feeling comfortable in both cases. Thus, as a coder, you should not note discomfort (d): Unless it is unambiguous that the interviewee experiences an emotion in a certain environment, it should not be coded.

Robustness of the Coding of Emotions

To explore the robustness of the emotions coding, an independent RA, who was unaware of the research hypothesis, and gpt-4o, coded the data again. I use the F1-score to assess the robustness of the labels. The F1-score is the harmonic mean of precision and recall. Precision measures how many of the emotions identified by the coder (gpt-4o or independent RA) are correct, while recall measures how many of the correct emotions (from the labelled dataset) the coder successfully identifies. All scores range from 0 to 1, with higher values indicating better performance. The average F1-score of the independent RA was 0.85 and that of gpt-4o was 0.78. These numbers are considered very satisfactory (Bermejo et al., 2025).

Robustness of the Analysis of Emotions

When I collapse the information at the interview level, I count each individual appearance of each emotion. Results remain qualitatively unchanged if I only count each emotion once per interview.

D.4 Second Interview - Prompts and Technical Details

This interview uses the same parametrization and temperature as in Chopra and Haaland (2023), and maintains the instructions for the summary agent, the probing agent and the security agent largely unchanged. Here I provide the English translation of the instructions given to the topic agent. I focus on the interview structure for those that bargained. The interview for those who did not bargain follows the same structure but is shorter, as it does not ask them about any negotiation (details can be sent upon request).

First Question

All the interviews with participants who bargained start with the same opening question:

```
In this interview, we would like to learn more about how you felt before and
during the negotiation.  || Let us begin by exploring how you felt beforehand.
|| When you found out that you were going to negotiate, how did you feel?
```

Topic 1

After the opening question, the chat asks participants about their feelings when they learned they were going to bargain. It asks participants two times about this. Here I reproduce the Topic with the instructions to the AI-interviewer:

```
The interview you are about to conduct is part of an academic research project
in behavioral economics.  In this project, we want to understand why people
decide to enter a negotiation, or avoid it, and how the communication channel
through which the negotiation takes place affects this decision.  To better
understand this question, we have designed an experiment.  You will be interviewing
the participants of this experiment, which I will now explain.  The participants
you will be interviewing have been assigned the role of Individual X or Individual
Y. Individuals X have to choose whether they want to enter a five-minute face-to-face
negotiation with Individual Y, and whether they want to enter a five-minute
electronic negotiation.  Individuals X have to decide what they want to do
in each case, face-to-face and electronically, whether to enter the negotiation
or not.  In both cases, if they choose to negotiate, they will have five minutes
to decide how to split 24,000 Argentine pesos with Individual Y. In the face-to-face
negotiation they will have to discuss it in person.  In the electronic negotiation,
they will have to discuss it via chat.  If they do not reach an agreement within
the five minutes, Individual X will receive 16,000 pesos and Individual Y will
```

```
receive 5,000 pesos (16,000 pesos plus 5,000 pesos equals 21,000 pesos, an
amount lower than the 24,000 pesos available to split in the negotiation).
If, on the other hand, Individual X decides not to negotiate, both will wait
five minutes in silence in front of their computer screen, and Individual X
will receive 16,000 pesos and Individual Y will receive 5,000 pesos.  Given
that:  1.  the amount they receive if they choose not to negotiate is the same
as the amount they receive if they fail to reach an agreement, and 2.  the
amount to be split (24,000 pesos) is greater than the sum of 16,000 pesos plus
5,000 pesos, economic theory predicts that all Individuals X should always
attempt to negotiate (in person and electronically):  they would not be risking
anything.  The goal of this research project is to understand why, in this
context where negotiating is weakly dominant, some participants prefer not
to negotiate, and how they feel when they are forced to do so.  Now that you
know what we are studying, in order to help you conduct the interview as effectively
as possible, I will outline four further important clarifications.  The first
clarification concerns who we are interviewing at this point.  We are now interviewing
participants who had to negotiate, electronically or face-to-face, in one of
the two ways.  In the context of this experiment, there are two reasons why
a participant may have had to negotiate:  a.  because Individual X chose to
do so, or b.  because they were randomly assigned to negotiate.  The second
clarification concerns how you should approach the interview and what language
you should use.  I want you to focus on understanding how participants felt
when they found out they were going to negotiate.  I want to know how they
received this news and why—what made them feel the way they did.  Furthermore,
throughout the entire interview I want you to use plain language, without making
direct reference to "economic theory" or technical terms such as "optimal"—you
are not speaking with an expert.  The third clarification is that you should
bear in mind that in this experiment, in terms of monetary payoffs, the optimal
strategy was to negotiate (for the reasons already explained).  Thus, any unwillingness
to negotiate on the part of participants must be explained by non-monetary
reasons—which are the ones we are interested in exploring—or by a misunderstanding
of the instructions.  Please keep this in mind throughout the interview.  The
fourth and last clarification is that non-monetary reasons are just as valid
as monetary ones.  I do not want you to pass judgment on participants' responses.
Our goal is to understand their motivations, whatever those may be.  In summary:
I want participants to explain how they felt when they found out they were
going to negotiate, and why they felt that way.  With particular emphasis on
understanding the reasons why they felt the way they did.  Furthermore, I want
your interview to always remain focused on the interviewee—I do not want you
to ask them what they think would happen to others in their situation.  You
must always focus on the interviewee, on their own experience.
```

Topic 2

The second topic explores participants' feelings during the negotiation. It asks participants two times about this. Here I reproduce the Topic with the instructions to the AI-interviewer:

```
"Now, I want you to focus on understanding how they felt during the negotiation,
and whether they felt better or worse than they had expected.  Remember that
the interview must always remain focused on the interviewee, not on what might
happen to others in their situation."
```

Topic 3

The third topic explores the role of the communication channel on participants' feelings. It asks participants twice about this. Here I reproduce the Topic with the instructions to the AI-interviewer:

```
"Finally, as researchers, we are also very interested in understanding the
role of the communication channel in participants' feelings.  I want you to
focus on exploring whether the communication channel, in person or via chat,
has influenced how they felt, and why.  Remember that participants have negotiated
only in one of the two ways—in person or via chat, not in both.  If it has
not become clear in the previous part of the interview, ask the interviewee
which communication channel they negotiated in, and how they think this influenced
their experience.  Remember that the interview must always remain focused on
the interviewee, not on what might happen to others in their situation."
```

Closing Question

Every interview finished with the same closing question:

```
Is there anything else you would like to share?
```

D.5 Second Interview - Emotions

In the second interview, for those participants who bargained, with the help of two RAs we coded: 1. whether the emotions that the interviewee felt when he learned he was going to bargain were positive, negative or neutral, and 2. whether the emotions during the negotiation were better, worse or the same as expected. Moreover, we list all the emotions mentioned by the interviewee. To label the data, the three of us read 500 questions and responses independently, discuss them together, and reach consensus on the labels. The two RAs repeated this process with the remaining questions and responses to complete the 3,736 questions and responses corresponding to the 232 interviews.

For those participants who did not bargain, we coded the emotions they felt when they learned they were not bargaining (positive, negative, neutral), along with the detailed list of emotions. To label the data, the three of us read 500 questions and responses independently, discuss them together, and reach consensus on the labels. The two RAs repeated this process with the remaining questions and responses to complete the 634 questions and responses corresponding to the 62 interviews.

Online Appendix E First experiment in Madrid

An earlier version of this paper with data gathered in Universidad Carlos III (Madrid) circulated as a chapter of my PhD thesis. As part of this thesis, I ran both the baseline and asymmetric treatments of Experiment 1. After the Covid-19 pandemic, in November 2021, I decided to run the baseline and asymmetric treatments again in Argentina, along with the new forced, dictator and conversation treatments. For readers who are interested, here I share the main findings from the first experiment in Madrid, which are aligned with those in Buenos Aires.

E.1 Experiment details

The functioning of the baseline and asymmetric treatments was exactly the same as in Argentina (payoffs in euros, see Table E.22). The experiment was run in the laboratory of experimental economics at University Carlos III (Madrid, Spain). Between late April and early May 2018, 406 participants from various degrees were recruited using ORSEE (Greiner, 2015). Deciders' gender was balanced across communication channels. The average duration of an experimental session was approximately 90 minutes. The average payment was 15€ including a 6€ show-up fee.

Table E.22

Round Payoffs – Madrid

	Money to Split (P)	Decider Out. Opt. (x)	Counterpart Out. Opt. (y)
Round A	20	15	2
Round B	20	12	2
Round C	20	12	5
Round D	16	12	1

Note: The first column reports the money to split in each round, the second column reports the decider's outside option, and the third column reports the counterpart's outside option. Payoffs denominated in euros. The outside options coincide with the disagreement payoffs.

E.2 Results

The results are generally aligned with those obtained in Argentina, although in this case, I do not find differences by communication channel in round B. While in Buenos Aires, the opt-out rate in round B is almost zero in the electronic condition (see Fig. 1), in Madrid, it remains around 10% (see Figure E.16). At the same time, for rounds A, C and D, the opt-out rate face to face is 36.1% in the baseline treatment, while it is 22% electronically (p=0.031). Reassuringly, these numbers are close to those found in Buenos Aires (see Fig. 3), and close to those in the asymmetric treatment (Fig. E.16). As in Buenos Aires, total monetary payoffs in the baseline treatment do not differ significantly across communication conditions (see Table E.23).

Looking at the results by gender, women in Madrid are also more reluctant to bargain than men (baseline treatment avoidance rate: 30% women, 19% men, p=0.063, see Fig. E.17). Following the same reasoning developed in section 5, I explore whether women are less generous in the asymmetric treatment, but I find no evidence in this direction (see Fig. E.18). This aligns with the findings in Buenos Aires and corroborates the role of confrontation costs in the gender opt-out gap.

After the experiment, participants took the TKI® test and the post-experimental survey. Like in Argentina, Table E.24 shows a significant positive correlation between being in a high percentile of the TKI® avoidance score and avoiding negotiations. Similarly, the results of the survey show that 62% of participants declared that negotiations entail emotional costs, a proportion that is significantly higher among women (69% vs. 55%, p=0.003). When I asked deciders about cost differences by channel, 95% declared that these costs are higher face to face.

Table E.23

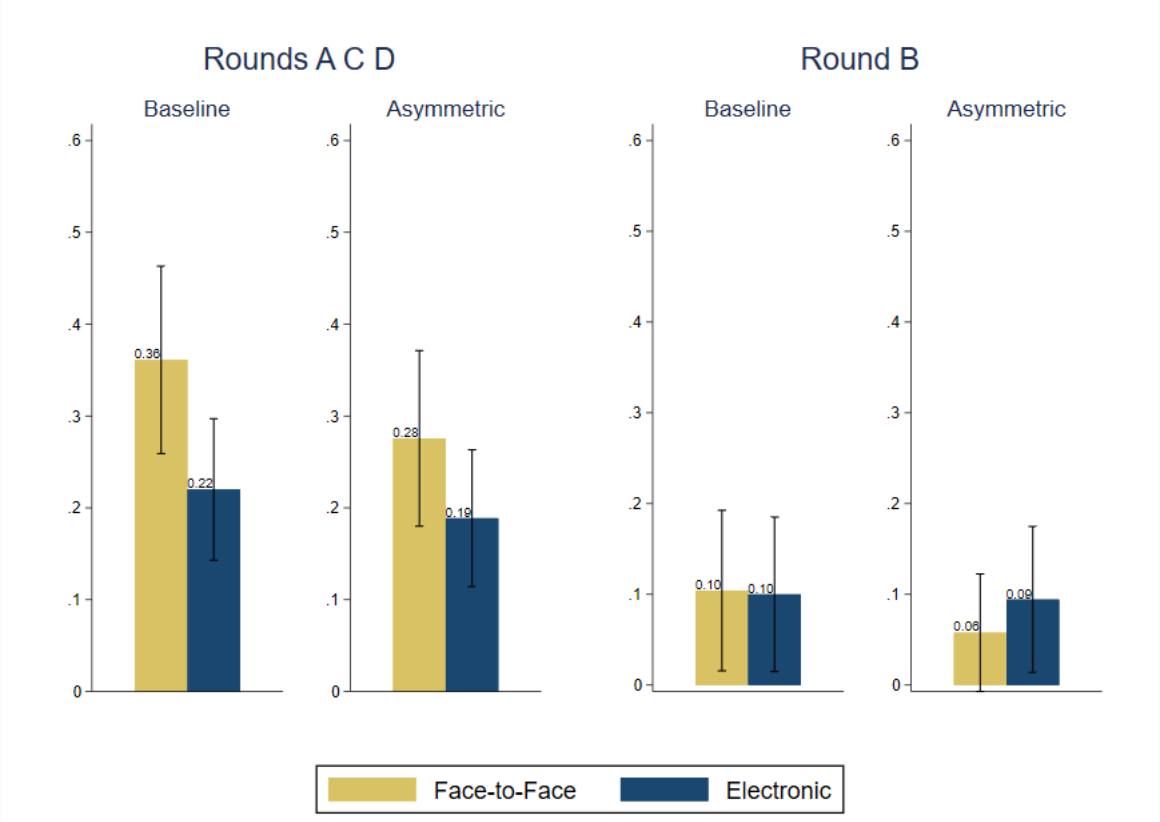
Baseline Treatment – Madrid

	Electronic	Face to Face	Difference
Avoidance (%)	19.0	29.7	-10.7*
	(3.3)	(4.6)	(5.6)
Agreement (%)	85.2	94.5	-9.3***
	(2.7)	(1.9)	(3.2)
Total Payoff (€)	71.3	71.5	-0.2
	(0.5)	(0.7)	(0.8)
Num. Deciders	50	48	

Note: This table reports the average avoidance rate across all rounds, the average rate of agreement for those negotiations that occur, the average total monetary payoff per pair expressed in euros, and the number of deciders in the baseline treatment for each communication condition. It also reports the difference between the two communication conditions. * p<0.1, ** p<0.05, *** p<0.01 correspond to the p-values of two-sided t-tests.

Figure E.16

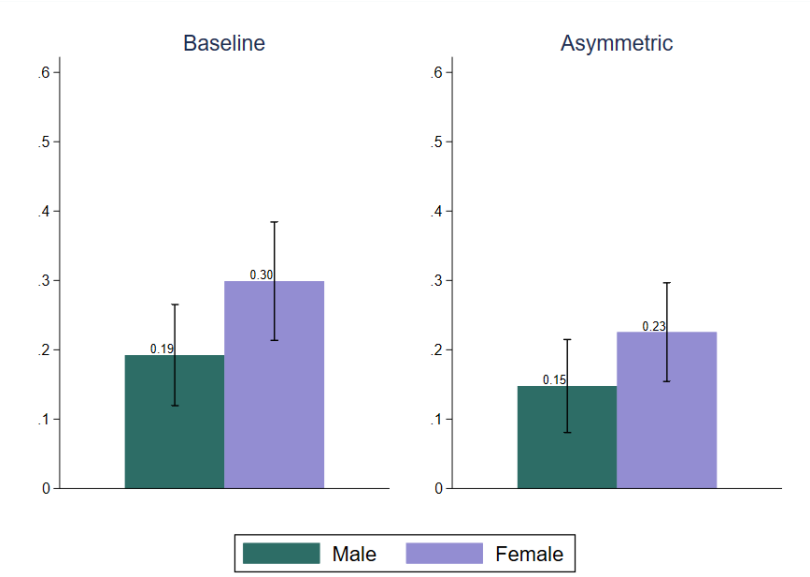
Avoidance Rate by Communication Channel – Madrid



Note: This graph shows deciders' avoidance rates and their 95% confidence intervals in the baseline and asymmetric treatments, both in the face-to-face and electronic conditions for rounds A, C and D together, and for round B.

Figure E.17

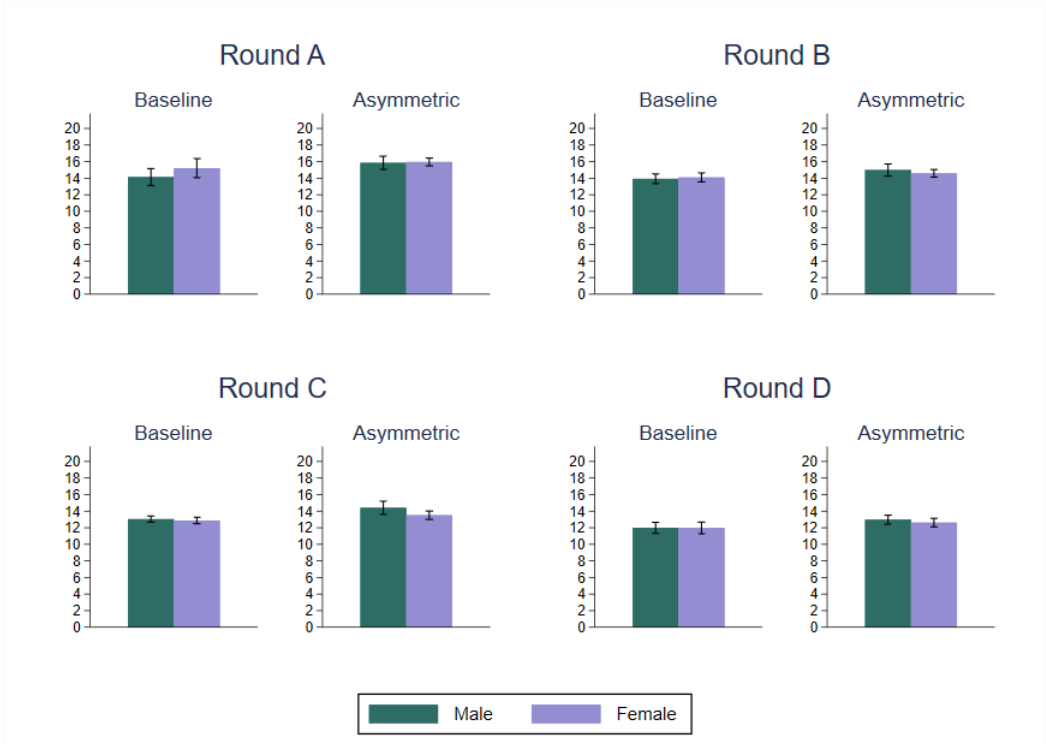
Avoidance Rate by Gender – Madrid



Note: This graph shows deciders' avoidance rates and their 95% confidence intervals in the baseline and asymmetric treatments by gender.

Figure E.18

Deciders' Earnings after an Agreement – Madrid



Note: This graph shows deciders' average earnings by gender and their 95% confidence intervals in the baseline and asymmetric treatments for rounds A, B, C and D.

Table E.24

TKI Avoidance Score – Madrid

	(1) Opt-Out	(2) Opt-Out	(3) Opt-Out	(4) Opt-Out
TKI Avoiding	0.225***	0.241***	0.238***	0.239***
	(0.0674)	(0.0676)	(0.0663)	(0.0666)
Face-to-Face		0.0980***	0.0948**	0.0928**
		(0.0369)	(0.0366)	(0.0368)
Female			0.0709*	0.0777**
			(0.0363)	(0.0363)
Fair Deal	-0.222***	-0.227***	-0.213***	
	(0.0281)	(0.0294)	(0.0289)	
Observations	800	800	800	800
Clusters	200	200	200	200

Note: OLS regression of a dummy that takes a value of 1 if an individual decides to avoid a negotiation on the individual's TKI avoidance score percentile, expressed on a scale from 0 to 1, and a dummy that takes a value of 1 if the individual implements a 50/50 agreement. Column 2 includes a dummy that takes a value of 1 if the negotiation is face to face. Column 3 includes also a dummy if the decider is female. Column 4 removes the dummy for a fair deal. Standard errors are clustered at the individual level. * p<0.1, ** p<0.05, *** p<0.01.

Online Appendix F Experiment 1: Instructions

This is an English version of the instructions I used in the baseline electronic treatment. Instructions in other treatments were adapted accordingly. The original set of instructions in Spanish can be sent upon request.

Thank you for participating. This experiment is part of a research project in economics. In this project, we are trying to understand how people make decisions. However, we are not expecting any specific behavior. This means that there is no such thing as one correct answer. We just want to learn how people make decisions.

From now on you may not use your cellphones, get up or communicate with other participants, unless you are told otherwise. If you have any questions, please raise your hand and an instructor will answer them in private. Please do not ask aloud!

This experiment is funded by Universidad Torcuato Di Tella. You will receive 400 pesos as a show-up fee. If you follow the instructions correctly, you may earn more, depending on your decisions.

At the beginning of the experiment, your computer will tell you whether you are "Individual X" or "Individual Y". Roles are randomly assigned and they are maintained throughout the experiment. Now, we describe the experiment for individual X.

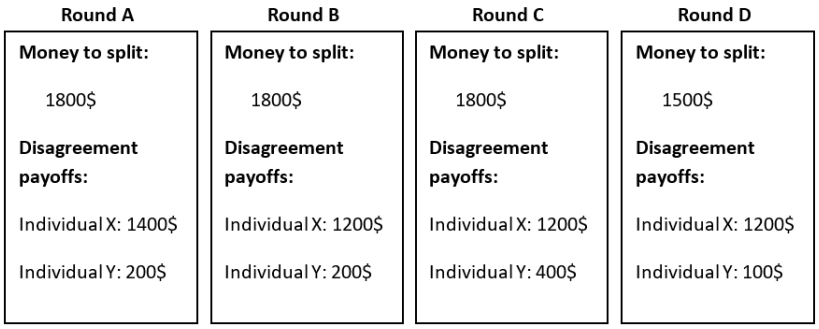
Individual X

The experiment has four rounds. If you are individual X, in each round you will be paired with a different anonymous individual Y. In each round, you and Y will receive some money as a payment for taking part in the experiment.

You (individual X) can decide whether you want to negotiate how to split this money, or whether you do not want to:

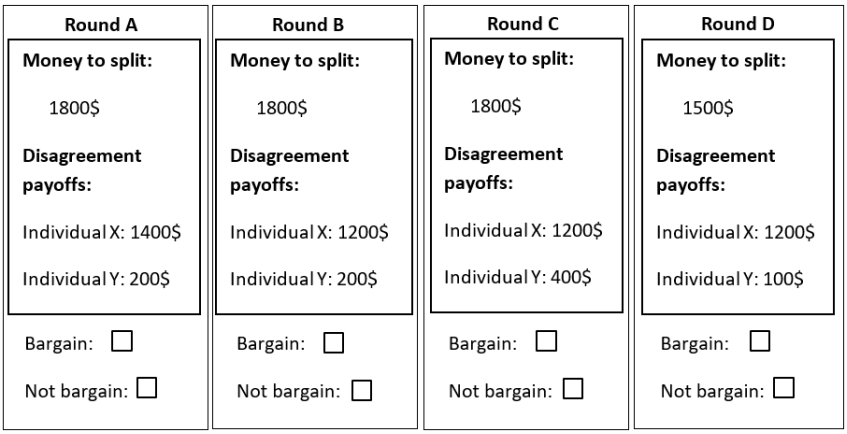
- If you choose to negotiate, you will have three minutes to bargain electronically with individual Y. If you do not reach an agreement, you will receive your disagreement payoffs.
- If you choose not to negotiate, you and individual Y will wait in your respective seats and directly receive your disagreement payoffs.

The following table shows the money that each pair receives in each round, as well as the disagreement payoffs:



The only interaction with individual Y occurs during the negotiation. You will have no other contact with Y before or after. If you choose not to bargain, you will have no contact at all.

At the beginning of the experiment, you must tick the box on your computer screen to indicate whether you want to bargain or not in each round:



Round order is randomly determined. This means that the order is not necessarily round A, then round B, then round C, and finally round D.

At the end of the experiment, you will be paid your earnings in one of the rounds chosen at random. Each round has the same probability of being chosen for payoff (25% probability). Your decisions in the experiment do not affect these probabilities.

Individual Y

If you are individual Y, you will be paired with four anonymous individual Xs. In each round, X chooses whether or not to bargain over how to split the money you have received in the corresponding round.

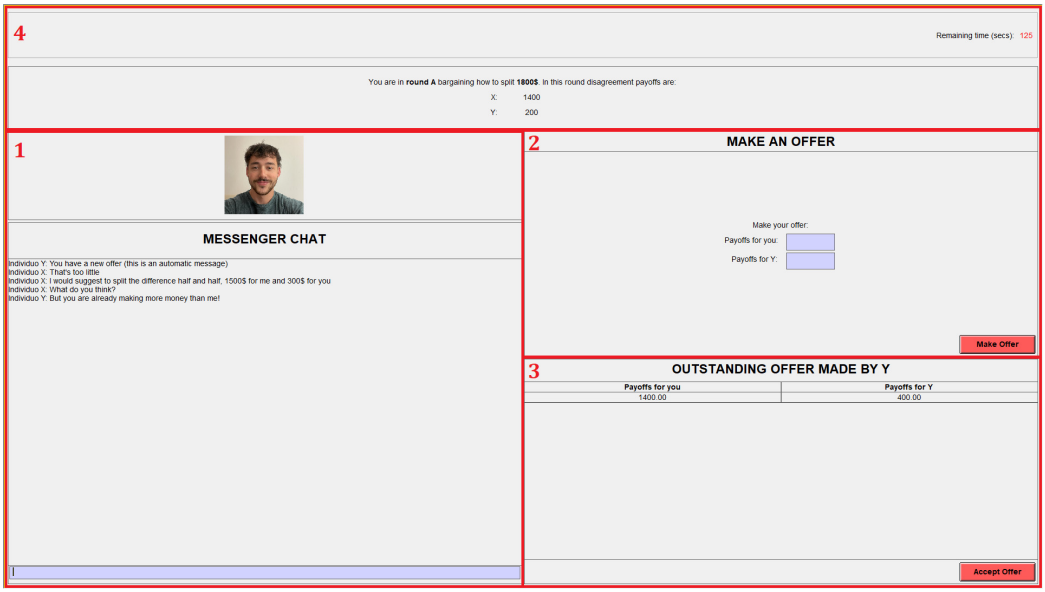
- If X chooses to bargain and you reach an agreement, you will receive the agreed payoff.
- If X chooses to bargain and you do not reach an agreement, you will receive the disagreement payoff.
- If X chooses not to bargain, you will receive the disagreement payoff.

In the next section, I give details of the functioning of the negotiation for those pairs in which X has chosen to bargain. Then, I will explain the payment method. Finally, I will give an example of the full functioning of the experiment.

Negotiation

Each pair has three minutes to bargain. The negotiation is unstructured. Any agreement in which the sum of payments is no higher than the total amount to split is possible.

The following items describe the elements on the bargaining screen:



1. In box number 1, you can see the chat. In the upper part, you will see the profile picture of your partner. In the blue bar on the bottom, you can type messages. To send a message, push "Enter". Messages will appear in the gray box, preceded by "Individual X" or "Individual Y" depending on who sent them.
2. In box number 2, you can enter your offers.
3. In box number 3, you can see your partner's offers. To accept an offer, select it on the screen and click "Accept Offer".
4. In box number 4, you see the time remaining, the round, the money to split, and the disagreement payoffs.

Both X and Y can make offers simultaneously. To withdraw an offer, you just have to click on it in box 2 and then click "Withdraw Offer". Once you have withdrawn your outstanding offer, you can make a new one.

The chat is open for three minutes. To close a deal, either X or Y has to accept the counterpart's offer. After the three minutes, there will be no more chances to close a deal for that round.

Please raise your hand if you have any doubts.

Payoffs

Everyone finishes the experiment at the same time, regardless of their role and their decisions. Once the experiment is over, participants must answer a questionnaire.

Remember that every round has the same probability (25%) of being chosen as the payment round.

Payment is done by wire transfer. I will proceed first with those who were individual X. After they have all checked their bank information, signed their receipt and left the lab, I will proceed with those who were individual Y.

All information will be anonymized and used only for scientific purposes. At no point will any participant be informed about the decisions of other participants.

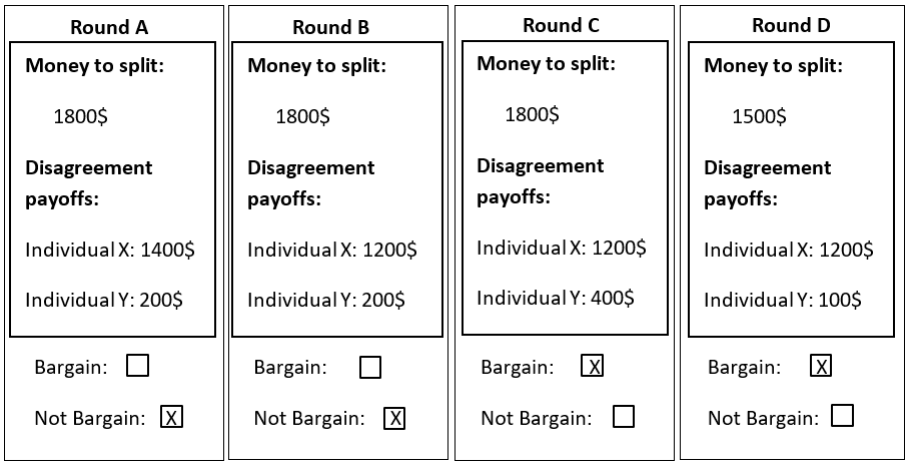
Example

The following example is illustrative and does not suggest how to make decisions.

- The computer randomly determines who is individual X and who is individual Y.
- Individual X makes the following choices:

- Round order is randomly determined:

    1. Round D



2. Round B
3. Round A
4. Round C

- Round D starts. Individual X has chosen to bargain, so X and Y view the negotiation screen. For three minutes, they bargain electronically about how to split the 1,500$. After three minutes, they do not reach an agreement. Payments for that round equal the disagreement payoffs: 1,200 AR$ for individual X, 100 AR$ for individual Y.
- Round B starts. Individual X has chosen not to bargain, so X and Y stay in their seats in front of a blank screen. Payments for that round equal the disagreement payoffs: 1,200 AR$ for individual X, 200 AR$ for individual Y.
- Round A starts. Individual X has chosen not to bargain, so X and Y stay in their seats in front of a blank screen. Payments for that round equal the disagreement payoffs: 1,400$ for individual X, 200 AR$ for individual Y.
- Round C starts. Individual X has chosen to bargain, so X and Y view the negotiation screen. They bargain electronically about how to split the 1,800$ for three minutes. They agree to split the amount so individual X gets Z AR$ and individual Y gets H AR$. Z AR$ and H AR$ could be any quantities as long as: Z AR$ + H AR$ = 1,800 AR$.

Questionnaire and payment

- Participants respond to a questionnaire.
- Round A is randomly chosen as the payment round. Individual X gets 1,400 AR$ plus the 400$ show-up fee: 1,800 AR$. Individual Y gets 200 AR$ plus the 400 AR$ show-up fee: 600 AR$.

Once the experiment is over, individual Xs will leave the lab first. Remember that any round can be chosen as the payment round. You will always receive the 400 AR$ show-up fee, irrespective of your decisions and the payment round.

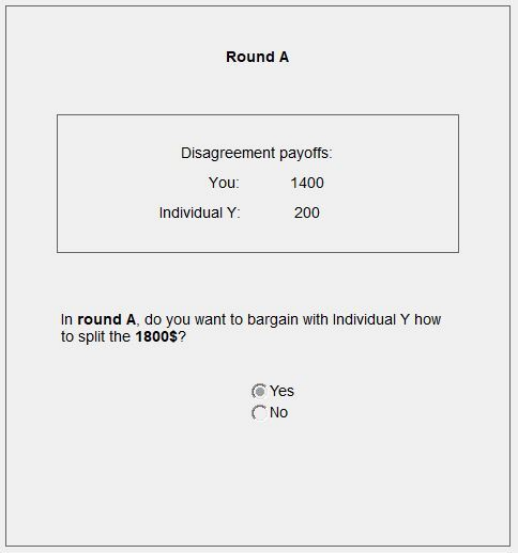
Online Appendix G Experiment 1: Control questions

This is an English version of the control questions that participants had to answer in the baseline treatment before the experiment started. Control questions in other treatments were adapted accordingly. The original control questions, administered in Spanish in z-Tree, can be sent upon request.

Welcome to the experiment! Before you start, you must correctly answer some control questions. The aim of these questions is to verify that you have correctly understood the instructions. If you have doubts, please raise your hand and a researcher will clarify them in private. Thank you!

The following examples are illustrative and do not suggest how to make decisions.

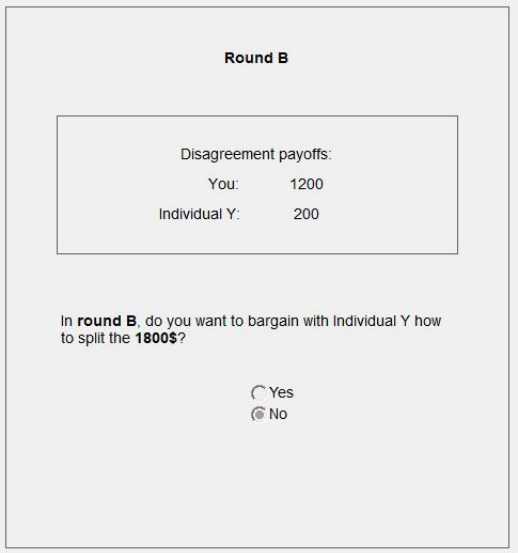
In round A, individual X has chosen the following:



Besides the 400 AR$ show-up fee:

- How much money will X and Y be able to split in the negotiation?
- How much will X get if they do not reach an agreement?
- How much will Y get if they do not reach an agreement?

In round B, individual X has chosen the following:



Besides the 400 AR$ show-up fee:

- How much money will X get in this round?
- How much money will Y get in this round?

Online Appendix H Experiment 2: Instructions

This is an English version of the instructions. The original set of instructions in Spanish can be sent upon request.

Thank you for participating in this experiment, which is part of a research project. In this experiment we seek to understand how decisions are made, but we do not expect any particular behaviour from you. This means that there is no such thing as a correct answer. We simply want to understand how people make decisions.

From now until the end of the experiment, the use of mobile phones, leaving your seat, or communicating with other participants is not allowed, unless otherwise indicated. If you have any questions, please raise your hand and one of the instructors will answer your questions in private. Please do not ask out loud!

This experiment is funded by Universidad Torcuato Di Tella. You will receive a fixed payment of 5,000 pesos just for showing up on time. Additionally, over the course of the experiment, you may earn more money, as detailed in the following instructions.

At the beginning of the experiment, you will be randomly assigned the role of Individual X or Individual Y. Your computer will indicate which role (X or Y) you have been assigned. You will keep this role throughout the whole experiment.

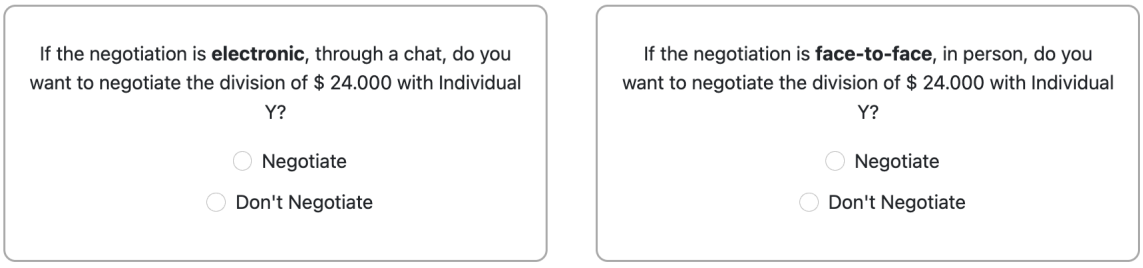
The experiment is now described for Individual X.

Individual X

If you are Individual X, you will be paired at random with an Individual Y. You can decide whether or not to negotiate how to split the $24,000 that has been assigned to you.

- If you choose to negotiate, you will have five minutes to bargain over how to split the $24,000. If after five minutes you have not reached an agreement, you will receive the disagreement payoffs:
  - Individual X: $16,000.
  - Individual Y: $5,000.
- If you choose not to negotiate, you and Individual Y will wait silently in front of your computers for five minutes. After five minutes, you will receive the disagreement payoffs:
  - Individual X: $16,000.
  - Individual Y: $5,000.

As Individual X, you will have to decide what you want to do if the negotiation is face-to-face—in person—and what you want to do if the negotiation is electronic—through a chat. Thus, when the experiment begins, you will have to indicate your choice for each case on your computer screen:



Afterwards, the computer will roll a die to determine which of your decisions is implemented:

- If the die comes up 1, your decision for the face-to-face negotiation will be implemented.
- If the die comes up 2, your decision for the electronic negotiation will be implemented.
- If the die comes up 3, you will have to negotiate face-to-face, regardless of what you have chosen.
- If the die comes up 4, you will have to negotiate electronically, regardless of what you have chosen.

That is, as Individual X, you will not be able to choose what number the die will show, and the decisions you make will not affect the die roll. However, you will be able to choose what happens if the die comes up 1 and what happens if the die comes up 2. Your decisions will remain anonymous at all times, and we will not inform other participants of what you have chosen for the different values of the die.

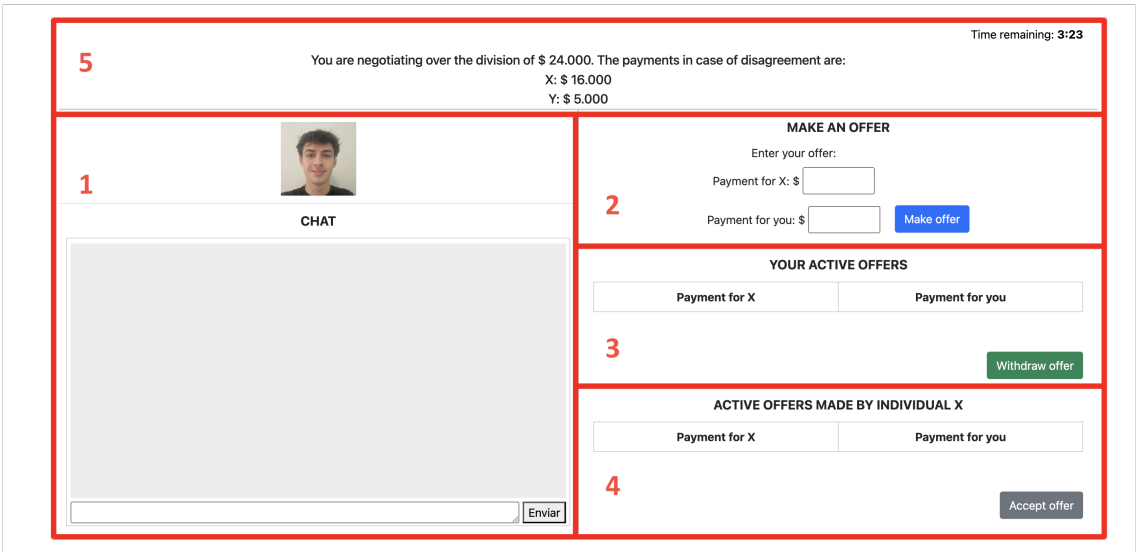
Regardless of the outcome of the die, the amount to split will always be $24,000, the disagreement payoffs will always be $16,000 for Individual X and $5,000 for Individual Y, and the negotiation will always last 5 minutes.

We now explain in detail how the electronic negotiation works, and how the face-to-face negotiation works.

Electronic Negotiation

For five minutes, each pair will be able to negotiate through a chat how to split the money they have received. The negotiation is completely unstructured, and any split in which the sum of the payments does not exceed $24,000 is possible.

The following shows the negotiation screen and describes its different elements:



1. Chat. At the top, you will see the profile picture of the person you have been paired with. In the white bar at the bottom, you can type your messages. Messages will be sent by pressing the "Enter" key and will appear in the grey box, preceded by "Individual X" or "Individual Y" depending on who sent them.

2. Here you can enter offers.

3. Here you will see if there is any outstanding offer you have made. You can withdraw an outstanding offer at any time by clicking the "Withdraw Offer" button. When you enter a new offer, it will replace the previous one. Thus, at any given moment, there will be at most one outstanding offer made by you.

4. Here you will see your partner's outstanding offers. You can accept them by clicking the "Accept Offer" button.

5. Here you will see the time remaining, the money to split, and the disagreement payoffs.

Both X and Y can make offers simultaneously. The chat will remain open for five minutes. To seal a deal, either X or Y must accept the other's offer. Once the other's offer is accepted, the deal is closed. After five minutes, if no agreement has been reached, you will receive the disagreement payoffs.

Face-to-Face Negotiation

Negotiations will take place in person in the negotiation room, adjacent to this laboratory. For five minutes, each pair will be able to discuss face-to-face how to split the money they have received. The negotiation is completely unstructured, and any split in which the sum of the payments does not exceed $24,000 is possible.

If an agreement is reached, the members of the pair must write down the terms of the agreement (how much X receives and how much Y receives) on the paper form that will be handed to them, sign the form, and wait for the five minutes to end. After five minutes, if no agreement has yet been reached, there will be no further opportunity to close a deal, and you will receive the disagreement payoffs.

Individual Y

If you are Individual Y, you will be paired at random with an Individual X. Individual X will be able to decide whether or not to negotiate the split of the $24,000 that has been assigned to you in a face-to-face negotiation, and whether or not to negotiate in an electronic negotiation.

After Individual X has made his decision, the computer will roll a four-sided die:

- If the die comes up 1, Individual X's decision for the face-to-face negotiation will be implemented.
- If the die comes up 2, Individual X's decision for the electronic negotiation will be implemented.
- If the die comes up 3, you will negotiate face-to-face.
- If the die comes up 4, you will negotiate electronically.

That is, as Individual Y, you will be subject to the decision made by Individual X and to the outcome of the die. In case there is a negotiation, any split of the $24,000 that you agree on within the five minutes will be valid.

Questionnaire and Payment

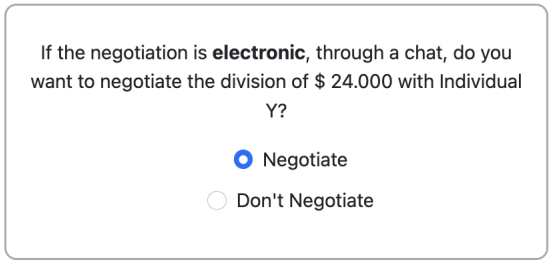
After completing some questionnaires, all participants will finish the experiment at the same time, regardless of their role and their decisions. Once all participants have finished, they will fill in their receipt and leave the laboratory one by one. Payment will be made by bank transfer within one week.

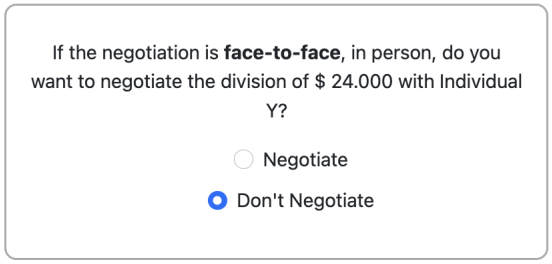
All information collected in this experiment will be anonymised and used only for scientific purposes. Only participants who negotiate will have any interaction with each other (Individual X with the Individual Y he has been paired with). Beyond what relates to this interaction, at no point will any participant be informed of the decisions made by the rest of the participants. An example is presented below.

Example

The following example is illustrative and is not meant to suggest how to make decisions.

- The computer randomly determines who is Individual X and who is Individual Y.
- Individual X makes the following choices:





What happens if...

- ...the die comes up 1? Since Individual X has indicated that he does not want to negotiate face-to-face, Individual X and the Individual Y he has been paired with will wait silently in front of their computers for five minutes. After five minutes, X receives $16,000 and Y receives $5,000.

- ...the die comes up 2? Since Individual X has indicated that he does want to negotiate electronically, Individual X and Individual Y will discuss how to split the $24,000 through a chat for five minutes. If after five minutes they have not reached an agreement, X receives $16,000 and Y receives $5,000.
- ...the die comes up 3? Individual X and Individual Y will discuss how to split the $24,000 face-to-face for five minutes. If after five minutes they have not reached an agreement, X receives $16,000 and Y receives $5,000.
- ...the die comes up 4? Individual X and Individual Y will discuss how to split the $24,000 through a chat for five minutes. If after five minutes they have not reached an agreement, X receives $16,000 and Y receives $5,000.

Once the experiment is over and the questionnaires have been completed, participants will remain silent until everyone has finished. Afterwards, they will leave the laboratory one by one. Remember that all participants will always receive the $5,000 punctuality payment, regardless of their role and their decisions.

Online Appendix I Experiment 2: Control questions

This is an English version of the control questions. The original control questions were in Spanish in oTree and can be sent upon request.

Welcome to this experiment! Before you start, you must correctly answer the following control questions. The purpose of these questions is to verify that you have correctly understood the instructions. If you have any questions, please raise your hand and one of the researchers will attend to you in private. Please do not ask out loud. Thank you!

The following examples are illustrative and do not suggest how to make decisions.

In this experiment, Individual X is going to make two decisions. He is going to decide whether he wants to enter a face-to-face negotiation or whether he prefers to avoid it, and he is going to decide whether he wants to enter an electronic negotiation or whether he prefers to avoid it. Afterwards, the computer will roll a die. Please select the correct answer for each value of the die.

If the die comes up 1…

a) Individual X's decision for the electronic negotiation will be implemented.

b) Individual X and Individual Y will negotiate face-to-face, regardless of what Individual X has decided.

c) Individual X's decision for the face-to-face negotiation will be implemented.

d) Individual X and Individual Y will negotiate electronically, regardless of what Individual X has decided.

If the die comes up 2…

a) Individual X's decision for the electronic negotiation will be implemented.

b) Individual X's decision for the face-to-face negotiation will be implemented.

c) Individual X and Individual Y will negotiate electronically, regardless of what Individual X has decided.

d) Individual X and Individual Y will negotiate face-to-face, regardless of what Individual X has decided.

If the die comes up 3…

a) Individual X's decision for the electronic negotiation will be implemented.

b) Individual X's decision for the face-to-face negotiation will be implemented.

c) Individual X and Individual Y will negotiate electronically, regardless of what Individual X has decided.

d) Individual X and Individual Y will negotiate face-to-face, regardless of what Individual X has decided.

If the die comes up 4…

a) Individual X and Individual Y will negotiate electronically, regardless of what Individual X has decided.

b) Individual X and Individual Y will negotiate face-to-face, regardless of what Individual X has decided.

c) Individual X's decision for the face-to-face negotiation will be implemented.

d) Individual X's decision for the electronic negotiation will be implemented.

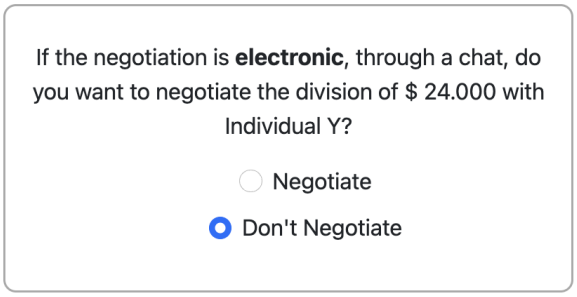
If, for the face-to-face negotiation question, Individual X has chosen the following:



If the die comes up 1…

- How much money will X and Y have to split during the negotiation?
- How much money will X receive if they do not reach an agreement?
- How much money will Y receive if they do not reach an agreement?

If, for the electronic negotiation question, Individual X has chosen the following:



If the die comes up 2…

- How much money will X receive?
- How much money will Y receive?

Online Appendix J Experiment 3: Instructions

This is an English version of the instructions I used in the electronic condition. Instructions in the face-to-face condition were adapted accordingly. The original set of instructions in Spanish can be sent upon request.

Thank you for participating. This experiment is part of a research project in economics. In this project, we are trying to understand how people make decisions. However, we are not expecting any specific behavior. This means that there is no such thing as one correct answer. We just want to learn how people make decisions.

From now on you may not use your cellphones, get up or communicate with other participants, unless you are told otherwise. If you have any questions, please raise your hand and an instructor will answer them in private. Please do not ask aloud!

This experiment is funded by Torcuato Di Tella University. You will receive a fixed payment of 600 pesos for showing up on time. Additionally, over the course of the experiment, you may earn more money, as detailed in the following instructions.

You will be randomly assigned the role of Buyer or Seller. Your computer will indicate which role (Buyer or Seller) you have been assigned. You will keep this role throughout the whole experiment.

Sellers

If you are a seller, you will receive this mug:



Then, you will be paired with an anonymous buyer. This buyer will receive 1,000 additional pesos on top of the 600 pesos for showing up on time (i.e., the buyer will receive a total allocation of 1,600 pesos). The buyer will decide whether to bargain with you electronically for your mug. If the buyer chooses to negotiate, you will have five minutes to try to reach an agreement.

- If you reach an agreement, the buyer will pay you the agreed price, and keep the mug and the difference between the 1,600 pesos and the agreed price.
- If you do not reach an agreement, you will keep the mug and the buyer will keep his allocation of 1,600 pesos.

As a seller, you will always receive the 600 pesos for showing up on time.

Buyers

If you are a buyer, you will receive an allocation of 1,000 pesos on top of the 600 pesos for showing up on time (i.e., you will receive a total of 1,600 pesos) and you will decide whether to bargain electronically with the seller to buy the mug.

If you choose to negotiate, you will have five minutes to bargain electronically and try to reach an agreement.

- If you reach an agreement, you will keep the mug plus the difference between your allocation of 1,600 pesos and the agreed price. The seller will receive the agreed price.
- If you do not reach an agreement, the seller will keep the mug, and you will keep your allocation of 1,600 pesos.

If you choose not to negotiate, you will receive a different amount of money depending on the result of a 10-sided die roll. The following table shows how much you will receive according to the result of the die. These amounts are final and include the punctuality payment.

If a 1 is rolled, those who choose not to negotiate will receive 900 pesos.
If a 2 is rolled, those who choose not to negotiate will receive 1,000 pesos.
If a 3 is rolled, those who choose not to negotiate will receive 1,100 pesos.
If a 4 is rolled, those who choose not to negotiate will receive 1,200 pesos.
If a 5 is rolled, those who choose not to negotiate will receive 1,300 pesos.
If a 6 is rolled, those who choose not to negotiate will receive 1,400 pesos.
If a 7 is rolled, those who choose not to negotiate will receive 1,500 pesos.
If an 8 is rolled, those who choose not to negotiate will receive 1,600 pesos.
If a 9 is rolled, those who choose not to negotiate will receive 1,700 pesos.
If a 10 is rolled, those who choose not to negotiate will receive 1,800 pesos.

You can choose whether or not to bargain for each outcome of the die. This means that you can choose whether or not to negotiate if the die comes up 1, if the die comes up 2, if it comes up 3, and so forth. The following picture shows the decision screen.

Once you (the buyer) have made your decision, the computer will roll a die, and your decision for that outcome of the die will be implemented. If you have chosen to negotiate, then you will have five minutes to negotiate electronically with the seller. If you have chosen not to negotiate, you and the seller will wait silently in front of your computers for five minutes.

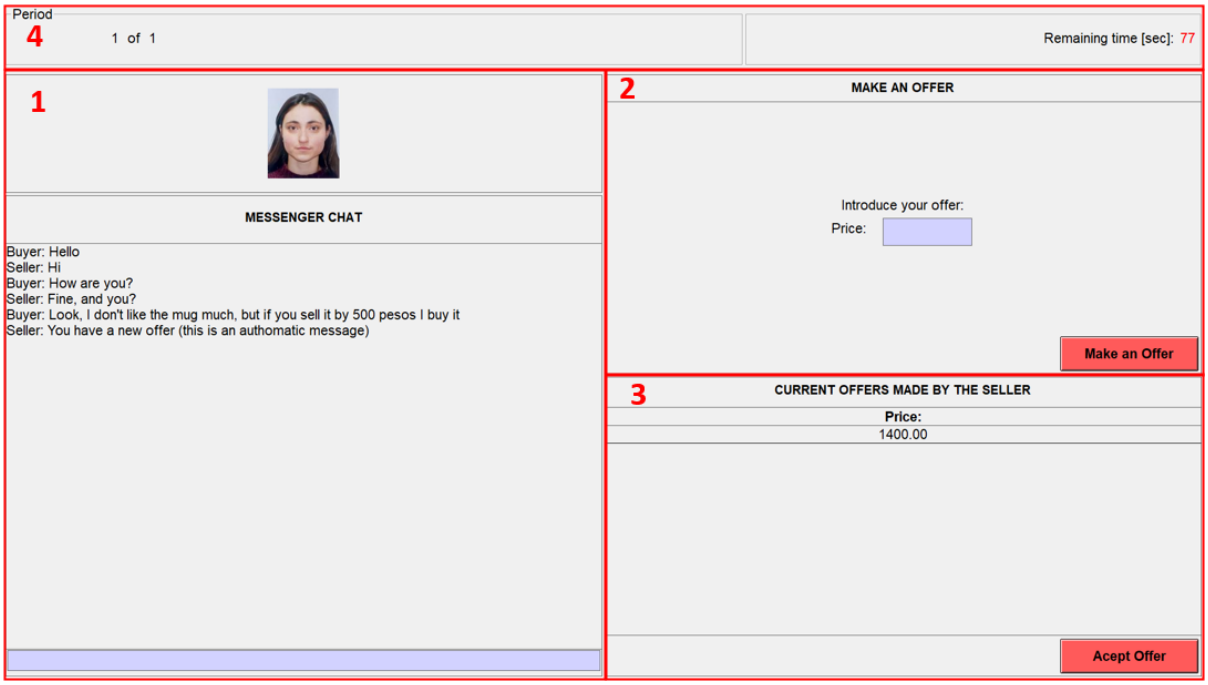
Remember that all decisions and pairings in this experiment are anonymous. Only when the buyer's decision for the die's outcome is to bargain will the seller be notified, and you will meet to bargain electronically. The experimenters will never inform the participants of other participants' choices.

Remember that, as a buyer, if you choose to negotiate and do not reach an agreement, you will keep 1,600 pesos. If you reach an agreement, you will take the mug, pay the price you agreed, and keep the change left from your 1,600 pesos.

Negotiation

Pairs bargain for the mug through an electronic chat for five minutes. The negotiation is completely unstructured, and any agreed price that does not exceed 1,600 pesos is suitable.

Below, we describe the different elements of the negotiation screen:



1. In box number 1, there is a chat. At the top, you will see the profile picture of your partner. In the blue bar at the bottom, you can type your messages. Messages will be sent by pressing the "Enter" key, and they will appear in the gray box, preceded by "Buyer" or "Seller" depending on who sent them.
2. In box number 2, you can enter offers (the price you offer if you are a buyer, or the price you demand if you are a seller).
3. In box number 3, you will see your partner's offers. By selecting them and pressing the "Accept Offer" button, you can accept them.
4. In box number 4, you will see the time remaining.

Both buyers and sellers can make offers simultaneously. You can withdraw your offers at any time by clicking on the "Withdraw Offer" button. Once you have withdrawn an offer, you can make a new one.

The chat will remain open for five minutes. To seal a deal, either the buyer or the seller must accept the offer of their counterpart. After five minutes, there will be no further opportunities to reach an agreement.

Please raise your hand if you have any questions and an instructor will assist you.

Payment

All participants will end the experiment at the same time, regardless of their role and their decisions. Once the experiment is over, they will be asked to complete a questionnaire.

The payment will be made by bank transfer during the course of this week, and the mug will be delivered immediately.

All the information collected in this experiment will be anonymized and used only for scientific purposes.

Examples

All the following examples are illustrative, and they do not suggest how to make decisions. Before knowing their role, all participants will choose what to do if they are randomly selected to be buyers. They will decide for each outcome of the die whether or not to negotiate.

Example 1

The sample participant chooses the following:

If the die shows a **1**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **900** pesos home?	**Don't Bargain**
If the die shows a **2**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1000** pesos home?	**Don't Bargain**
If the die shows a **3**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1100** pesos home?	**Don't Bargain**
If the die shows a **4**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1200** pesos home?	**Don't Bargain**
If the die shows a **5**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1300** pesos home?	**Don't Bargain**
If the die shows a **6**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1400** pesos home?	**Don't Bargain**
If the die shows a **7**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1500** pesos home?	**Don't Bargain**
If the die shows a **8**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1600** pesos home?	**Don't Bargain**
If the die shows a **9**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1700** pesos home?	**Don't Bargain**
If the die shows a **10**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1800** pesos home?	**Don't Bargain**

Roles are randomly assigned, and the participant happens to be a buyer.

The die comes up 7. For a value of 7, the participant has chosen not to negotiate, and so both the participant in this example and the seller with whom he has been paired wait silently in front of their computers for five minutes. The buyer takes home 1,500 pesos. The seller keeps the mug and receives 600 pesos.

These payments are final and include the 600 pesos for punctuality.

Example 2

The sample participant chooses the following:

If the die shows a **1**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **900** pesos home?	**Bargain**
If the die shows a **2**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1000** pesos home?	**Bargain**
If the die shows a **3**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1100** pesos home?	**Bargain**
If the die shows a **4**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1200** pesos home?	**Don't Bargain**
If the die shows a **5**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1300** pesos home?	**Don't Bargain**
If the die shows a **6**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1400** pesos home?	**Don't Bargain**
If the die shows a **7**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1500** pesos home?	**Don't Bargain**
If the die shows a **8**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1600** pesos home?	**Don't Bargain**
If the die shows a **9**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1700** pesos home?	**Don't Bargain**
If the die shows a **10**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1800** pesos home?	**Don't Bargain**

Roles are randomly assigned, and the participant happens to be a buyer.

The die comes up 9. For a value of 9, the participant has chosen not to negotiate, and so both the participant in this example and the seller with whom he has been paired wait silently in front of their

computers for five minutes. The buyer takes home 1,700 pesos. The seller keeps the mug and receives 600 pesos.

These payments are final and include the 600 pesos for punctuality.

Example 3

The sample participant chooses the following:

If the die shows a **1**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **900** pesos home?	**Bargain**
If the die shows a **2**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1000** pesos home?	**Bargain**
If the die shows a **3**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1100** pesos home?	**Bargain**
If the die shows a **4**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1200** pesos home?	**Don't Bargain**
If the die shows a **5**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1300** pesos home?	**Don't Bargain**
If the die shows a **6**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1400** pesos home?	**Don't Bargain**
If the die shows a **7**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1500** pesos home?	**Don't Bargain**
If the die shows a **8**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1600** pesos home?	**Don't Bargain**
If the die shows a **9**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1700** pesos home?	**Don't Bargain**
If the die shows a **10**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1800** pesos home?	**Don't Bargain**

Roles are randomly assigned, and the participant happens to be a buyer.

The die comes up 3. For a value of 3, the participant has chosen to negotiate, so the participant in this example and the seller meet in the electronic chat.

They reach an agreement to exchange the mug for X pesos before the five minutes are over. Therefore, the buyer takes home 1,600 pesos minus the agreed price (1,600-X pesos) and the mug, while the seller takes home 600 pesos plus the agreed price (600+X pesos).

These payments are final and include the 600 pesos for punctuality.

Example 4

The sample participant chooses the following:

If the die shows a **1**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **900** pesos home?	**Bargain**
If the die shows a **2**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1000** pesos home?	**Bargain**
If the die shows a **3**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1100** pesos home?	**Bargain**
If the die shows a **4**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1200** pesos home?	**Bargain**
If the die shows a **5**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1300** pesos home?	**Bargain**
If the die shows a **6**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1400** pesos home?	**Bargain**
If the die shows a **7**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1500** pesos home?	**Bargain**
If the die shows a **8**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1600** pesos home?	**Bargain**
If the die shows a **9**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1700** pesos home?	**Bargain**
If the die shows a **10**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1800** pesos home?	**Bargain**

Roles are randomly assigned, and the participant happens to be a buyer.

The die comes up 10. For a value of 10, the participant has chosen to negotiate, so the participant in this example and the seller meet in the electronic chat.

They do not reach an agreement within the five minutes. Therefore, the buyer takes home 1,600 pesos, and the seller takes home 600 pesos and the mug.

These payments are final and include the 600 pesos for punctuality.

Example 5

The sample participant chooses the following:

If the die shows a **1**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **900** pesos home?	**Bargain**
If the die shows a **2**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1000** pesos home?	**Bargain**
If the die shows a **3**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1100** pesos home?	**Don't Bargain**
If the die shows a **4**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1200** pesos home?	**Don't Bargain**
If the die shows a **5**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1300** pesos home?	**Don't Bargain**
If the die shows a **6**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1400** pesos home?	**Don't Bargain**
If the die shows a **7**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1500** pesos home?	**Don't Bargain**
If the die shows a **8**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1600** pesos home?	**Don't Bargain**
If the die shows a **9**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1700** pesos home?	**Don't Bargain**
If the die shows a **10**, do you want to bargain **electronically** to purchase the mug, or do you prefer not to bargain and take **1800** pesos home?	**Don't Bargain**

Roles are randomly assigned, and the participant happens to be a seller.

The mug is delivered to him. The participant waits to find out whether the buyer he has been paired with has chosen to bargain for the mug.

Remember that all participants have to decide what to do if they happen to be buyers. Afterwards, the computer randomly chooses who is a seller and who is a buyer. Finally, the computer rolls a die, and depending on what the buyer has chosen, the experiment continues.

Online Appendix K Experiment 3: Control questions

This is an English version of the control questions of Experiment 3. The original control questions were in Spanish in zTree and can be sent upon request.

Welcome to this experiment! Before you begin, you must correctly answer the following control questions. The purpose of these questions is to check that you have correctly understood the instructions. If you have any questions, please raise your hand and one of the researchers will attend to you in private. Thank you!

We will ask you some hypothetical questions to make sure that you have understood the instructions.

Remember that buyers choose what to do for each possible value of the die. Buyers have to make this decision before the computer throws the die and, therefore, before they know the value of the die.

Imagine that the buyer has chosen the following:

If the dice shows a **1**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **900** pesos?	○ Don't Bargain ◉ Bargain
If the dice shows a **2**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1000** pesos?	○ Don't Bargain ◉ Bargain
If the dice shows a **3**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1100** pesos?	◉ Don't Bargain ○ Bargain
If the dice shows a **4**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1200** pesos?	◉ Don't Bargain ○ Bargain
If the dice shows a **5**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1300** pesos?	◉ Don't Bargainr ○ Bargain
If the dice shows a **6**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1400** pesos?	◉ Don't Bargain ○ Bargain
If the dice shows a **7**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1500** pesos?	◉ Don't Bargain ○ Bargain
If the dice shows a **8**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1600** pesos?	◉ Don't Bargain ○ Bargain
If the dice shows a **9**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1700** pesos?	◉ Don't Bargain ○ Bargain
If the die shows a **10**, do you want to bargain **electronically** to purchase of the cup, or do you prefer not to bargain and take **1800** pesos?	◉ Don't Bargain ○ Bargain

- If the die shows a 10, will there be a negotiation?
- If the die shows a 4, will there be a negotiation?
- If the die shows a 1, will there be a negotiation?

Online Appendix L Prolific survey

This is a transcription of the survey that I administered to the Prolific representative sample. I ran this survey in Qualtrics. Questions 3, 7 and 11 are attention checks. Questions 12 and 13 are related to a separate project.

Thank you for taking this survey. In this survey, we want to learn about your personal opinions and habits. Please take the time to read all the questions carefully, and reflect before you give us an answer. Thoughtful answers help us to do good science. Thank you for helping us do good science.

1. Please consider your bargaining skills and rate them on a scale from 1 (very low) to 5 (very high).



2. Would you say that sometimes bargaining produces in you some negative feelings that you would rather avoid? (e.g. it makes you feel uncomfortable / awkward / nervous / anxious…)

   - Yes
   - No

3. In this survey, we want to learn about your personal opinions. However, to get meaningful answers, it is important for us that, while completing the survey, you are attentive and read everything carefully. Therefore, sometimes we will ask you really easy quiz questions that we call attention checks. An example of such an attention check is the following question: In Supermarket A, the gallon of milk costs 3 dollars, while in Supermarket B, it costs 4 dollars. Where is the gallon of milk cheaper?

   - Supermarket A
   - Supermarket B
   - They cost the same

4. If you had to bargain, do you think that you would reach a more favorable agreement bargaining face to face, or bargaining through an electronic chat?

   - Face to face
   - Electronic chat
   - No difference

5. Think on the negative feelings that one might feel during negotiations (anxiety, awkwardness, nervousness…). Are those feelings more intense when you bargain face to face, or when you bargain through an electronic chat?

   - Face to face
   - Electronic chat
   - N/A

6. Please indicate which of the following consumer-to-consumer marketplaces you have ever used to buy, sell or rent stuff. You can select more than one option:

   - Facebook Marketplace
   - Craigslist
   - Other (please specify which one):____
   - Not sure (please elaborate):____
   - None

7. Please complete this sentence. “Tomatoes are…":

   - Red
   - White
   - Black
   - Black and white

8. Consumer-to-consumer marketplaces, such as Craigslist, Facebook Marketplace, and others, typically allow for electronic communication. In the last five years, how many times have you used the chat in any of these platforms to ask for a discount?

    - None
    - Between 1 and 5
    - Between 6 and 10
    - More than 10

9. Imagine that you find an item that you need (for example, a second-hand bike, a couch or an apartment to rent). Suppose that this item is significantly out of your budget. Please read the possible answers very carefully, and select the one that best describes you:

    - If I like the item, but it is significantly out of my budget, I would probably not try to haggle. I would just continue the search until I find something that fits my budget.
    - If I like the item, it is likely that I will make an aggressive offer and try to haggle, as long as I can do it electronically. But I would not do it if I had to do it face to face.
    - If I like the item, it is likely that I will make an aggressive offer and try to haggle, but only if I can do it in person, not electronically.
    - If I like the item, even if it is significantly out of my budget, it is likely that I will make an aggressive offer and try to haggle, no matter whether I have to do it through an electronic chat or in person.

10. Imagine that you find an item you are looking for, but it is slightly out of your budget. Please rate how likely it is that you will try to make an offer, instead of continuing the search. (1 - very unlikely; 5 - very likely)

    Make an offer if the negotiation is electronic (chat).

    1  5

    Make an offer if the negotiation is in person (face to face).

    1  5

11. Imagine that you are in the middle of a negotiation. Suppose that in this negotiation you are bargaining with Susan, who is 5 feet tall, and with Mark, who is 6 feet tall. Who is taller, Mark or Susan?

    - Susan is taller
    - Mark is taller
    - They are equally tall
    - None of the above

12. Imagine that you find an item that you like and that fits your needs, but it is significantly out of your budget. You can decide to haggle face to face, asking for a high discount, or you can continue searching until you find something that is within your budget. Consider two different situations. In both of them, the owner, who apparently loves the item, is making the final decision. But in one case you are bargaining directly with the owner, and in the other, you are bargaining with an agent who is selling the good on behalf of the owner. Would it make any difference?

    - I would make an aggressive offer and try to haggle in any case.
    - I would try to haggle if an agent is selling the good, but I would not ask for a high discount directly from the owner.
    - I would try to haggle if I could do it directly with the owner, but I would not ask for a high discount from a selling agent.
    - I would not try to haggle—I would continue the search until I found something closer to my budget.

13. Imagine that you find an item you are looking for, but it is slightly out of your budget. Please rate how likely it is that you will try to make an offer, instead of continuing the search. (1 - very unlikely; 5 - very likely)

Make an offer if the negotiation is face to face with a selling agent.

1  5

Make an offer if the negotiation is face to face with the owner.

1  5

14. Imagine that you are selling an item (for example, a second-hand surfboard). Would you rather allow potential buyers to reach you both electronically (chat, WhatsApp, text...) and by phone call?
    - Yes, I would allow them both
    - No, I would only allow them to reach me by phone calls
    - No, I would only allow them to reach me electronically
15. In the last five years, how many times have you used consumer-to-consumer platforms (e.g. Craigslist, Facebook Marketplace...) to advertise something:
    - Never
    - Between 1 and 5 times
    - Between 6 and 10 times
    - More than 10 times
16. Please indicate your gender:
    - Male
    - Female
    - Other (please, specify):____
    - N/A
17. Indicate your Prolific ID. Please make sure that you indicate it correctly, so we can verify that you took the survey and pay you in a timely manner. _____
18. Do you have any additional comments?_____